

Master Thesis

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Macroeconomic and Bank-specific Drivers of Non-Performing Loans and the EBA's EU-wide Transparency Exercise: An Empirical Investigation.

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Abstract

*The objective of this paper is to analyze the impact of macroeconomic and bank-specific factors on Non-Performing Loans (NPL) in the EU-banking sector using an extensive, granular dataset on major European banks gathered by the European Banking Authority (EBA) in the context of the EU-wide Transparency Exercise for the period 2014-2019. The analysis takes into consideration disaggregated bank's portfolio characteristics in terms of different exposure types (households, non-financial corporations, financial corporations other than credit institutions), geographic distribution of exposures (worldwide-country of destiny), intensity of cross-border diversification and initial levels of NPL. Moreover, the characteristics of the legal system and the regulatory status under the Single Supervisory Mechanism (SSM) are also analyzed as potential determinants of NPL. Traditional panel data methodologies (Random Effects/Fixed Effects/GMM) and quantile regression methods for panel data (Pooled/Penalized Fixed Effects) have been applied for the empirical approach. **The empirical results show the importance of bank-specific and macroeconomic factors as well as the characteristics of the legal system for explaining the development of NPL in the European banking system.** In this context, Common Equity Tier1 Ratio (CET1) and Returns on Risk adjusted Capital (RORAC) are identified as the main bank-specific determinants of NPL, whereas inflation rates, government gross debt as percent of GDP and unemployment rates are found to be the main macroeconomic determinants of NPL. With regard to the characteristics of the legal system, recovery rates after NPL resolution proceedings are also found to be an important determinant of NPL. **The "new" findings also show that the relative importance of the determinants of NPL depends greatly on the exposure type under consideration, the original amount of NPL and the intensity of cross-border diversification in a bank's portfolio.** In this sense, this thesis provides some new empirical evidence that further recognizes the importance of understanding the heterogeneity in the European banking sector for explaining the development of NPL in Europe. **Furthermore, the "new" findings also suggest that regulation and supervision under the Single Supervisory Mechanism (SSM) contributes significantly to a reduction of NPL in the European banking system.** In light of these results, the supervisory authorities should further concentrate efforts on procedures and systems for improving risk management tools at a disaggregated level, thus accounting for the large heterogeneity in the European banking system, while promoting at the same time a further integration of the European financial sector within the established European institutions. Moreover, the presented results are also relevant for the design of recommendations to banks on NPL as published by the ECB and other national supervisory authorities regarding the development and implementation of adequate NPL strategies and Early Warning Indicators (EWI). **In this context, the present analysis provides a new example of the potential use of the EBA's data and a new application of the quantile regression methodology for panel data.***

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List of Abbreviations

AMC	Asset Management Company
BCBC	The Basel Committee on Banking Supervision
BIS	The Bank for International Settlements
CBDI	Cross-Border Diversification Index
CET1	Common Equity Tier 1
EBA	The European Banking Authority
ECB	The European Central Bank
ECOFIN	The European Council for Economic and Financial Affairs
EEA	The European Economic Area
EU	The European Union
EWI	Early Warning Indicators
FSB	The Financial Stability Board
FD	First Differences
GMM	Generalized Method of Moments
HC	Heteroscedasticity-consistent
HHI	Hirschmann-Herfindahl Index
IMF	The International Monetary Fund
LSDV	Least Squares Dummy Variable
QR	Quantile regression
NPL	Non-Performing Loans
OLS	Ordinary Least Squares
PFEQR	Penalized Fixed Effects Quantile Regression
SRM	The Single Resolution Mechanism
SSM	The Single Supervisory Mechanism

1 Introduction

Since the aftermath of the financial crisis that started in 2008, many European banks were left facing large accumulations of Non-Performing Loans (NPL) in their balance sheets. During the last years, this development have been a very important issue for banks, supervisors and market participants in the European Union.

In 2014, a harmonized definition of NPL across EU Member States was introduced by the European Banking Authority (EBA), representing the basis for an adequate monitoring and comparability of NPL across European banks. By 2015, around 1 trillion Euros were reported by European banks to the European Banking Authority (EBA). In 2017, the European Council for Economic and Financial Affairs (ECOFIN) recognized the importance of providing a bold response to these accumulations of Non-performing Loans (NPL) in the European banking sector, thereby pointing out the need of a coordinate action plan both to reduce current levels and to avoid their re-emergence in the future. In this sense, the Council highlighted the need for solutions at European level for a problem that has until recently been addressed at national level (European Banking Authority, 2020a). Moreover, the expected dramatic economic consequences of the Covid-19 pandemic that started in 2020 have added new concerns on the development of NPL in the European Union, reawakening the interest in this topic.

In this context, a proper understanding of the determinants and evolution of NPL is critical to perform an adequate microprudential monitoring of the banking system. It becomes even more relevant for macroprudential purposes as the size and evolution of NPL may increase concerns in terms of systemic risks, such as a threat to the overall financial stability and to the basic role of the banking sector as financial intermediary and main provider of credit into the real economy (Suarez and Sánchez Serrano, 2018; Fell et al., 2018; Bending et al., 2014). The accumulation of NPL may thus involve severe social costs, specially during and in the aftermath of financial crises. For instance, increasing levels of NPL for longer periods in the banking system are generally followed by a reduction of the credit supply. In this situation, banks are characterized by having higher funding and recovery costs as well as limited capability to raise new capital (Suarez and Sánchez Serrano, 2018). Moreover, the presence of large stocks of NPL in a bank's portfolio can also generate negative externalities in relation to borrowers' payment culture. For instance, an increase in the number of borrowers defaulting their loans may create individual incentives for other healthy borrowers not to repay their debts if they perceive that the bank's capabilities to pursue defaulted borrowers is limited due to the lack of resources for loan monitoring (Suarez and Sánchez Serrano, 2018).

Since 2016 the EBA has also been disclosing for the public detailed bank-by- bank supervisory data gathered in the context of the EU-wide Transparency Exercise on an annual basis. This data enhances the comparability of problem loans across European banks by using an harmonized definition of NPL and providing an extensive granularity (exposures are broken down by banks, asset classes, the largest countries of counterparty, default status and calculation methods, etc...). Additionally, the quality controls undertaken by the EBA in the context of its supervisory competencies ensure high reliability of the provided data (Kerbl and Döme, 2017; Turk-Ariss, 2017).

In the academic research, scholars and international organizations have also pointed out the presence of some challenges in the NPL literature when using individual bank data despite the need of more disaggregated analyses of NPL drivers (International Monetary Fund, 2019; Cerutti and Schmieder, 2012). These challenges can be summarized in the following points:

- Data is usually provided on a consolidated basis and its availability is limited due to data protection regulation and banking law.
- Private platforms like Dealscan, Datastream, Bankscope or Debtwire suffers from a representativeness bias and data quality issues
- The use of macroeconomic information from the bank's country of origin does not totally account for cross-border portfolio diversification in the banking sector when analyzing its impact on non-performing loans by using consolidated data at bank-level.
- The lack of comparability of NPL across banks due to divergences concerning the definition of non-performing loans and the method of compilation of the data across countries.

Therefore, the main purpose of this thesis is to contribute to a further understanding of the determinants of NPL by analyzing the impact of macroeconomic and bank-specific factors on non-performing loans in the EU-banking sector using a granular dataset on major European banks gathered by the European Banking Authority (EBA) in the context of the EU-wide Transparency Exercise for the period 2014-2019. Motivated by the granularity of this dataset, important aspects of a bank's portfolio, such as different exposure types (households, non-financial corporations, financial corporations other than credit institutions), geographic distribution of exposures (worldwide-country of destiny), intensity of cross-border diversification and initial levels of NPL will be taken into consideration. Moreover, the characteristics of the legal system and the regulatory status under the Single Supervisory Mechanism (SSM) are also analyzed as potential determinants of NPL.

The use of the EBA data in this thesis is even more relevant as there are rarely any academic paper in the Non-Performing Loans (NPL) literature that made any use or reference to the EBA

transparency exercises Database despite its high quality standards, the novelty of its compilation and its usefulness to handle certain difficulties concerning the analysis of NPL. More details about its characteristics, advantages and the institutions behind its compilation will also be provided in the following sections. For the empirical analysis quantile regression models for panel data will also be estimated. These methods provide new insights into the determinants of NPL by allowing a disaggregated estimation of their impact at different NPL levels along the conditional distribution of NPL in the EU banking sector.

In summary, the motivation for this present topic stems from the following reasons:

- EU authorities and the academic literature claimed that any policy measure and regulatory approach in the European Union with respect to problem loans required a deep understanding of their underlying determinants
- To explore the factors that contribute to this understanding, whether those factors have a macroeconomic or bank-specific nature and their impact on NPL
- The availability of disaggregated supervisory data on major European banks disclosed by the EBA allows new contributions in the NPL literature, for instance, a detailed analysis of disaggregated NPL-portfolios at bank-level split by exposure type, geographic allocation of exposures and a bank's supervisory status
- To provide an empirical example of the potential benefits from using the data of the EBA's EU-wide Transparency Exercise for supervisory and academic purposes with respect to certain challenges presented in the NPL literature
- New developments in the quantile regression methodology allow a deeper exploration of the impact of these factors at different quantiles along the conditional distribution of NPL in the EU banking sector

In light of these considerations, the following research questions are formulated:

General Question:

- *What is the impact of macroeconomic and bank-specific factors on Non-Performing Loans (NPL) in the EU-banking sector?*
- *Which of the factors have a significant and positive correlation with NPL?*
- *Which of the factors have a significant and negative correlation with NPL?*

Specific Questions:

- *What is the impact of Supervision under the Single Supervisory Mechanism (SSM) on Non-Performing Loans?*
- *What is the impact of geographic diversification on Non-Performing Loans?*
- *Are there different conclusions given:*
 - ◊ Different Exposure Types:
 - Financial corporations other than credit institutions
 - Households
 - Non-financial corporations
 - ◊ Different initial levels of NPL in portfolio

In the context of these general questions, the following research hypotheses will be analyzed in this thesis:

- H_1 : NPL are mainly determined by bank-specific and macroeconomic factors
- H_2 : Business cycle indicators are significantly related to NPL
- H_3 : Inflation rates are significantly related to NPL
- H_4 : Public finance indicators are significantly related to NPL
- H_5 : Terms of trade indicators are significantly related to NPL
- H_6 : The size of a bank is significantly related to NPL
- H_7 : Legal aspects of credit access and NPL resolution are significantly related to NPL
- H_8 : Bank profitability and cost efficiency indicators are significantly related to NPL
- H_9 : Bank capital adequacy ratios are significantly related to NPL
- H_{10} : The level of geographic diversification in portfolio is significantly related to NPL
- H_{11} : Supervision under the SSM has significant effects on NPL
- H_{12} : Magnitude and direction of the effects on NPL varies across exposure typesThe effects on NPL varies conditional on the initial level of NPL
- H_{13} : The effects on NPL vary conditional on the initial level of NPL

Additionally, the findings in this thesis might also provide interesting insights for supervisory and regulatory authorities in the EU banking system as well as scholars interested in the NPL literature. Firstly, European supervisory authorities and Central Banks might find the results relevant for the definition of Regulatory requirements or general standards with respect to lending policies, capital requirements and overall banking operations. In particular, the results might also be interesting for the preparation of recommendations on NPL strategies, Early Warning Indicators (EWI) or bank-stress testing methodologies drafted by the competent supervisory

bodies over banks operations in any of the EU member states. Finally, the findings also provide new contributions in the academic research on NPL in terms of the granularity and novelty of the data (EBA's EU-wide Transparency Exercise), the empirical methodology (quantile regression methods) and the focus on the disaggregated aspects of loan portfolios.

2 NPL Literature and Institutional Background

2.1 NPL Literature Review

Many studies have shown particular interest in the identification of the relationships between bank-specific and macroeconomic determinants and the bank's asset quality, using non-performing loans data as main indicator. In particular, the literature is divided in two lines of research: one line of research exclusively focusing on the macroeconomic determinants of non-performing loans and other line of research that also analyze the impact of bank-specific indicators. In this context, macroeconomic conditions are considered as external events that are likely to directly affect the debtor capacity to service loan obligations, while bank-specific determinants allows a focus on the factors explaining the variability of non-performing loans across banks.

2.1.1 Linkages between Macroeconomic Indicators and NPL

Regarding the macroeconomic linkages to bank's loan quality, there is a significant amount of studies. In general, these studies could be subdivided in studies including general macroeconomic indicators related to the Business cycle (GDP growth, unemployment rates), Terms of trade(current account deficits/surplus, balance of payments) and Price levels (inflation), while other studies also include government related indicators such as sovereign debt and fiscal deficit levels into their models. Finally, a reduced number of papers also analyzes the impact of indicators related to the overall strength of the legal system, legal protection of creditors/debtors and supervisory regulation.

In this context, most researchers identified the importance of the macroeconomic environment in explaining the performance of loan portfolios in the banking sector (Makri et al., 2014; Dimitrios et al., 2016; Louzis et al., 2012). For instance, Makri et al. (2014) and Dimitrios et al. (2016) based their hypothesis in the idea that an expansionary phase in the economy in terms

of GDP growth and low unemployment levels is assumed to improve the ability of economic agents to service their loans as a sufficient stream of income is provided for the household sector. This situation improves the ability to repay bank loans as well as boosting private demand, with also positive consequences for the corporate sector. Once the expansionary phase comes to an end, lines of credit are also extended to lower-quality debtors, leading to a progressive increase in defaulted loans. Moreover, if this situation is followed by higher unemployment rates, more borrowers will be unable to serve current loan obligations, further increasing the amount of defaulted loans in the bank's portfolio.

The relation between price levels and non-performing loans have also been researched in several papers (Salas and Saurina, 2002; Klein, 2013; Nkusu, 2011). Inflation rates may affect the capacity of borrowers to service their loan obligations in a positive or a negative way, depending on the transmission channel under consideration. On the one hand, higher inflation rates tend to cut down the real value of outstanding loans, improving the borrower's capacity to repay loans. On the other hand, these higher rates may also have negative effects on real income under the assumption that nominal wages are sticky. This last situation may make debt-repayment for borrowers more difficult as real wages and real income are cut down. In this context, the causal relation between inflation rates and non-performing loans can be negative or positive.

A small number of authors, such as De Bock and Demyanets (2012) and Kaminsky and Reinhart (1999), also discuss the impact of trade related indicators (current account deficits, balance of payments, exchange rates, etc...) on the level of non-performing loans as potential drivers of bank asset quality. For instance, persistent current account surpluses or deficits could be indications of unbalanced economies, relying too much on savings or consumer spending. These indicators can also signal the lack of competitiveness in the economy of a country in relation to their neighbors. These issues regarding asymmetries and macroeconomic imbalances among European countries in terms of trade and their effects on the economic and financial instability in the European Union have gained importance in the last years.

The relation between sovereign debt and financial crises in the banking sector have also been identified in many studies after the severe financial crisis in 2008 and further sovereign debt developments, in which two main channels of transmission are considered. First, a worsening of the finances in the public sector sets limits on the credibility of central banks in the market. This increases the pressure on banks for additional liquidity, leading to a reduction in lending and loan refinancing. Moreover, an excessive increase in sovereign debt may lead to the need of severe fiscal measures, potentially affecting social expenditure and public wages. These negative shocks in income have impacts on the household and corporate sector by causing a general decrease in demand. These circumstances may reduce the ability of the economic

agents to meet loans obligations (Reinhart and Rogoff, 2011; De Bock and Demyanets, 2012)

2.1.2 Rule of Law, Regulation and Supervisory practices

In recent years, some important studies focused on the question of how regulation and supervisory practices promote banking development, banking performance and banking stability. For instance, supervisory organizations pointed out the important role of the regulation and monitoring of bank activities as well as the definition of rules regarding capital ratios as important buffers against financial instability and banking crises. This line of research also estimate the effects of membership to an overall official supervisory power on non-performing loans as the characteristics of the supervisory body have important consequences in bank's risk-taking behaviour and practices. For instance, the independence of the supervisor when defining rules for loan classification, liquidity requirements, definition of guidelines and restrictions on loan practices is an important characteristic to consider as a potential determinant of non-performing loans (Barth et al., 2004; Boudriga et al., 2009).

In addition, the legal aspects concerning lending activities and the overall quality of law enforcement as well as the relative protection of creditors and debtors in loan contract obligations within the legal system may also have important consequences for debt performance and NPL resolution. In this regard, Mora-Sanguinetti et al. (2017) and Rodano (2021) argued that differences in the quality of the enforcement institutions may impact the levels of non-performing loans in a banks' portfolio due to the fact that fewer loans are defaulted in countries where efficient declaratory procedures are in place. For instance, variations in the length of time required to recover debt or in the final recovered loan amount, allowing collateral or other assets of the borrower to be repossessed once the procedure for the resolution of non-performing loans concluded, may have important effects on the creditor behaviour for the loan decision process as well as on the willingness of debtors to meet their contractual obligations. However, these factors may also affect debtor behaviour when deciding to serve its loan contract obligations. For instance, differences in the debt restructuring frameworks and in the laws regulating private sector insolvency in regard to the intensity of the creditor/debtor legal protection may have important consequences for determining the levels of non-performing loans. Furthermore, complex legal frameworks or congested judicial procedures may also discourage the undertaking of legal actions at resolving NPL by foreclosing on available collaterals despite potential losses as mentioned by Suarez and Sánchez Serrano (2018). Additionally, the existence of an integrated or unified legal framework regulating the accessibility and disposal of credit information and the availability of data on borrowers' loan repayment behaviour, which is provided by credit

bureaus or public credit registries and used by banks for loan underwriting purposes, are also relevant factors that may affect the amount of risk in a bank's portfolio and the levels of non-performing loans.

In regard to the resolution of NPL, the existence and conditions prevailing in secondary markets for NPL have also been given special attention in recent times (Avgouleas et al., 2021). A secondary market for NPL, where banks can easily sell their NPL to investors, may contribute to accelerate the NPL adjustment process as banks try to restructure portfolios with problem loans. However, some authors argue that one of the main challenges facing this kind of markets is the presence of asymmetric information. In the case of a secondary market for NPL, these asymmetries mean that potential buyers tend to have less information on the quality of NPL than the banks selling their NPL stocks and therefore, the buyers will try to underbid the fair value of the portfolio, with the result that the bank will either continue the selling operation at depressed prices, which lead to potential losses, or withdraw the selling offer (Suarez and Sánchez Serrano, 2018). In this context, the need of an adequate institutional and legal framework is also essential to reduce some of the negative consequences that may lead to forced sales of NPL portfolios at loss or prevent that buyers and sellers reach an agreement to sell these at fair value. In other words, the reduction of information asymmetries by establishing a proper regulatory framework for secondary markets of NPL is an important instrument for the reduction of NPL in the banking sector. In this regard, Suarez and Sánchez Serrano (2018) and Avgouleas et al. (2021) suggested that the implementation of an adequate legal framework for the creation and operation of (public and private) asset management companies (AMCs) specialized in buying and managing NPL in secondary markets represents a desirable development. In fact, the sales or securitizations in the secondary market of NPL have been shown to be one of the main factors behind the reduction of problem loans, specially for those countries with high levels of NPL (European Banking Authority, 2020a).

2.1.3 Linkages between Bank-specific Indicators and NPL

A significant group of papers in the NPL literature also analyze the impact of bank-specific indicators on the proliferation of non-performing loans. In general, this line of research examined the intersection between a bank's asset quality and other important indicators of bank performance such as cost efficiency, lending policy and bank capitalization. These factors are important elements to consider when evaluating the quality of a bank management and the level of risk that the organization is prepared to accept for business decisions. In this context, following the initial study of Berger and DeYoung (1997), several lines of reasoning or hypotheses

are presented in the literature to explain the causal relation between bank-specific factors and the amount of non-performing loans at bank level, namely bad management, skimping, moral hazard, diversification and too big to fail.

Under the "bad management" hypothesis, an increase in non-performing loans is found to be positively related with low cost efficiency as a signal of poor senior management practices. Failures in practicing an adequate loan underwriting, poor performance in credit scoring or excessive risk-appetite may lead to grant loans to an elevated proportion of customers with higher risks of default. Moreover, this hypothesis also covers the reasoning behind the identification of a correlation between banks' profitability and NPL as it is assumed that highly profitable banks have fewer incentives to enter into riskier activities, thereby showing a lesser amount of problem loans. In most papers, financial soundness indicators such as return on assets (ROA), return on equity (ROE) or net interest margins (NIM) are used as measures of bank-profitability. Furthermore, the implementation of poor practices in monitoring and controlling borrowers after loans are issued can also be considered as a bad management practice, for instance, the implementation of poor collection and/or refinancing measures in order to avoid that loans in arrears surpass the usual threshold of 90 days past-due required for their classification as non-performing loans (Berger and DeYoung, 1997).

Nevertheless, the relationship between cost efficiency and non-performing loans may also be described by a two-way causality mechanism. This idea leads to the "skimping" hypothesis, which highlights the importance of providing additional resources to improve the underwriting, monitoring and control mechanisms in an organization despite the potential increase in measured costs. In this context, high cost efficiency will be positively correlated with the accumulation of non-performing loans. This line of reasoning recognizes the presence of a trade-off between cost efficiency and the allocation of additional resources for underwriting and monitoring. Banks that put fewer efforts to ensure a better loan quality may be more cost-efficient in the short-term but show higher levels of non-performing loans in the long-term (Berger and DeYoung, 1997; Louzis et al., 2012).

Regarding the "moral hazard" hypothesis, it refers to the classical "moral hazard" problem when a party takes excessive risks and part of this risk is to be carried by another party. In the literature, this problem is usually linked to concerns regarding bank capitalization. In this line of reasoning, managers in low-capitalized banks have more incentives to undertake riskier lending activities regardless of underwriting and monitoring standards. Furthermore, better capitalized banks are considered to operate more efficiently than less capitalized banks because capital adequacy ratios may serve as instruments to control for improper risk-taking behaviour. In resume, a negative causal relation between the degree of bank capitalization and the levels

of non-performing loans may be identified in this context (Berger and DeYoung, 1997; Louzis et al., 2012; Sinkey and Greenawalt, 1991; Ghosh, 2015)

Nevertheless, Keeton and Morris (1987) firstly suggested that this relationship may also show a two-way causality mechanism by arguing that managers in higher capitalized banks also have the possibility of implementing more flexible lending policies given a compliance with the respective supervisory regulation. The implementation of more flexible lending policies is also explained by the presence of a greater buffer in capital adequacy for undertaking riskier lending activities, which may lead to a major accumulation of non-performing loans in the portfolio, without excessively increasing the risk of insolvency. For the contrary, low-capitalized banks with a capital adequacy ratio under the regulatory minimum may be requested to apply capital adjustments to meet supervisory regulation that restricts their flexibility in defining lending policies. These adjustments will then take place either by increasing capital, if assuming constant (risk-weighted) assets, or reducing (risk-weighted) assets, if assuming constant capital. In this situation, banks will likely have to implement more restrictive measures, and therefore, show less willingness to engage in riskier lending activities that may increase the levels of non-performing loans in their portfolio. Under this last line of reasoning, the causal relation between the degree of bank capitalization and the levels of non-performing loans will be positive as considered in some papers (Boudriga et al., 2009; Ghosh, 2017; Kwan and Eisenbeis, 1997; Osei-Assibey and Asenso, 2015).

The strategic aspects of a bank's loan portfolio are also important elements to take into account when identifying determinants of non-performing loans at bank-level as mentioned by Ghosh (2017) and Zhang et al. (2016). For instance, strategic considerations are in place when decisions regarding credit-portfolio diversification and specialization are to be made. In general, diversification opportunities in the banking sector are considered to be positively related with loan quality as more diversified portfolios are assumed to carry less credit risks. Some authors also use bank size as a proxy for diversification opportunities, identifying a negative relation between bank size and non-performing loans (Salas and Saurina, 2002).

Regarding bank size, Berger and DeYoung (2001) pointed out the potential gains in terms of bank efficiency with greater economies of scale as it may be the case for larger banks. In this sense, larger banks are expected to show lower ratios of non-performing loans in their portfolios.

However, bank size has also been related to concerns on the presence of moral hazard in the too-big-to-fail problem, which represents another potential channel to explain non-performing loans at bank-level. In this logic, larger banks have more incentives to take excessive risks under the assumption that due to their size, complexity and interconnectedness with other economies,

they are more likely to be bailed out by the government in case of bank failure or insolvency.

In these cases, traditional market penalties imposed by creditors and other market participants are not properly taken into consideration when forming business decisions. This situation leads to an excessive willingness to engage in risky lending practices under the belief that a potential failure is to be avoided by government protection due to the threat that it represents for the integrity of the financial system and of the national economy as a whole. In other words, banks, whose failure or insolvency may represent a systemic risk for the overall financial stability, may also show higher proportions of non-performing loans in their portfolios (Louzis et al., 2012). In this view, bank size is assumed to be positively related to non-performing loans.

2.1.4 Geographic Diversification and Disaggregated Loan Aspects

There are few papers that assess the effects of geographic diversification on bank efficiency. According to traditional portfolio theory, the main advantage of portfolio diversification is the reduction of potential losses due to over concentration. Geographic diversification is a technique that aims at reducing risks under the assumption that resources allocated in different regions will react differently to the same event (Berger and DeYoung, 2001). For these reasons, an increase in geographic diversification is expected to lead to a reduction of the risk of financial loss and, therefore, to lower levels of non-performing loans in the portfolio. Moreover, geographic diversification in banking may facilitate the flow of internal cash through the organization, thereby reducing the need of external funding costs (Morgan and Samolyk, 2016). In addition, geographic diversification may also benefit from the presence of economies of scale as explained by the fact that it is easier for larger banks to allocate resources beyond their own country of origin. In a regulatory context, the intensity of geography diversification is also important because the existence of heterogeneities in the bank supervisory and regulatory framework may create incentives to diversify or concentrate loan portfolios in one or several countries/regions. In this situation, the presence of different regulatory and risk-control requirements for banks regarding loan exposures in countries other than the country of origin may have implications for the overall bank's asset quality in a consolidated basis.

However, geographic diversification may also create additional costs due to problems related to the learning curve and agency costs that may cause potential inefficiencies in the business decision process as well as in the monitoring and controlling of borrowers in cross-border economies. As banks tend to diversify their portfolio by entering into new markets, the collection of soft information becomes more difficult because the distance between headquarters

and subsidiaries increases and the transmission of information across management layers becomes more challenging, thereby deteriorating the effectiveness of banks' monitoring activities and increasing overall agency costs (Morgan and Samolyk, 2016). Furthermore, Hayden et al. (2006) suggested that banks' decision to enter into a country with an already well-developed banking sector may cause an adverse selection problem in the pool of borrowers. For instance, foreign banks in already well established banking sectors may have more difficulties to differentiate new borrowers from old borrowers, who have been rejected by their previous bank, than national banks fully operating in their home market. In this sense, banks with greater geographical diversification are more affected by the presence of asymmetric information and adverse selection problems with respect to borrowers than less geographically diversified banks. In this line of reasoning, an increase in the cross-border diversification of a bank's portfolio may be positively correlated with a higher amount of non-performing loans (Hayden et al., 2006).

Despite the fact that the largest part of the academic literature explored the determinants of NPL from an aggregate view, Nkusu (2011), Ghosh (2017) and Manz (2019) also recognized the importance of considering different loan types when analyzing the determinants of non-performing loans. This consideration is justified by the fact that different weights of one or more specific sectors in a bank's portfolio composition may play an important role in explaining the amount of NPL in a consolidated basis (Keeton and Morris, 1987). Different loan types face different credit risks and react differently to specific sector developments, thus also reflecting differences in the potential amount of non-performing loans due to external and internal events (Salas and Saurina, 2002). In this context, an analysis of the impact of NPL determinants in a bank portfolio from a disaggregated perspective in terms of loan type is also favoured by a part of the NPL literature.

2.1.5 Empirical methods and Regional Scope

With regard to the empirical methods, there is a vast amount of models and methodologies available in the literature. The most common applied methodologies cover static and dynamic panel models such as Random/Fixed Effects and Generalized Method of Moments (GMM) estimations, Granger-causality techniques and logit regressions. Additionally, the use of Vector Autoregression Regression methods (VAR) in this line of research has also increased during the last years given that long panel datasets with many time periods are also getting more easily available as observed by Manz (2019). Most recently, some papers have also been published that make some use of quantile regression methods in related topics, although very few in nature. In this sense, Sánchez Serrano (2021) analyzed the impact of NPL on bank lending by using

pooled quantile regression methodology and data from the EBA Transparency Exercises. Also in recent times, Karadima and Louri (2020) applied a penalized quantile regression model for dynamic panel data in an attempt to analyze the impact of market power in NPL with data obtained from the Orbis BankFocus (BankScope) Database. The available literature also provides many studies covering different regions and countries, with some papers focusing on specific countries while others expanding their analyses to a group of countries or regions. There also are differences in the source and level of aggregation of the data, which vary greatly among papers. Most studies have tended to use aggregated data, mainly driven by the fact that aggregated data at country level is more easily available than data at bank level, whose use for academic research is subject to more restrictions due to data protection regulations and bank law. In this context, studies using data at bank level are either compelled to subscribe private database providers subject to the payment of fees such as the Orbis BankFocus (BankScope), Dealscan, Datastream among others, or to use data gathered by national central banks or institutions, which is usually not public and only available to specific circle of researchers, central bank employees or subject to authorization procedures. In this situation, the replicability of these studies is quite challenging and, therefore, any assertion about their reliability under scientific standards may easily be put in doubt. Furthermore, the NPL literature also varies greatly in terms of the time period under consideration (pre-crisis and post-crisis) as well as the periodicity of the data (yearly, quarterly, semesterly).

2.2 The European Banking Authority (EBA) and the Single Supervisory Mechanism (SSM)

2.2.1 The European Banking Authority (EBA)

The European Banking Authority (EBA) is an independent EU Authority established on 1 January 2011 as part of the European System of Financial Supervision (ESFS) to ensure effective and consistent prudential regulation and supervision across the European banking sector.

With the EU-wide Transparency Exercise, the EBA has been disclosing supervisory data on an annual basis since 2016 that provides the public and market participants with updated (quarterly/semesterly) information on the financial conditions of EU banks at the highest level of consolidation. This data is characterized by its extension and granularity with portfolio-by-portfolio financial items with respect to major European banks. In particular, the EU-wide Transparency Exercise contains detail information about loan exposures, asset classes, default status and important balance sheet items broken down by bank, and it covers between 80% and

85% of banking assets in the European Union (European Banking Authority, 2019; European Central Banking Supervision, 2017). These exercises mainly aim to foster market discipline in the European single market and enhance transparency in the EU banking sector.

Furthermore, the data from the EU-wide Transparency Exercise makes use of a harmonized definition of NPL across European countries implemented by the EBA technical standards and published in 2014 by the European Banking Authority (EBA) for supervisory reporting of non-performing loans and forbearance (European Central Banking Supervision, 2014). These technical standards represent a first time attempt to create a hard-law instrument that establishes a common definition of NPL by providing a common framework across the EU to accommodate and to harmonize differing national practices without modifying accounting and regulatory standards. The new common framework significantly improves the comparability of NPL across individual banks operating in the European banking and financial sector.

According to the EBA's EU-harmonized definition, a non-performing exposure is in general an exposure that is 90 days past-due (material exposure) or unlikely to be repaid in full without collateral realization (irrespective of any past-due amount or of the number of days past-due), or impaired/defaulted according to the applicable accounting or regulatory framework (European Central Banking Supervision, 2014).

In recent times, the lack of comparability of NPL across countries has been a major concern as expressed by many international organizations such as the Bank for International Settlements (BIS), the Financial Stability Board (FSB), the Basel Committee on Banking Supervision (BCBS) and the International Monetary Fund (IMF). These concerns are explained by the presence of significant divergences regarding the classification system, scope, and contents of NPL across countries and by the fact that the criteria defined in domestic laws and applied by banks for designating a loan as non-performing is to a large extent discretionary. In general, the main factors explaining divergences in NPL definitions across domestic laws are related to (1) the required number of past-due dates before a loan is considered a non-performing loan (2) the "extension" of affected loans, that is whether the categorization of a loan as non-performing loan is to be extended to all outstanding loans from the same borrower in the same bank regardless of their performing status (3) the role played by collaterals in the non-performing criteria and valuation (4) the minimum amount of time for a "restructured or cured" non-performing loan before the reclassification as performing loan takes place again (Monokroussos and Gortosos, 2017).

After the financial and sovereign crises, there were many concerns regarding asset quality and non-performing loans management across the EU-banking sector due to the lack of compara-

bility of NPL figures as reported by the national supervisory authorities. This situation made difficult the achievement of a common understanding, the implementation of an adequate risk assessment and the design of solving strategies for banks with cross-border exposures in the European Union (EU). In this context, the implementation of the EBA technical standards in 2014 meant a step forward towards larger levels of convergence and comparability within the EU-banking sector.

2.2.2 The Single Supervisory Mechanism (SSM)

The Single Supervisory Mechanism (SSM) represents the system of banking supervision for all European Union (EU) member states that are part of the Economic and Monetary Union (EMU), which became fully operational on November 4th 2014, as the first pillar of the European banking union, and it is composed of the European Central Bank (ECB) and the national supervisory authorities of the EU member states.

The SSM seeks to protect financial stability and assist the integration of the European financial sector with the support of the established European institutions and in conjunction with the Single Resolution Mechanism (SRM). The Single Resolution Mechanism (SRM) complements the SSM by centralizing key competences and resources for managing the failure of banks subject to SSM supervision in case of facing serious financial difficulties (European Central Banking Supervision, 2018).

The SSM also provides the legislative and institutional framework that gives the European Central Bank (ECB) sole licensing authority and prudential supervisory competencies over all banks operating in any of the EU member states (except branches of banks from non-European Economic Area (EEA) countries), directly for the most important credit institutions ("significant" banks) and indirectly for the rest of credit institutions ("less significant" banks). The significance of a bank is assessed based on size, significance for the economy of any participating Member State and importance of cross-border activities as established in the SSM Regulation. All countries in the euro area are required to participate, while participation is voluntary for non-eurozone EU member states (The Council of the European Union, 2013; European Central Banking Supervision, 2018).

One of the main priorities for the European Central Bank (ECB) banking supervision is to address asset quality issues. The main focus of the ECB on non-performing loans started in 2014 after a comprehensive assessment that established as main pillars the implementation of asset

quality reviews and stress tests. The ECB banking supervision has since then continued to intensify its supervisory activities regarding NPL monitoring. In this context, the ECB also publishes guidances to banks on NPL that provide recommendations on the development of NPL strategies for all "significant" banks supervised directly under the Single Supervisory Mechanism (SSM) (European Central Banking Supervision, 2017).

In general, these guidances point out that the development and implementation of an NPL strategy should consider internal factors such as a correct assessment of internal capabilities and its capital implications, external factors such as the operating and macroeconomic environment and regulatory factors such as the legal and judicial framework (European Central Banking Supervision, 2017).

3 Data and Variables

All data used in this thesis has been taken from the following public sources:

- * *EBA Transparency Exercise: Databases published between 2015 and 2020*
<https://eba.europa.eu/risk-analysis-and-data/eu-wide-transparency-exercise>
- * *IMF Data*
<https://data.imf.org>
- * *World Bank Data*
<https://datacatalog.worldbank.org/>
- * *World Bank Doing Business Report*
<https://www.doingbusiness.org/>

3.1 Data Preparation

As mentioned in previous sections, an extensive, granular dataset on major European banks from the EBA transparency exercises published between 2015 and 2020 will be used in this thesis along with World Bank and IMF data for the macroeconomic information. The (unbalanced) sample consists of over 130 banks, at the highest level of consolidation, from 27 countries in the EU and the European Economic Area (EEA) on an annual basis between 2014 and 2019. This dataset is very rich in dimensionality in terms of banks, exposure types, geographic allocation of exposures and performing status among others.

Once the full databases from the EBA's EU-wide Transparency Exercise were downloaded for every annual exercise and template class (credit risk -cre-, market risk -mrk-, sovereign exposures -sov-, other templates-oth-), an intensive data preparation process (collecting, cleaning, and consolidation) has been exercised to aggregate the data in .csv files provided by the EBA website for each year into an aggregated database. These .csv files are split by credit risk (tre_cre.csv), market risk (tre_mrk.csv), sovereign exposures (tre_sov.csv) and other templates (tre_oth.csv), covering all available data from the transparency exercises that took place between 2015 and 2020.

In the same way, metadata and mapping files were also downloaded for each EBA transparency exercise that took place between 2015 and 2020. These files were used to:

- * Identify mapping rules for the different data dimensions (performing status, exposure type, portfolio class, etc...)
- * Merge data from different exercises for each bank and item in order to identify common LEI-codes and item numbers (same LEI code for each bank and same item number for each item for the total period 2015-2020). This is necessary because these identifiers and the specific banks participating in each exercise may change from year to year.

For the disaggregation in exposure types, the following item is provided by the EBA:

- **Item Name:** Exposure type [exposure]
- **Definition:** Dimensions of exposure types:
- **Source:** EBA transparency exercises Databases
 - Group:* cre (Credit Risk and Asset Quality)
 - Sheet:* NPE
 - Dimensions (exposure) from EBA 2020:*
 - ◇ 101=Central banks
 - ◇ 102=Credit institutions
 - ◇ 201=General governments
 - ◇ 202=Financial corporations other than credit institutions
 - ◇ 301=Households
 - ◇ 401=Non-financial corporations

However, only the following exposure types provided NPL data:

- Households
- Financial corporations other than credit institutions
- Non-Financial Institutions

The rest of available exposure types (Central Banks, Credit Institutions, General Governments) do not provide NPL data in most cases and, therefore, these types will not be included in the analysis. Furthermore, their relevance to understand the determinants of NPL is usually secondary compared to the previous ones. In other words, the disaggregated analysis in this thesis will focus on households, non-financial institutions and financial corporations other than credit institutions.

The selection of the bank-specific variables was based on using the most common variables analyzed in the NPL literature as long as these were also available in the EBA Transparency

Exercise database for the period 2014-2019. The selection of macroeconomic variables were also based on taking variables commonly examined in NPL analyses. In order to obtain some bank-specific indicators the application of simple calculations using individual items from the EBA transparency exercises was also needed due to the fact that the values of some relevant financial ratios were not directly included in the databases provided by the EBA. These calculations have been validated by comparing the final values at country level with those provided by the EBA in several publicly available reports and dashboards, which also used the EBA transparency exercise databases for analyses at country-level, specifically, the NPL Report 2019 (European Banking Authority, 2020a), key statistics for EBA's EU-wide Transparency Exercise spring-2020 (European Banking Authority, 2020c) and Risk Dashboard for Q4 2019 (European Banking Authority, 2020b). A detailed description of the formulas applied are also available in the next sections.

As leverage exposures were not provided for the EBA Transparency Exercise of 2016, the generated missing values in December 2015 for some bank-specific variables that needed leverage data for the calculation were linearly extrapolated using all available data. In order to obtain a valid extrapolation, all banks with less than four periods of available data (semesterly or quarterly) were excluded from the sample. In other words, linear imputation methods are applied over these missing values for December 2015. Additionally, observations with missing non-performing loan ratios or non-plausible values in any of the regressors were also excluded from the sample.

3.2 Variables

Table 1 present a list of the macroeconomic and bank-specific indicators used in this thesis. As previously mentioned, the selection of these indicators is based on the common practice in the NPL literature as well as their availability in the EBA Transparency Exercise database.

More detailed information regarding variable definitions, formulas, item IDs and sources is available in the appendix. Furthermore, Figures 14 and 15 provide the distribution and the basic scatterplot on NPL ratios of each explanatory variable in the sample.

	Variable	Explanation
Bank-Specific	NPL	Non-Performing Loans to total gross loans
	BankSize_tran	Total Bank Size
	CET1_ratio_tr	Common Equity Tier 1 Ratio
	Cost_to_Income	Cost to Income ratio
	LEVER_tr_ratio	Leverage ratio - using a transitional definition of Tier 1 capital
	NIM_tr	Net Interest Margin ratio
	RORAC	Return on Regulatory (Risk-Adjusted) Capital
	CBDI_index	Cross-Border Diversification Index (CBDI)
Macroeconomic	infl_perc	Inflation, consumer prices (annual %)
	Stgth_legal_index	Strength of legal rights index (0=weak to 12=strong)
	GDP_growth_perc	GDP growth (annual %)
	Reco_rate_dol	Recovery rate (cents on the dollar)
	unempl_rate_imf	Unemployment rate
	curr_acc_bal	Current account balance, percent of GDP
	gross_public_debt	General government gross debt, percent of GDP
	SSM	Single Supervisory Mechanism Dummy

Source: Author's compilation

Table 1: List of Macroeconomic and Bank-specific Indicators

3.3 Cross-Border Exposure-Weighted Macroeconomic Indicators

To determine the macroeconomic drivers on NPL for banks with a cross-border portfolio using data provided on a domestically controlled consolidation basis, the use of exposure-weighted macroeconomic indicators, which account for the geographic composition of exposures in portfolio according to their country of destiny, was required as suggested by Quagliariello (2009) and Fuchs (2020). This is possible because the EBA Transparency Exercise also provided data on original exposures up to the 10 largest countries of counterparty at bank level, covering over 85% of bank assets.

For the calculation of cross-border exposure-weighted macroeconomic indicators, the calculation approach presented by Quagliariello (2009) was followed:

$$y_{it}^* = \sum_{l=0}^{N_{it}} w_{il}^t Y_{lt} \quad \text{with} \quad \sum_{l=0}^{N_{it}} w_{il}^t = 1$$

where w_{il}^t is the cross-border exposure-weight of bank i from country l at time t , Y_{lt} is the macroeconomic variable of country l at time t and y_{it}^* is the cross-border exposure-weighted macroeconomic indicator for bank i at time t . The value of N_{it} varies depending on the number of countries with cross-border exposures for each bank i at time t with $\max(N) = 10$.

Detailed information regarding the item used for determining cross-border exposures:

- **Item Name:** Original Exposure - by exposure class (SA_and_IRB)
- **Source:** EBA Transparency Exercises Databases
 - Group:* cre (Credit Risk)
 - Sheet:* Credit Risk
 - Item Number from EBA 2020:*
 - 2020502 [Original Exposure - by exposure class (SA_and_IRB)]
- **Dimensions:**
 - Country_rank:
 - Taking values from 1 (1st-largest country exposures) up to 10 (10th-largest country exposure)
 - Country/ISO_code:
 - Country identification number and ISO country codes to designate every country as a two-letter combination.

3.4 Cross-Border Diversification Index ($CBDI$)

A geographic diversification index of bank exposures is also introduced into the model in order to account for the effects of cross-border diversification on non-performing loans. The Cross-Border Diversification Index ($CBDI$) has been calculated as the inverse of the Hirschmann-Herfindahl Index (HHI) for each bank i at time t as suggested by Morgan and Samolyk (2016). Formally,

$$CBDI_{it} = 1 - HHI_{it}$$

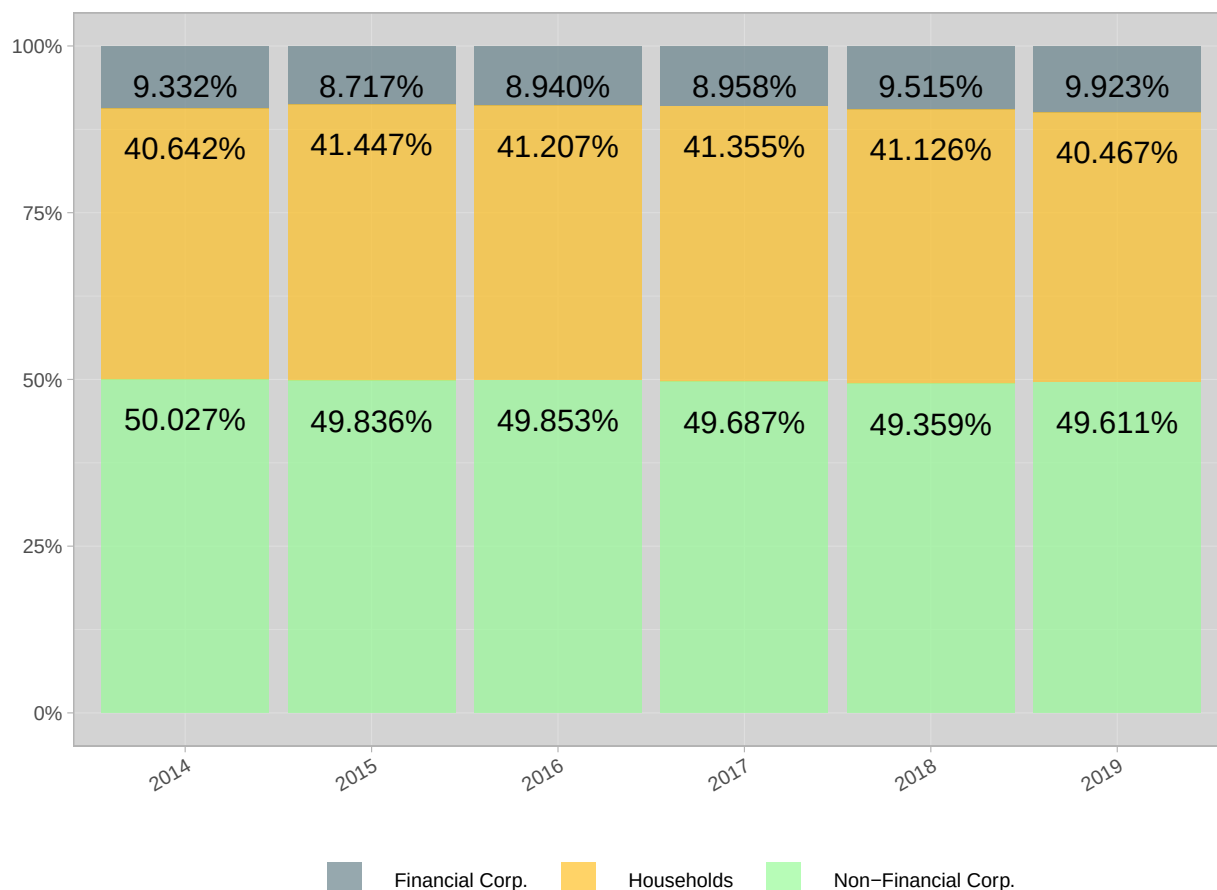
with

$$HHI_{it} = \sum_{j=0}^{N_{it}} \left(\frac{X_{ijt}}{X_{it}} \right)^2$$

Where X_j measures the original exposure towards country j and N_{it} represents the number of countries with the largest cross-border exposures for each bank i at time t with $\max(N) = 10$. The Hirschmann-Herfindahl Index is bounded by $1/N_{it} \leq HHI_{it} \leq 1$. A Hirschmann-Herfindahl Index (HHI) close to one means that the bank's original exposures are very concentrated on certain countries whereas an index close to its minimum means that the banks' original exposures are very diversified across countries (Schertler et al., 2006; Hayden et al., 2006; Berger and DeYoung, 2001). The Cross-Border Diversification Index ($CBDI$) is bounded by $0 \leq CBDI_{it} \leq (N_{it} - 1)/N_{it}$. A Cross-Border Diversification Index ($CBDI$) close to its maximum means that the bank's original exposures are very diversified across countries whereas an index close to zero means that the bank's original exposures are very concentrated on certain countries (Morgan and Samolyk, 2016). The calculations are based on data taken from the same item used for determining cross-border exposures as described in the previous point.

4 Data Visualization

4.1 Non-Performing Loans (NPL)

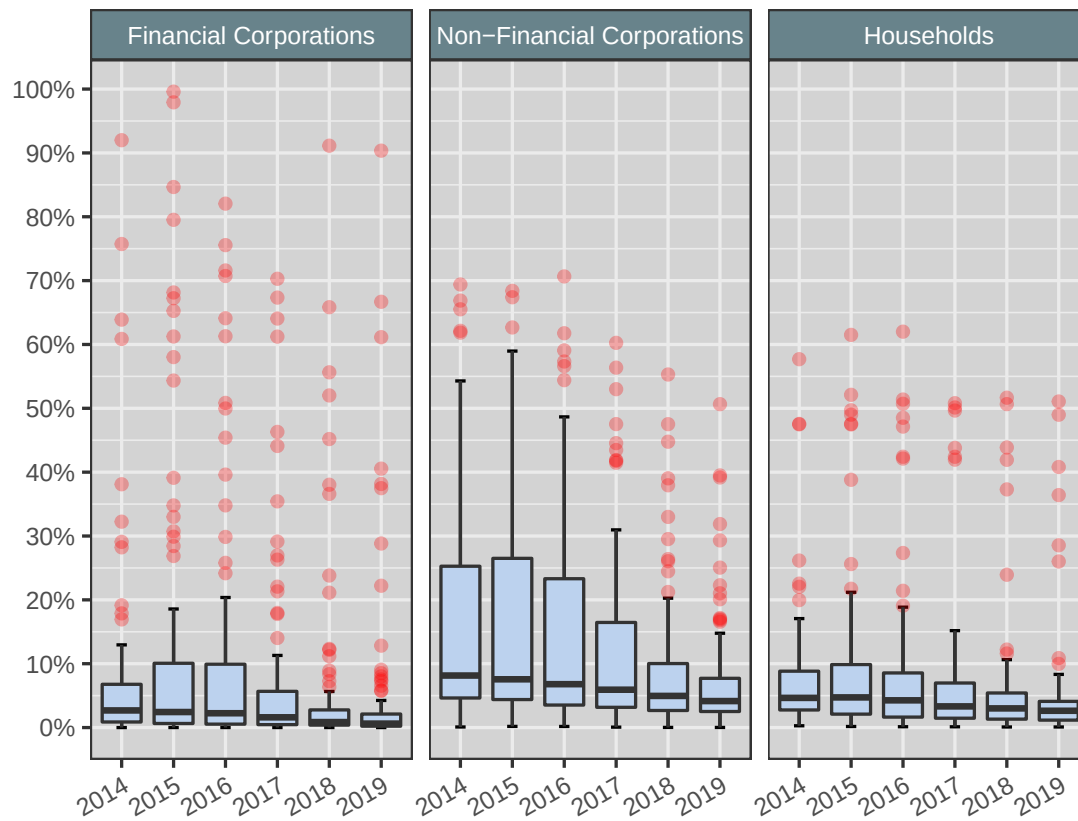


Source: EBA Transparency Exercises, authors' calculations

Figure 1: Total Gross Loans Portfolio per Exposure Type (%)

Figure 1 shows the total gross loans portfolio distribution per exposure type for the period 2014-2019. The distribution illustrates that non-financial corporations and households represent around 90% of the total loans in the sample while financial corporations other than credit institutions only represent around 10%. In particular, approximately 50% of gross loans are granted to non-financial corporations followed by households with around 40%. This distribution among exposures types seems to be quite constant during the period considered.

Figure 2 shows the distribution in the sample of non-performing loans per exposure type for the period 2014-2019. A decline in non-performing loans over the years is observable, with NPL ratios ranging between 0% and 13%. Moreover, there are many banks showing high NPL ratios over the 75th quantile, suggesting the presence of heterogeneity across observations and a positive (right) skewness of the NPL distribution. Furthermore, there are important disparities in NPL when comparing exposures types as observed Table 3. In particular, the average NPL



Source: EBA Transparency Exercises, authors' calculations

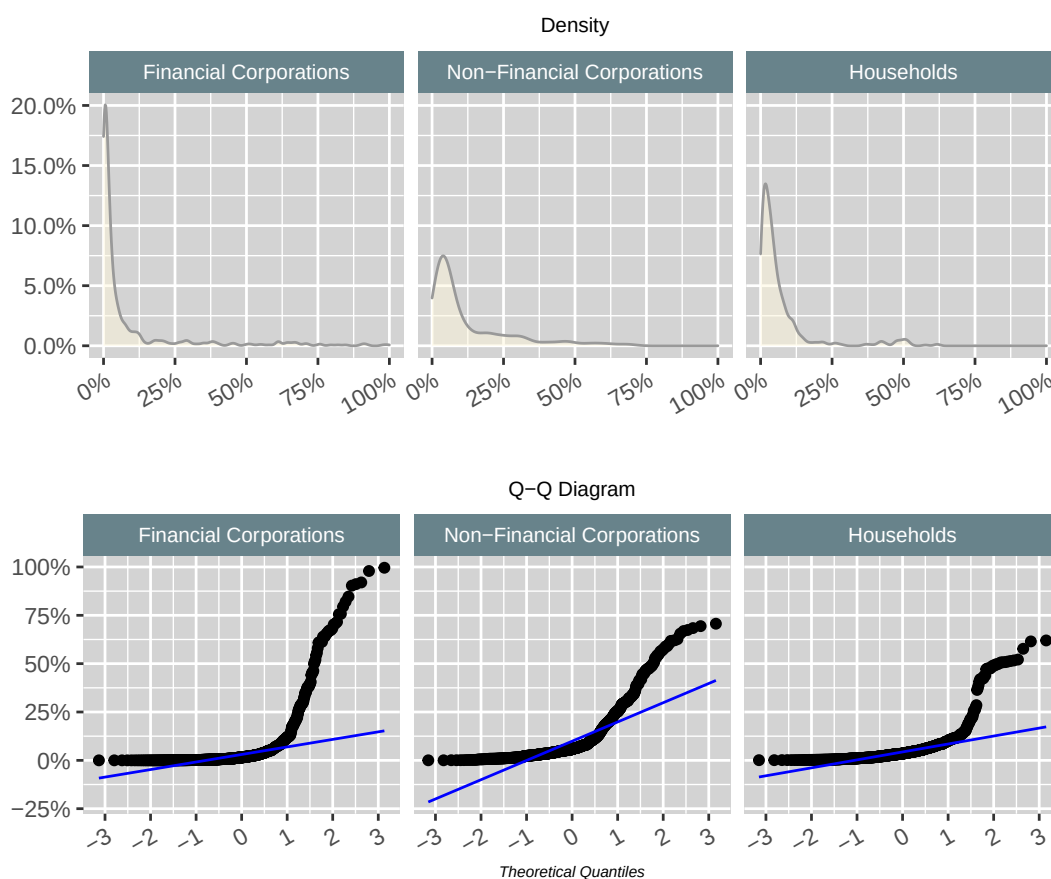
Figure 2: NPL: Distribution per Year

	Financial Corporations	Non-Financial Corporations	Households
Mean	8.339	12.306	6.932
Median	1.532	5.807	3.502
Pctl(25)	0.398	3.177	1.575
Pctl(75)	5.659	16.603	7.119
Standard Deviation	17.432	14.575	10.617
Skewness	2.998	1.874	3.147
Kurtosis	8.839	3.046	9.898

Figure 3: NPL: Distribution-Basic Information

ratio for non-financial corporations stood at 12,3%, compared with 6,9% for households and 8,3 % for financial corporations other than credit institutions for the period 2014-2019. Comparatively, the exposure type financial corporations other than credit institutions shows the greatest

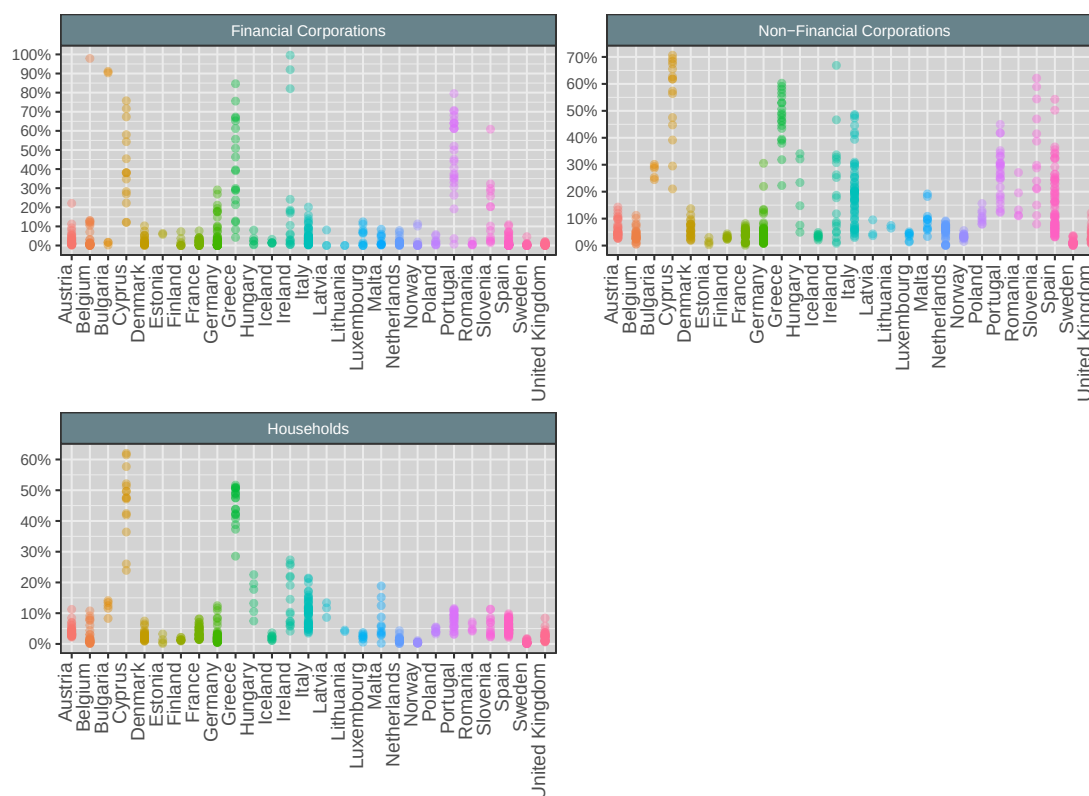
number of observations with NPL classified as outliers in the sample. Moreover, non-financial corporations have the highest NPL on average and the largest distance between the lower (25th) and the upper quantile (75th). Interestingly, Table 3 also contains skewness and kurtosis metrics for the sample. In this regard, the value for the skewness, which measured the degree of symmetry in the distribution and the relative size of the tails, is large and positive for all exposure types, confirming the presence of positive (right) skewness and non-symmetry in the distribution. In addition, the value for the kurtosis, which measures the degree of flatness in the distribution, is similarly large and positive for all exposure types, signaling the presence of a tail-heavy and slightly flat distribution. Given the presence of outliers and the non-normal distribution of NPL, the additional use of quantile regression methodology seems reasonable. Similar conclusions are provided observing the density and Quantile-Quantile diagrams in Figure 4.



Source: EBA Transparency Exercises, authors' calculations

Figure 4: NPL: Density Q-Q Diagram

Figure 5 represents the NPL distribution per bank's country of origin. At country level, Greece and Cyprus stand out as they have the highest NPL ratios for all exposure types. Other countries with heightened NPL from non-financial corporations are Portugal and Slovenia followed by Spain and Italy. For the case of NPL from financial corporations, Ireland and Portugal show comparatively the highest levels. In general, Figure 5 illustrates some differences in NPL



Source: EBA Transparency Exercises, authors' calculations

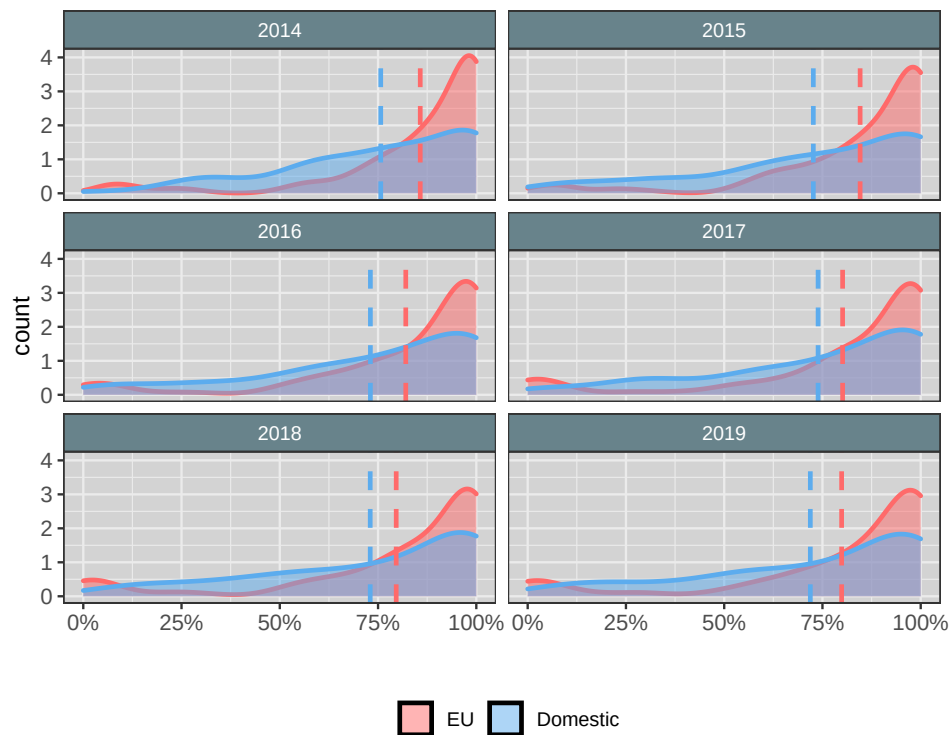
Figure 5: NPL per Country

between EU member states in line with the often recognized core-periphery structure in the distribution of NPL in the European Union.

4.2 Original Exposures and Cross-Border Diversification

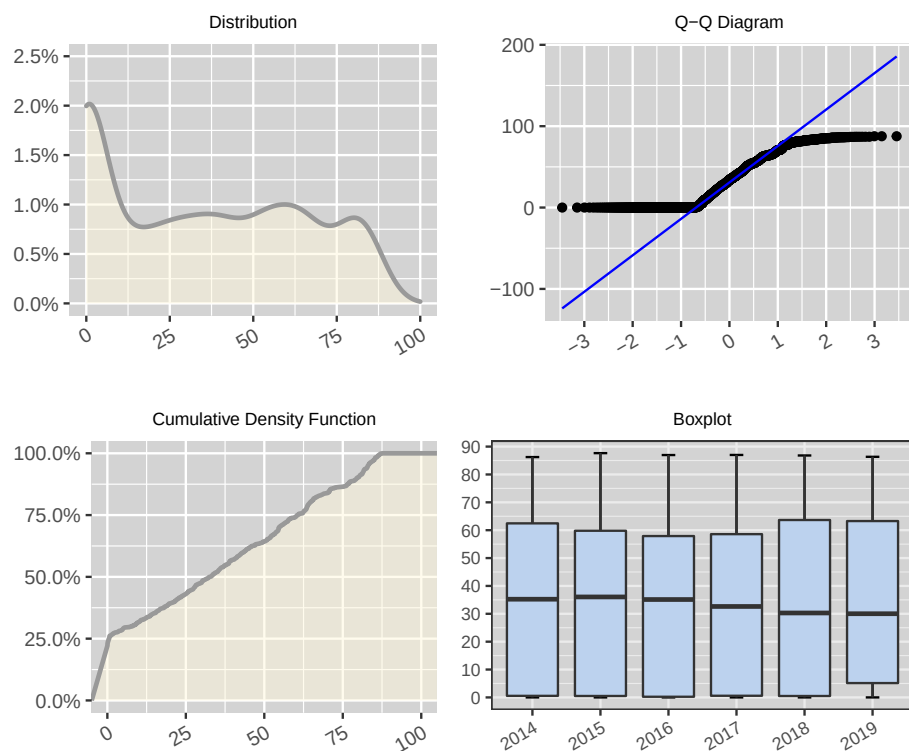
Figure 6 shows the distribution in the sample of original exposures in the bank's country of origin and original exposures with destination in an EU country in relation to the bank's total original exposures. Although the average is around 75% of total original exposures both in homecountry and EU exposures, the distribution tends to be right skewed and slightly flat for homecountry exposure ratios. In the case of EU ratios, the distribution shows a more clear left skewness with important tails to the left. In light of these observations, the use of weighted exposures per country of destination in the calculation of macroeconomic indicators seems to be reasonable.

Figure 7 shows the distribution of the Cross-Border Diversification Index, which represents the geographic concentration of the bank's original exposures, and its respective quantile-quantile plot. A Cross-Border Diversification Index(CBDI) close to its maximum (100) means that the bank's original exposures are very diversified across countries whereas an index close to zero



Source: EBA Transparency Exercises, authors' calculations

Figure 6: Exposures Ratio Distribution: EU & Domestic



Source: EBA Transparency Exercises, authors' ca

Figure 7: Cross-Border Diversification Index

means that the bank's original exposures are very concentrated on certain countries. In this figure, it is also observable some flatness in the total distribution around a CBDI value of 50, which also provides some evidence in favor of the use of macroeconomic indicators weighted by the country of destination of exposures. Furthermore, Figure 13 (appendix) illustrates the diversity and the frequency of countries included in the top 10 country of destiny for bank's exposures. In this regard, approximately 70 different countries of destiny around the world are presented in the data, although most frequent countries of destiny are located in Europe except for the United States, which shows a frequency of 9,6%.

4.3 Country of Origin and the Single Supervisory Mechanism (SSM)

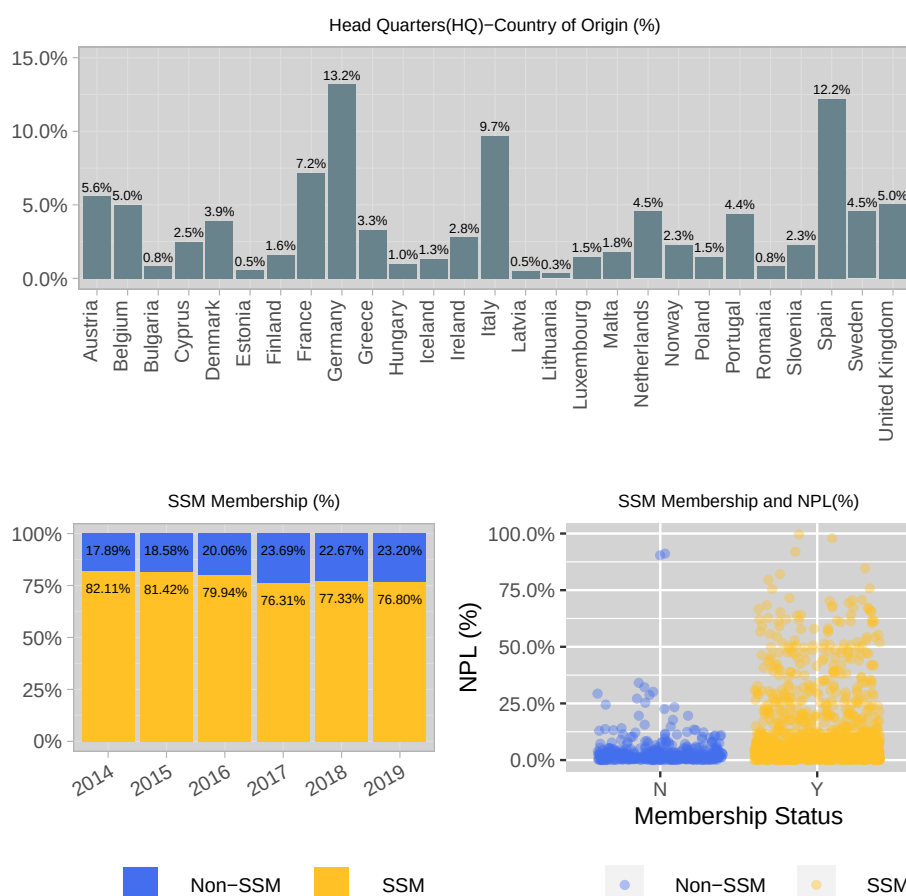


Figure 8: Single Supervisory Mechanism (%) & Country of Origin (%)

Source: EBA Transparency Exercises, authors' calculations

Figure 8 provides basic information regarding the percentage distribution of the sample according to its SSM membership status during the period 2014-2019 as well as the distribution of banks per country of origin. Regarding the SSM membership most banks in the sample belong

to EU countries, which are members of the Single Supervisory Mechanism, with a relation of approximately 80%-20% between SSM members and non-SSM. Despite these asymmetries, there are enough observations for an empirical analysis of the effects of SSM membership on NPL ratios. In particular, NPL ratios seems to be more dispersed for SSM members. The disproportional presence of SSM members with respect to non-SSM members in the sample motivates the need of some caution during the analysis of empirical results. The distribution of banks per country of origin is also asymmetric, with a higher concentration of banks located in Europe's largest economies.

4.4 Descriptive Statistics and Correlation Diagrams

Summary descriptive statistics are reported in Table 4 (appendix). Particularly, the observation of important disparities in the metrics related to NPL both between and within exposure types must be noted. Correlation diagrams are reported in figures 16,17, 18 (appendix). As observed in these figures, the correlation of the explanatory variables with NPL are similar between exposure types and, in general, low for the bank-specific variables while increasing in the case of macroeconomic indicators. Interestingly, Gross public debt (`gross_public_debt`), Unemployment rates (`unempl_rate_imf`) and Recovery rates (`Reco_rate_dol`) show the highest correlation with NPL in the macroeconomic variables under consideration. In regard to the bank-specific variables, the highest correlation is reported by leverage rates (`LEVER_tr_ratio`), Return on Risk-adjusted capital (`RORAC`) and Net interest margin (`NIM_tr`). Additionally, a Variance Inflation Factor (VIF) test is performed for each exposure type to measure the amount of multicollinearity in the context of multivariate regressions (see Table 5) with NPL as response variable. As observed in Table 5, the coefficients are under 10 for all variables, providing some evidence against the presence of multicollinearity in a regression with all these variables.

5 Empirical Approach

5.1 "Standard" Panel Data Methods

A (balanced) panel data set differs from a regular time-series or cross-section regression in that multiple observations on each individual over time are provided in the sample (Baltagi, 2005), that is:

$$y_{it} = \alpha + X'_{it}\beta + u_{it} \quad i = 1, \dots, N; \quad t = 1, \dots, T$$

with i denoting individuals and t denoting time. In our case, it is the observation corresponding to the i th bank in year t , thus the i subscript represents the cross-section dimension and the t represents the time-series dimension. The dependent variable y_{it} represents the non-performing ratio for bank i in year t , α is a scalar, β is $K \times 1$ and X_{it} is the it th observation on K explanatory variables. The K explanatory variables consist of bank-specific and macroeconomic variables.

The models used in the literature for defining the disturbance term u_{it} are classified as one-way error component model or two-way error component model. In the present case, a one-way error component model for the disturbance term u_{it} will be assumed.

A one-way error component model is defined as:

$$u_{it} = \mu_i + v_{it}$$

where μ_i denotes the unobservable individual bank-specific effects and v_{it} denotes the remainder disturbance (Baltagi, 2005). This is the case in most panel data applications.

The use of specific panel models in this paper presents several benefits over other standard models such as cross-section or time series (Hsiao, 2010; Klevmarken, 1989; Baltagi, 2005):

1. *These models explicitly address unobserved individual heterogeneity across banks and periods and thereby allowing for controls over unobserved state- and time-invariant variables that possibly lead to biased estimators whereas this is not the case in time-series or cross-sectional models*

2. *Micro panel data at bank level may provide more precise measurements than similar variables at the macro level. Potential bias due to aggregation over individual data may decrease or be fully excluded*
3. *Panel data provided more information, less collinearity among the variables and more variability, which lead to an improvement of efficiency*
4. *Panel data is more adequate to study the dynamics of adjustment over time than cross-sectional data because cross-sectional distributions may hide a multitude of interesting aspects*
5. *Panel data is better to identify and measure certain effects not detectable in mere cross-sectional or time-series analysis*
6. *Panel data models allow the construction and testing of more sophisticated models using microdata with multiple time periods*

The description of the below models corresponds to the standard case where the panel data is assumed to be complete, that is in the case where all individuals are observed over the entire sample period. Nevertheless, incomplete data is also handled in the present analysis, in so far as the exact same set of banks are not necessarily covered for each time period because the individual requirements and circumstances that a bank has to meet for its participation in the EBA transparency exercise may change from year to year. This typical scenario leads to an "unbalanced" or "incomplete" panel (Baltagi, 2005). In general, an unbalanced panel data set is one in which individuals may be observed different numbers of times. This can be notated as a block of observations T_i where the total number of observations are defined as $\sum_{n=1}^N T_n$. Although there are certain technical differences that require more tedious matrix algebra, the models described above can also be implemented for the case of unbalanced panel data (Green, 2012; Croissant and Millo, 2019).

5.1.1 Fixed Effects (FE) and Random Effects (RE) Models

In this approach the time-invariant individual effects (μ_i, λ_{it}) are explicitly introduced into the econometric specification as fixed parameters. For our analysis it implies that these unobserved effects corresponding to each individual bank may or may not correlate with the independent variables used for the estimation, leading to unbiased and inconsistent estimators. The remainder disturbances (v_{it}) are assumed to be independent and identically distributed $\text{IID}(0, \sigma_v^2)$. The X_{it} are assumed to be uncorrelated with the stochastic disturbances v_{it} for all i and t (Baltagi, 2005). In order to remove these unobserved individual effects across banks, an FE model can be defined through different techniques (LSDV Methods, First-Differences or FD estimator)

(Croissant and Millo, 2019).

In the LSDV estimation, the time-invariant individual components (μ_i) are explicitly estimated in the model in the form of individual intercepts given the presence of a second dimension (time). In this case, the degrees of freedom will decrease to $NT - N - K$ due to the extra parameters. This method may suffer a large loss of degrees of freedom for cases where N are very large as it is the case in the present analysis. The estimation of $N - 1$ extra parameters and several dummies may worsen the problem of multicollinearity among the independent variables (Baltagi, 2005). Another FE method is the so called First-Differences or FD estimator, in which the state-invariant components can be eliminated by first differencing the data, that is by lagging and subtracting the variables the intercept and the individual error components (μ_i) are removed.

As mentioned previously, the LSDV Method suffers from a potentially large number of estimates in the basic regression. A statistically equivalent and more compact method for obtaining the same estimators is achieved by performing a data transformation, in which the average over time per individual is subtracted to each variable. This represents the so called Within regression, and it has become the standard method to perform estimations under a fixed effects model with individual effects. This method is the one selected for the Fixed Effects estimations with static panel data. It is defined as:

$$y_{it} - \bar{y}_i = (x_{it} - \bar{x}_i)\beta + (u_{it} - \bar{u}_i)$$

where $\bar{y}_i$ and $\bar{x}_i$ are the individuals means of y and X . This is equivalent to the regression model:

$$y_{it} = \beta'x_{it} + u_{it}$$

where the individual intercepts are free to vary while still being considered valid parameters for estimation. These are also easily recovered from the OLS estimation of the within transformation data.

A typical issue in the Fixed Effects models is the danger of a large loss of degrees of freedom due to the incorporation of too many parameters. This issue can be avoided by assuming that the individual effects μ_i are random events. In that case, $\mu_i \sim \text{IID}(0, \sigma_\mu^2)$, $v_{it} \sim \text{IID}(0, \sigma_v^2)$ and μ_i are independent of v_{it} . Furthermore, X_{it} are independent of μ_i and v_{it} for all i and t , whereas in the Fixed Effects model the unobserved individual effects (μ_i) are allowed to correlate with the

included variables (X_{it}). The Random Effects model is appropriate for cases where N is large, as in the present analysis, because it greatly reduces the number of parameters to be estimated compared to the other model, which may lead to a large loss of degrees of freedom. In this case, the inference of the individual effects is assumed to be random realizations of a sample taken from a larger population (Baltagi, 2005). The potential cost for this model would be the possibility of inconsistent estimators in case the assumption of randomness turns out to be inappropriate (Green, 2012).

Additionally, several tests for individual effects, serial correlation, correlated effects and multicollinearity will also be applied in the empirical analysis.

In order to test whether individual effects are present, the hypothesis test should be constructed in such a way that the null hypothesis represents the assumption that there are not significant individual effects across observations and, therefore, there is no ground to a preference of panel data techniques over simple Ordinary Least Squares (OLS) estimation. In this paper, the Breusch-Pagan Lagrange Multiplier Test will be used to test for individual effects. The main idea is to start with the OLS regression model and infer whether unobserved time-invariant effects can be detected from the OLS residuals (Croissant and Millo, 2019). More specifically, the Breusch-Pagan Lagrange Multiplier corresponds to test:

- $H_0: \sigma_\mu^2 = 0$
- $H_1: \sigma_\mu^2 \neq 0$

In this case, the null hypothesis H_0 means that there is not significant variances across individuals (absence of individual effects). Moreover, as the presence of panel effects can not be identified, there are not justification for the use of panel techniques over OLS in order to control for individual heterogeneity.

Additionally, an F test for the presence of individual effects that compares the Ordinary Least-Squares and the Fixed Effects model is also useful. Under the null of no individual effects, the statistic is distributed as F with degrees of freedom equal to the number of individual banks minus 1 on the numerator and to the degrees of freedom of the Fixed Effects Model on the denominator.

Furthermore, the errors in models with individual effects are always serially correlated due to the presence of time-invariant unobserved effects in the error components that do not decline over time. Additionally, serial correlation that die out over time may also be present in the errors

in the sense of an **AR(1)** process. In order to test for this aspect, Unobserved Effects and Fixed Effects (FE) test will be applied.

In fact, the main argument that determines whether a random or fixed effects model is more appropriate for a specific econometric analysis is the possibility of correlation between the covariates and the individual effects. Maintaining the assumption that the stochastic disturbances v are uncorrelated with the covariates ($\mathbb{E}[X'v] = 0$), two situations may take place (Croissant and Millo, 2019):

- $\mathbb{E}[X'\mu] = 0$: the individual effects are uncorrelated and both models are consistent but a Random Effects estimator is more efficient than a Fixed Effects estimator.
- $\mathbb{E}[X'\mu] \neq 0$: the individual effects are correlated. This implies that only a Fixed Effects model provides consistent estimates because the within estimation eliminates all unobserved time-invariant effects.

In this paper, the Hausman test approach will be used in order to test for correlated effects. The general principle of the Hausman test in panel models consists in comparing the estimates of the Fixed Effects (FE) model and the Random Effects (RE) model by performing the following hypothesis test:

- Under H_0 : FE and RE are consistent, but RE is more efficient than FE.
- Under H_1 : Only FE is consistent

The idea of the test is that if the null hypothesis H_0 is true, then the estimated coefficients from FE and RE should not be significantly different, whereas under the alternative hypothesis the estimated coefficients would diverge. Therefore, the test is based on $\hat{\beta}_{FE} - \hat{\beta}_{RE}$ where under H_0 the variance is $\text{Var}(\hat{\beta}_{FE} - \hat{\beta}_{RE}) = \text{Var}(\hat{\beta}_{FE}) - \text{Var}(\hat{\beta}_{RE})$ (Croissant and Millo, 2019). The common hypothesis test will be the version where:

$$\hat{q} = \hat{\beta}_{FE} - \hat{\beta}_{RE}$$

with variance:

$$\text{Var}(\hat{q}) = \text{Var}(\hat{\beta}_{FE}) + \text{Var}(\hat{\beta}_{RE}) - 2\text{Cov}(\hat{\beta}_{FE}, \hat{\beta}_{RE})$$

The hypothesis test is then formalized as:

- $H_0: \hat{q} = 0$
- $H_1: \hat{q} \neq 0$

Moreover, a standard assumption in the estimation is that the explanatory variables are not perfectly correlated with each other. If two independent variables are very highly correlated, then the variance of the corresponding coefficient is very high, tending to infinite if the correlation is perfect. This is known as a problem of multicollinearity, in which coefficients may have very high standard errors and low significance even if these are jointly significant and the R^2 of the estimation is very high or some of the coefficients may show implausible or negative values (Green, 2012).

In this paper, the Variance Inflation Factor (VIF) will be used to test for multicollinearity. The Variance Inflation Factor (VIF) is determined by $1/(1 - R_k^2)$ where R_k^2 is the R^2 in the regression of the independent variable x_k on all other independent variables included in the model. Thus, the VIF will be determined for each independent variable x_k in the main model. The standard benchmark is that VIF values higher than 10 are a signal of the presence of multicollinearity (Green, 2012).

5.1.2 Generalized Method of Moments (GMM) Models

To understand better the dynamics of non-performing loan ratios, certain dynamic panel models developed in the framework of the Generalized Method of Moments (GMM) approach will also be used in the econometric analysis (Baltagi, 2005). These models are dynamic because one of the covariates is the lagged dependent variable, in our case the lagged values of NPL. The dynamic relationship of NPL ratios over time are estimated for a large sample of individual banks observed over a small period of years (2014-2019). The dynamic panel models are very well suited for this kind of microdata where $N \gg T$ (Croissant and Millo, 2019). This model is usually formalized as:

$$y_{it} = \delta y_{i,t-1} + x'_{it} \beta + u_{it} \quad i = 1, \dots, N; \quad t = 1, \dots, T$$

where δ is a scalar, x'_{it} is $1 \times K$ and β is $K \times 1$, and u_{it} follows a one-way component model:

$$u_{it} = \mu_i + v_{it}$$

where $\mu_i \sim \text{IID}(0, \sigma_\mu^2)$ and $v_{it} \sim \text{IID}(0, \sigma_v^2)$, and the error disturbances v_{it} are assumed to be mutually uncorrelated $\mathbb{E}(v_{it} v_{is}) = 0 \forall s \neq t$ and not correlated with the individual effects $\mathbb{E}(\mu_i v_{it}) = 0$. In our analysis, these models include time-invariant individual effects represented derived from the heterogeneity among individual banks as well as autocorrelation related to the use of a lagged endogenous variable (NPL ratio) in the regressors (Baltagi, 2005; Croissant and Millo, 2019).

Given that y_{it} is a function of μ_i it can be derived that $y_{i,t-1}$ is also a function of μ_i . The result is a biased and inconsistent OLS estimator. The bias of the OLS is caused by the correlation between the lagged endogenous variables and the disturbances due to the existence of individual effects. In the OLS estimation, the estimator is proved to be positively biased while in the Fixed Effects estimation, the estimator is negatively biased (Croissant and Millo, 2019). In this situation, a consistent coefficient is expected to be comprised between that of a OLS and a Fixed Effects estimation (Croissant and Millo, 2019; Baltagi, 2005). Several consistent estimation methods for dynamic panel models are presented in the literature (Anderson and Hsiao, 1982; Ahn and Schmidt, 1995). In this paper, the method proposed by Arellano and Bond (1991) and eventually its extended versions (Arellano and Bover, 1995; Blundell and Bond, 1998) will be considered due to their larger efficiency in this analysis.

The main method is the so-called Difference GMM Estimator, which was firstly developed by Arellano and Bond (1991) as the further extension of the instrumental variable (IV) approach developed by Anderson and Hsiao in 1982. Anderson and Hsiao (1982) suggested differencing of both the dependent and independent variables to get rid of μ_i and then instrument the lagged endogenous variables in differences, $\Delta y_{i,t-1} = (y_{i,t-1} - y_{i,t-2})$ with the endogenous variables in levels lagged by two periods $y_{i,t-2}$. These instruments will not be correlated with $\Delta y_{i,t-1} = (y_{i,t-1} - y_{i,t-2})$ as long as the v_{it} are not serially correlated. Although this method generates consistent estimates, these are not necessarily efficient as observed by Baltagi (2005) because:

- * It does not make use of all the available moment conditions
- * It does not account for the correlation introduced into the errors $_{it}$ by first-differencing
- * There are more valid instruments available

In this context, Arellano and Bond (1991) suggested a Generalized Method of Moments (GMM)

procedure that is more efficient than the estimator proposed by Anderson and Hsiao (1982). This procedure considers the fact that the number of valid instruments is growing in t . Because of the dynamic aspect of the estimation and the use of first-differencing, the first and second observations are not usable. For this reason, the model can be written as:

$$y_{i3} - y_{i2} = \rho(y_{i2} - y_{i1}) + (v_{i2} - v_{i1})$$

where y_{i1} is the only valid estimator. For y_{i4} the error is $(v_{i4} - v_{i3})$ with y_{i2} and y_{i1} as valid instruments. In other words, an additional valid instrument is available each time t is increased by 1. The use of additional valid instruments should improve the efficiency of the estimates. In fact, Arellano and Bond (1991) developed two versions of the GMM estimator; the standard version is called the one-step version and was designed under the assumptions that the errors are actually heteroscedastic and/or autocorrelated, and an extended version called the two-step version that solves some problems in model consistency that arise when the previous assumptions are violated. The main problem with the Difference GMM estimator is that the lagged first differences are very often weakly correlated with the lagged levels of the dependent variable used as instruments.

An extension of the difference GMM estimator known as System GMM Estimator will be used in this thesis. The system GMM estimator appeared as a solution to the weakness of the instruments in the difference GMM estimator. In this sense, Arellano and Bover (1995) and Blundell and Bond (1998) showed that under a weak hypothesis about the process that generates the data, another moment condition can be added from the following levels equation:

$$y_{it} = \rho(y_{i,t-1}) + u_{it}$$

with

$$u_{it} = \mu_i + v_{it}$$

where it can be showed that the lagged differences, $\Delta y_{i,t-s}$ with $s = 1 \dots t-1$, are valid instruments for $y_{i,t-1}$.

This estimator that adds the previous mentioned valid instruments from the equation in levels

into the difference model is called the system GMM estimator. This estimator uses lagged differences of $y_{i,t-1}$, $\Delta y_{i,t-s}$, as instruments for equations in levels, in addition to lagged levels of $y_{i,t-1}$, $y_{i,t-s}$, as instruments for equations in first differences (Baltagi, 2005).

The system GMM estimator appeared to provide an improvement in efficiency compared to the difference GMM estimator as $\rho \rightarrow 1$ and σ_μ^2/σ_u^2 increases and a reduction in the finite sample bias. According to the findings presented by Blundell and Bond (1998), these improvements in efficiency and precision in the system GMM estimator are larger in "short" panels, which contain data with small T ($T \ll N$) and persistent series (Baltagi, 2005).

The GMM procedure presents two type of complications to take into account when inferencing:

1. If the estimator is consistent, the variance-covariance matrix of coefficients may not, and therefore, the use of robust covariates matrices would be required.
2. GMM estimation is only consistent under the assumptions of no error correlation and of moments' validity.

Several tests are used to determine whether the assumptions of no error correlation and of moments' validity are justified in the corresponding GMM estimation, the following tests must be considered:

* Overidentification Test (Sargan-Hansen):

In order to determine whether the assumption of moments' validity really holds, a test for overidentification proposed by Sargan in 1958 and applied to the case of GMM models by Hansen in 1982 will be considered in this paper (Croissant and Millo, 2019). The null hypothesis in this test is the validity of the moment conditions.

* Error Serial Correlation Test (Arellano-Bond):

The GMM estimator is also not consistent unless the assumption of serially uncorrelated errors is correct. For this purpose, Arellano and Bond (1991) proposed a test of error serial correlation that will also be considered in this paper. In this test the null hypothesis is that the errors are not serially correlated.

Depending on the results delivered by the previously mentioned test, the use of robust covariance estimators may be required in order to perform a valid inference while relaxing the hypothesis of independence and homoscedasticity of the errors. Besides the general concerns regarding the potential present of heteroscedasticity in the errors, there also are concerns related to the so-called "clustering" as individuals and time periods are considered "natural" clusters. These clusters may appear as multiple observations corresponding to the same individual bank

or to the same time period tend to share common characteristics that can lead to a violation of the assumption of independent and identically distributed errors and, therefore, potentially to a biased estimation (Croissant and Millo, 2019). In the present paper, the model estimations for the standard panel data approach make use of heteroscedasticity- and clustering-robust covariance matrices.

5.2 Quantile Regression Methods

The application of quantile regression methods in this thesis is reasonable when looking at the distribution of NPL, which is characterized by its skewness and the large amount of outliers. The NPL show a negative skewness(left skewed distribution) and the residuals may be non-normal distributed. In samples with many outliers, quantile regression also shows more robustness than OLS, and it is semiparametric as it does not make assumptions about the parametric distribution of the errors. Additionally, it is interesting the use of quantile regressions for the analysis of the impact of the regressors at different points of the conditional distribution of NPL.

In Ordinary Least Squares, the average effect of a set of explanatory variables on the dependent variable is summarized by the conditional mean function ($E(y|x)$). Similar to the conditional mean function of a linear regression, the quantile regression considers the relationship between the explanatory variables and the dependent variable is defined by the conditional quantile function $Q_\tau(y|x)$. In other words, quantile regression seeks to apply an estimation of conditional quantile functions in which the quantile of the conditional distribution of the dependent variable is modelled as a function of observed covariates x (Koenker and Bassett, 1978). In the same manner that the mean is used to minimize a sum of squared residuals in a standard OLS, this problem is solved in a quantile regression by minimizing a weighted sum of absolute residuals (Fitzenberger et al., 2002; Koenker, 2005). Quantile regression methods are a supplement to the conditional mean approach in OLS as these also allow the estimation of a whole family of conditional quantiles. In this sense, these methods provide more flexibility to analyze the impact of a set of explanatory variables on the entire conditional distribution of the dependent variable (Fitzenberger et al., 2002).

The optimization problem in the quantile regression is dependent on the specific quantile in evaluation. In this context, negative errors suffer higher penalties for higher quantiles, while positive errors are penalized more for lower quantiles. In other words, $Q_y(\tau|x)$ represents the conditional τ -quantile of y given (x) . The quantile $\tau \in (0, 1)$ determines the point of y in which the data is split into proportions under τ and above $1 - \tau$, that is $F(y_\tau = \tau)$ and $y_\tau = F^{-1}(\tau)$.

Quantile regression seeks to minimize the errors ε_i with asymmetric penalties, $(1 - \tau)|\varepsilon_i|$ for overprediction and $\tau|\varepsilon_i|$ for underprediction. In this sense, the quantile regression estimator in quantile τ is the one that minimizes the following objective function:

$$Q(\beta_\tau) = \sum_{y_i \geq x_i' \beta} \tau |y_i - x_i' \beta_\tau| + \sum_{y_i < x_i' \beta} (1 - \tau) |y_i - x_i' \beta_\tau|$$

The estimator has been shown to be asymptotically normal, and bootstrap standard errors are often used instead of traditional standard errors.

5.2.1 Pooled Quantile Regression Methods

In a panel Quantile regression, there are pairs of dependent and explanatory variables (y_{it}, x_{it}) with $i = 1, \dots, N$; $t = 1, \dots, T$, where i denotes the i th individual and t denotes time. Furthermore, there also are unobservable individual-specific variables μ_i with $i = 1, \dots, N$, representing individual characteristics. In this method, the quantile regression is based on the conditional quantile of y_{it} given (y_{it}, μ_i) and seeks to estimate the partial effect of x_{it} on the conditional quantile. Formally, $Q_{y_{it}}(\tau | x_{it}, \mu_i)$ denotes the conditional τ -quantile of y_{it} given (x_{it}, μ_i) , where $\tau \in (0, 1)$ is a quantile index (Koenker et al., 2017). The basic model is then:

$$Q_{y_{it}}(\tau | x_{it}, \mu_i) = \alpha_i(\tau, \mu_i) + X_{it}' \beta(\tau)$$

where $\alpha_i(\tau, \mu_i)$ is some unknown function of τ and μ_i , and $\beta(\tau)$ is a dimensional vector of unknown parameters to be estimated in the model. In a more familiar form, the model can also be written as:

$$y_{it} = \alpha_i(\tau, \mu_i) + X_{it}' \beta(\tau) + u_{it}(\tau)$$

where $u_{it}(\tau)$ has a zero conditional τ -quantile given (x_{it}, μ_i) . Under the standard assumption, the conditional quantile restriction is equivalent to:

$$P(u_{it}(\tau) \leq 0 | x_{it}, \mu_i) = \tau$$

A panel quantile regression model with individual effects is necessarily nonlinear and for this reason individual heterogeneity can not simply be removed by first-differencing or transforming the data as it is usually the case in other panel regression models (Koenker et al., 2017).

5.2.2 Fixed Effects Quantile Regression (Penalized FE Method)

The penalized least squares interpretation of the classical random effects estimator was developed by Koenker (2004) with an estimation technique that offers the possibility of implementing quantile regression models that account for the existence of individual specific effects. This allows the analysis of panel data under less stringent distributional assumptions. These models are also fit for "short" panel data, where numerous cross-sectional individuals are observed over short periods of time ($T \ll N$). In general, the research on Quantile Regression methodologies for panel data has been experiencing steady growth during the last years as observed by Koenker et al. (2017). Formally, the model is defined as:

$$Q_{y_{it}}(\tau | x_{it}, \alpha_i) = \alpha_i + X'_{it}\beta(\tau) \quad i = 1, \dots, N; \quad t = 1, \dots, T$$

where the individual effect α_i does not vary over quantiles (no scalar effect). Furthermore, the following penalized estimator $(\hat{\alpha}(\lambda), \hat{\beta}(\lambda, \tau))$ is considered:

$$\min_{(\alpha, \beta)} \frac{1}{nT} \sum_{k=1}^q \sum_{i=1}^n \sum_{t=1}^T \omega_k \rho_{\tau}(y_{it} - \alpha_i - X'_{it}\beta(\tau_k)) + \lambda \sum_{i=1}^n |\alpha_i|$$

where $\omega_1, \dots, \omega_q$ are nonnegative weights and $\lambda \geq 0$ is the penalty level.

The idea in this model is to estimate several τ simultaneously and restricts the individual effect coefficients α_i to be the same for all quantiles. Moreover, the estimated individual effects will shrink toward zero as the penalty level λ tends to increase (Bache et al., 2013). In this sense, the parameter λ allows a calibration of the importance of accounting for individual effects in the model.

In general, when $\lambda \rightarrow 0$ the model is similar to a (weighted) dummy-variable quantile regression and when $\lambda \rightarrow \infty$ the model is more like a (weighted) cross-section quantile regression. The penalty allows a "control" over the many fixed effects coefficients by not letting them reach

unrealistic large numbers (Bache et al., 2013).

This estimator is expected to deliver results that are between those of a pure dummy-variable estimator and a pure cross-section estimator. In general, the estimator tends to perform worse in cases where the fixed effects are assumed to have a scalar effect, that is when the fixed effects show a variation over quantiles (Bache et al., 2013).

6 Results and Discussion

6.1 Regression Analysis and Test Estimates

Table 3 contains the results of the estimations and Table 6 (appendix) provides the results of multiples tests for the linear panel models. As observed in Table 6, the Hausman test for the static panel models suggests that a random effects approach is more appropriate for households and financial corporations [$\text{Prob (Hausman Test)} \geq 0.05$], whereas a fixed effects approach seems to fit better for non-financial corporations [$\text{Prob (Hausman Test)} < 0.05$]. Moreover, the Breusch-Pagan Lagrange Multiplier and the F tests for individual effects ($p\text{-value} \geq 0.05$) suggest the existence of panel effects for all exposure types, confirming that the use of panel techniques over simple OLS estimation is also appropriate. As observed in the results of the Wooldridge's tests, which are specific tests for serial correlation and unobserved individual effects under null hypothesis of no serial correlation and unobserved effects, there is evidence of serial correlation and the present of heteroscedasticity in the errors for all exposure types ($p\text{-value} \geq 0.05$). The heteroscedasticity in the errors justifies the use of Heteroscedasticity-consistent (HC) robust standard errors in the static panel models.

The Sargan tests, which aim at testing the joint validity of the instruments used for the dynamic panel models, indicates that the null hypothesis of instruments validity can not be rejected at a significant level of 0.01% [$\text{Prob (Sargan Test)} \geq 0.01$] for all exposure types. Moreover, the results of the Arellano-Bond tests for second-order autocorrelation [$\text{Prob (AR(2))} \geq 0.05$] confirm that the assumption of serially uncorrelated errors also holds. Given the results of the Sargan an Arellano-Bond tests, there is evidence that the GMM estimators are also consistent. Tables 14 and 15 are also provided with the purpose of providing additional robustness checks for the main results in "standard" panel methods. Table 14 show the results when only including bank-specific variables in the models and Table 15 show the results of the estimations when only considering in the sample banks with a domestic exposure greater or equal to 75%.

With regard to the estimations with quantile methods, Figure 9 show the results of the pooled quantile regression, whereas Figures 10, 11 and 12 show the results of the penalized fixed effects quantile regressions for λ equal to 1, 2.5 and 5, respectively. In the appendix, some tables regarding the results of the pooled quantile regressions for each exposure type are also available (see 7,8,9) as well as several joint tests for slope equality and symmetric quantiles. The results of the individual slope tests for each variable are also available in Table 10. In general, the joint and individual slope equality tests for the equality of slope coefficients also provide evidence that the coefficient values differ across quantiles for most explanatory variables. Similarly, the

joint symmetric quantile tests show that there is no evidence of symmetries in the coefficient values across quantiles. Tables regarding the results of the penalized fixed effects quantile regressions for each exposure type are available in 11, 12 and 13. Figures 19, 20, 21 and 22 show the results of the pooled quantile and penalized fixed effects quantile regressions for the case of considering only bank-specific variables as regressors. Similarly, Figures 23, 24, 25 and 26 show the results of the pooled quantile and the penalized fixed effects quantile regressions when only considering in the sample those banks with a domestic exposure greater or equal to 75%.

6.2 Contribution of explanatory variables to NPL

As observed in the empirical results, the bank-specific and macroeconomic factors considered in this analysis are jointly significant in explaining NPL for all exposures types. These results are also consistent with the case in which only banks with domestic exposures higher or equal to 75% are taken into consideration. Additionally, the very low p-value of the F-statistics in Tables 3, 14 and 15 indicates that the null hypotheses are to be rejected, meaning that the joint effects of the explanatory variables on NPL are also significant. In other words, it can be affirmed that the bank-specific and macroeconomic factors included in the analysis show statistical significance in all models, regression methods and exposure types under consideration, although the explanatory power largely decreases in models where only bank-specific factors are considered as observed in their adjusted R-squares (Figures 3 and 14).

6.2.1 Macroeconomic indicators

With respect to business cycle indicators, the unemployment rates and the GDP growth are included in the regressions as explanatory variables. The estimated coefficients for the unemployment rates appear to be very significant and positively related to NPL for household and non-financial corporations for all models but insignificant for the case of financial corporations. Moreover, their impact is quite constant for all NPL quantiles as illustrated in the results of the quantile regressions. Regarding the estimated coefficients of GDP growth (GDP_growth_perc), these also seem to be positively related with NPL but with low significance for all models and exposure types. Thus, it can be concluded that unemployment rates are very relevant determinants of NPL with large and significant coefficients, whereas GDP growth plays a secondary role in these results (Table 3 and Figures 9, 10). Moreover, the magnitude of the business cycle indicators remains quite flat along the distribution of NPL. This characteristic suggests that the

impact of macroeconomic indicators on NPL is not affected by the amount of NPL at portfolio level.

When looking at price levels, the estimated coefficients for the inflation rates are very significant and negatively related with NPL for all models and exposures types, suggesting that the price level is also very important determinant of NPL and that its impact is marginally increasing along the distribution of NPL in the portfolio (Tables 3, 15 and Figures 9, 23). In this sense, these results give empirical support to the economic argument affirming that higher inflation rates tend to reduce problem loans by cutting down the real value of outstanding loans.

Similarly, the general government gross debt as a percentage of GDP (*Gross_public_debt*) is also included in the analysis as an indicator of the efficiency in the public sector. The estimated coefficients are significant and positively related to NPL, suggesting that the performance of the general government also has a large impact on the development of problem loans at bank level (Tables 3, 15 and Figures 9, 10, 23, 24). Interestingly, the estimated coefficients also show a very significance marginal increase along the conditional distribution of NPL as demonstrated in the quantile regressions. This marginal increase indicates that the effect of the public sector on NPL tend to be larger for banks with higher NPL in their portfolios. In this regard, the estimated impact is in line with the theoretical considerations that see a relation between sovereign debt developments and the performance of loan portfolios at bank-level. With respect to indicators regarding Terms of trade, the estimated coefficients of the current account balance as percentage of GDP (*curr_acc_bal*) are significant and positively related to NPL for households and non-financial corporations but not for financial corporations other than credit institutions (Figures 9, 10, 23, 24). This last result suggests that persistent current account surpluses at national level during the period 2014-2019 may be followed by an increasing amount of problem loans at bank level, whereas the impact along the different NPL quantiles remains quite constant. One possible interpretation is that this relation may be driven by the efforts of many European economies, which excessively accumulated problem loans in their banking sectors, in improving their balance of payments at national level during the last years.

Regarding the characteristics of the legal system, two variables have been included into the models, the strength of legal rights index (*Stgth_legal_index*), which measures the strength of collateral and bankruptcy laws that positively influence the access to credit (ex-ante), and recovery rates (*Reco_rate_dol*) calculated as recovered cents per dollar in debt. The strength of legal rights index (*Stgth_legal_index*) shows estimated coefficients that signal a positive relation to NPL with low significance. In the quantile regressions, it shows a close to constant impact along the difference quantiles and statistically significance in the static panel models (Figures 9, 10, 23, 24). This positive relation holds in all models except for the case of financial corpora-

tions other than credit institutions, and it suggests that the legal systems of countries with more friendly regulations for the extension of credit have also been experiencing the larger accumulation of problem loans during the period 2014-2019. On the other hand, the recovery rates [Reco_rate] show a negative and very significant correlation with NPL in all models and exposure types, and with a similar impact along the conditional distribution of NPL as observed in the quantile regressions (Tables 3 and Figures 9, 10). In this sense, the negative relation between recovery rates and NPL also confirms theoretical considerations arguing that judicial court systems in countries with more efficient declaratory procedures in terms of the final recovered amount are also characterized by a lesser amount of NPL in the portfolios of banks operating in those countries. In general, these results illustrate the importance of the legal aspects with regard to the regulation of lending activities and NPL resolution to explain the development of NPL at bank-level.

6.2.2 Bank-Specific variables

With respect to the variables regarding the profitability at bank-level, Returns on Risk adjusted Capital (RORAC) and Net Interest Margin (NIM_tr) provide support for the identification of a causal relation between bank-profitability and problem loans. Returns on Risk adjusted Capital is found to be significant and negatively related to NPL in most models in line with the "management" assumption presented in the literature, by which highly profitable banks have fewer incentives to enter into riskier activities and, therefore, highly profitable banks tend to show fewer loans classified as non-performing (Tables 3, 14, 15 and Figures 9, 19, 23). In the case of net interest margins, the coefficients are found to be mostly positive but with lower significance levels. Furthermore, the impact of net interest margins strongly increases along the distribution of NPL. These coefficients may be explained by the effects of lending policies setting very high (net) interest rates that are not necessarily justified by higher costs of funding on NPL (Figures 9, 19, 23). In this sense, higher net interest margins in a bank's balance sheet signal an increase of problem loans in the long run despite any positive effect in the short run. In this context, the causal relation between bank-profitability and problem loans is negative and in line with the "bad management" hypothesis as long as any improvement in the bank's profitability is not exclusively explained by higher (net) interest rates over the cost of funding. As indicator of a bank's cost-efficiency, the cost to income indicator (Cost_to_Income) is also incorporated in the models. In this case, the estimated coefficients of the cost to income indicator show a negative correlation with NPL and low significance levels for all models and exposure types (Tables 3, 14, 15 and Figures 9, 19, 23). As observed in these results, it can be concluded that the cost to income indicator is not a good determinant of NPL despite previous theoretical considerations in this respect.

With regard to the relation between the size of a bank in terms of total assets and the amount of NPL in the portfolio, the estimated coefficients for the logarithm value of Bank Size (Bank-Size_tran) are large, negatively related to NPL and statistically very significant for all models and exposures types. Moreover, the negative impact also shows very large marginal increase along the conditional distribution of NPL, in which the largest impacts are to be found in the highest quantiles as observed in the results of the quantile regressions (Tables 3, 14 and Figures 9, 10, 19, 20). Particularly, these results offer some empirical support for theoretical considerations arguing that the potential gains in terms of bank efficiency and economies of scale for large banks are expected to have a greater impact on NPL than the effects related to the too-big-to-fail problem explained in previous sections. For this reason, larger banks will tend to show lower NPL ratios in their portfolios compared to smaller banks despite any size-related incentive to take additional risks.

As indicators related to bank capitalization, the Common Equity Tier 1 Ratio -CET1- (CET1_ratio_tr) and the leverage ratio (LEVER_tr_ratio) are also incorporated in the models as explanatory variables. The estimated coefficients of CET1 are moderately significant and negative for households and non-financial corporations and not significant for financial corporations, whereas the estimated coefficients of the leverage ratio are positive and mostly insignificant for all models and exposure types (Table 14 and Figures 9, 10, 19, 20). Interestingly, CET1, which is the main core indicator of capitalization for supervisory purposes, is significant and negatively related to NPL, while leverage ratios, which are mostly subject to the bank's discretion, seem to play a secondary role in explaining NPL. In this context, it must also be pointed out that the statistical significance and the values of the estimated coefficients for CET1 significantly increase for the models that do not include macroeconomic variables (Table 14 and Figures 19, 20). Furthermore, the estimated coefficients of CET1 and leverage ratios also present quite constant impact and significance along the conditional distribution of NPL for the quantile regressions with macroeconomic variables. Interestingly, the marginal impact tend to increase at the highest NPL quantiles for the quantile regressions that do not include macroeconomic variables. In general, the results of the estimated relation between CET1 and NPL is in line with the "moral hazard" hypothesis commented in previous sections by which managers in low-capitalized banks have larger incentives to undertake riskier lending activities that lead to higher NPL in their portfolios. However, the results regarding the impact of leverage ratios on NPL may appear to be explained by the reasoning that higher capitalized banks also have the possibility of implementing more flexible lending policies given previous compliance with the respective supervisory regulation, and, therefore, a positive relation between the level of capitalization and NPL can also be found.

Additionally, the dynamic model for panel data (GMM) also delivers the estimated coefficients

for the lagged values of NPL by one period. As expected, these estimates are positive, large and statistically significant, demonstrating the importance of the relation between past and present NPL ratios. Particularly, this relation represents the presence of a persistent process, which implies that, all else equal, an increase (decrease) in the NPL ratio in a bank's portfolio for one period would likely be followed by another increase (decrease) in NPL ratios for the next period. In this sense, the dynamic models identified the presence of a temporal interdependence of NPL from different periods within a certain lag.

6.2.3 Supervision under the Single Supervisory Mechanism (SSM)

As explained in previous sections, the Single Supervisory Mechanism (SSM) represents the system of banking supervision as one of the first pillars of the European banking union. The estimated coefficients for the dummy coefficient of membership under the SSM (SSMY) are negatively related to NPL, large and statistically significant for non-financial corporations in the case of random and fixed effects models and also for households in the quantile regressions, whereas the estimated coefficients for financial corporations remain insignificant for all models. Furthermore, the value and sign of the estimated coefficients are quite constant for all quantiles along the conditional distribution of NPL, suggesting that the impact of this variable is similar to all banks regardless of the current NPL in their portfolios (Figures 9, 10, 23, 24). To some extent, these results present empirical evidence in favour of the statement that banks under supervision in the context of the Single Supervisory Mechanism (SSM) tend to show lower NPL ratios. Moreover, the results may also be interpreted as evidence of the effectiveness of the recommendations and guidances published by the ECB banking supervision on the development and implementation of NPL strategies as well as indicators of the relevance of the ECB for addressing asset quality issues in Europe. However, despite the validity of the estimation in statistical terms, some caution is needed when interpreting these results because of the asymmetries presented in the sample (80%-20% distribution between SSM members and non-SSM).

6.2.4 Geographic Diversification Index

When considering the effects of geographic diversification on asset quality, the estimations results vary among the models under consideration. The estimated coefficients of the Cross-Border Diversification Index (CBDI_index) show low and positive values with statistical significance for all exposure types in the random and fixed effects approach for the models including all explanatory variables, whereas the same are statistically insignificant in the dynamic models

(GMM). Once the macroeconomic indicators are no longer considered in the models, the estimated coefficients of the CBDI index show different direction and magnitude with negative and lower values for all exposure types (Tables 3, 14, 15).

With regard to the results of the quantile regression, the estimated coefficients of the Cross-Border Diversification Index (CBDI_index) are positive but mostly low and statistically insignificant for all exposures types and quantiles in the models with all variables. However, once the macroeconomic variables are no longer considered as explanatory variables in the models, the results changed in terms of sign and value of the estimated coefficients. In this last case, the estimated coefficients of the quantile regression show a negative and statistically significant correlation between the geographic diversification and NPL for households and non-financial corporations, whereas in the case of financial corporations these values are close to zero and insignificant. Interestingly, the impact remains constant along the conditional distribution of NPL except for the case of non-financial corporations in which the marginal impact largely increases for higher quantiles (Figures 9, 10, 19, 20).

In general, the overall effects of geographic diversification on NPL are therefore not clearly identified given the divergences in the empirical results obtained for this indicator. In this sense, an analysis of the effects of geographic diversification on asset quality greatly depends on the degree of incorporation of external factors into the models. However, if the models only include bank-specific factors, the positive effects of geographic identification on NPL are evident in the case of exposures of non-financial corporations. To some extent, this ambiguity is also in line with the literature as the benefits of cross-border portfolio diversification must be compared against the additional costs derived from the learning curve and other agency problems attached to banking activities abroad.

6.2.5 Disaggregated loan Aspects: Exposure Types and NPL levels

As observed in the random/fixed effects and GMM models for panel data, the estimated coefficients in households and non-financial corporations present in general similarities in direction (+/-) but differences in magnitude, whereas the estimated coefficients in financial corporations present the greatest divergences in comparative terms with low values and low significance levels. Moreover, there are significant differences in the explanatory power according to the R squared. While non-financial corporations and households are quite good explained by the data with higher R squared, financial corporations show lower explanatory power with low R squared. Interestingly, the largest estimated coefficients in all models are found by non-financial corporations, followed by households and financial corporations, meaning that the relation be-

tween bank-specific/macroeconomic indicators and NPL is more intense for corporates and businesses operating in the non-financial sector (Tables 3 , 14, 15).

Similarly, the results of the quantile regressions also present significant differences in the estimated coefficients across exposures types. The estimated coefficients in households and non-financial corporations are similar in sign but not in value with non-financial corporations showing the highest impact for almost all explanatory variables. On the other hand, the estimated coefficients in financial corporations seem to be less reliable, showing very divergent results in terms of direction and values (Figures 9, 10, 23, 24).

Moreover, the quantile regressions allow us to assess the effects of the explanatory variables across our sample of banks with heterogeneous levels of non-performing loans in their portfolios. In this case, the marginal impact of most explanatory variables tend to increase along the conditional distribution of NPL for all portfolio types, showing that the magnitude of the estimated coefficients is greater for banks with more non-performing loans at quantile level in a non-linear fashion (Figures 9, 19, 23). In particular, the increase in the marginal impact is quite strong in the case of macroeconomic indicators. In this context, the magnitude of the estimated coefficients is found to be largest at the upper quantiles. Interestingly, when only the sample of banks with domestic exposures greater or equal 75% is considered in the estimations, the change of the marginal impact along the conditional distribution of NPL diminishes. This last result explains the somehow flatter curve for the estimated coefficients in comparison to the estimations that include the full sample. With regard to the levels of significance, the results seem to be the same for each quantile in all models and sample types, where the statistical significance remains similar for lower, median and upper quantiles regardless of any divergences in the estimated coefficients.

In general, these results may be explained by the fact that the exposures types considered in this thesis react differently to specific sector developments and heterogeneity in credit risks that are also reflected in the NPL. Moreover, non-financial corporations and households are better explained by the selected bank-specific/macroeconomic factors than financial corporations. Given that the selection of the explanatory variables is based on theoretical considerations and other published empirical research studying the determinants of NPL, it is also possible to conclude that these studies are adequate for explaining problem loans in the non-financial sector and households, but not for other type of loans such as loans to financial corporations other than credit institutions. Nevertheless, the previous statement does not affect the overall validity of the results due to the fact that exposures in the non-financial sector and households usually make up most loan exposures in a traditional bank's portfolio.

Furthermore, the different results at quantile level point out the presence of large divergences with respect to the determinants of NPL across a sample of European banks at portfolio level, also characterizing the heterogeneity in the European banking sector in terms of risk appetite, management strategies and participated markets.

In summary, the presented divergences in the results among exposure types and quantiles illustrate the importance of an analysis of the determinants of NPL on a disaggregated loan portfolio at bank level in order to obtain a proper understanding of the drivers and evolution of NPL in the European banking system.

6.3 Research Hypotheses

Given the empirical results presented in this thesis, the research hypotheses formulated in the first section are evaluated as follows:

H₁: NPL are mainly determined by bank-specific and macroeconomic factors

Both bank-specific and macroeconomic factors are jointly significant in explaining NPL for all exposures types as observed in the empirical results and, therefore, the inclusion of this class of data for studying the determinants of NPL is vastly justified in our analysis.

H₂: Business cycle indicators are significantly related to NPL

Statistical evidence is found in support of the claim that the business cycle also plays an important role in explaining NPL for the non-financial and household sectors but not for financial corporations other than credit institutions. As indicators of the business cycle, two variables are considered in this analysis: GDP growth (annual %) (GDP_growth_perc) and unemployment rates (unempl_rate_imf). GDP growth (annual %) is only statistical significant for the case of non-financial corporations, in which the estimated coefficients are positive or negative depending on the model under consideration. With regard to unemployment rates, a positive correlation with NPL is identified for the non-financial and household sectors, in which the values are found to be statistically significance.

H₃: Inflation rates are significantly related to NPL

Inflation rates are found to be statistically significant and negatively correlated for all exposure types. The results point out the relevance of taking a closer look into the price level when analyzing NPL in the banking sector. In this case, the empirical evidence suggests that an increase in the price level may contribute to a reduction of problem loans probably by cutting down the real value of outstanding loans.

H₄: Public finance indicators are significantly related to NPL

As an indicator of the efficiency in the public sector, an assessment of the effects of Gross Debt as a percentage of GDP (*gross_public_debt*) on NPL is also included in this analysis. According to the empirical results, the estimated coefficients are found to be positively correlated with NPL for all exposure types, and statistically significant for non-financial corporations, households and non-financial corporations other than credit institutions. In other words, it can be affirmed that public finance indicators at country level such as the gross debt as percentage of GDP are significantly and positively related to NPL at bank level given their respective country of origin.

H₅: Terms of trade indicators are significantly related to NPL

With regard to the terms of trade, the current account balance as a percentage of GDP (*curr_acc_bal*) is also included as a regressor in the models. The estimated coefficients of the current account balance as a percentage of GDP are only statistically significant for the case of non-financial corporations and households. Furthermore, the estimated coefficients are positive, suggesting that trade related indicators may also play an important role as determinants of NPL in the non-financial and household sectors, but a secondary role in the case of financial corporations other than credit institutions. In this sense, trade surpluses at country level are found to be statistically related with larger amounts of NPL in the banking sector.

H₆: The size of a bank is significantly related to NPL

As an indicator of the size of a bank, a variable for the total assets at bank level (*BankSize_tran*) is incorporated in the models. The estimated coefficients are in this case negative and statistically significant for all exposure types under consideration. In other words, the size of a bank in terms of assets is significantly related to NPL at bank level. In summary, banks with a larger amount of assets in their balance sheets tend to show lower NPL in their portfolios.

H₇: Legal aspects of credit access and NPL resolution are significantly related to NPL

As indicators of the effects of the legal framework on NPL the following variables were included in our analysis: The Strength of Legal Rights index (*Stgth_legal_index*) as a measure of the strength of collateral and bankruptcy laws to favour the extension of new credit and Recovery Rate -cents on the dollar- (*Reco_rate_dol*) as a measure of final recovered amount from NPL after debt restructuring, debt execution (foreclosure) or bankruptcy proceedings. The estimated coefficients of the strength of Legal Rights index (*Stgth_legal_index*) show a positive but statistically insignificant correlation with NPL for all exposure types. Therefore, it can be concluded that the strength of the collateral and bankruptcy laws does not seem to play a significant role in explaining the amount of NPL at bank level. All estimated coefficients for the variable Recovery Rate -cents on the dollar- (*Reco_rate_dol*) show a negative value for all exposure types but only statistical significance for the case of non-financial corporations and

households. For this reason, recovery rates appear to be an important determinant of NPL in the non-financial and household sectors. In this sense, an increase in the average recovered amount after the conclusion of proceedings in the context of NPL resolution appears to be related with a reduction of NPL in the portfolios of banks. In summary, it can be concluded that certain legal aspects in terms of NPL resolution are significantly related to NPL, while other legal aspects concerning laws favouring the access to new credit appeared not to be significantly related.

H₈: Bank profitability and cost efficiency indicators are significantly related to NPL

Returns on Risk adjusted Capital (RORAC) and Net Interest Margin (NIM_tr) are included in this analysis as indicators of a bank's profitability, while the cost to income (Cost_to_Income) is used as an indicator of a bank's cost-efficiency. The estimated coefficients of Risk adjusted Capital (RORAC) are statistically significant and negatively correlated with NPL for all exposure types and estimation models. The estimated coefficients of Net Interest Margin (NIM_tr) show a positive correlation with NPL for all exposure types with statistical significance for non-financial corporations and households in most models. In this sense, a general improvement of a banks' profitability appears to be related to a reduction of NPL at bank-level once the effects as long as the improvement is not motivated by higher net interest margins. Furthermore, the estimated coefficients of the cost to income (Cost_to_Income) are negative and statistically insignificant for all exposure types. In summary, the empirical evidence suggests a significant relationship between banks' profitability indicators and NPL, while with respect to the indicator of cost-efficiency a significant relationship could not be identified.

H₉: Bank capital adequacy ratios are significantly related to NPL

Common Equity Tier1 Ratio-CET1- (CET1_ratio_tr) and the leverage ratio (LEVER_tr_ratio) are included in this analysis as indicators of a bank's capital adequacy. The estimated coefficients of CET1 (CET1_ratio_tr) are negative and statistically significant for non-financial corporations and households, and insignificant for the case of financial corporations other than credit institutions. Specially, the magnitude and significance of the estimates greatly increase in models with only bank-specific variables, where the macroeconomic effects are explicitly excluded from the regressions. Regarding the leverage ratio (LEVER_tr_ratio), the estimated coefficients are positive but mostly insignificant for all exposures types. Given these results, it can be concluded that bank capital adequacy ratios are significantly related to NPL in as much as the regulatory capital is concerned. In this sense, an increase in the regulatory capital requirements contributes to a reduction of NPL.

H₁₀: The level of geographic diversification in portfolio is significantly related to NPL

In order to assess the effects of geographic diversification on NPL, a Cross-Border Diversification Index (CBDI_index) is calculated based on the available data on the country of destiny

of the exposures at bank level. The estimated coefficients of the Cross-Border Diversification Index (CBDI_index) are either positive or negative depending on the model under consideration. Furthermore, the estimates are statistically significant for non-financial corporations and households but mostly insignificant for financial corporations other than credit institutions. Therefore, although the empirical results suggest that the level of geographic diversification in a bank's portfolio is a relevant determinant of NPL, there is no conclusive evidence of whether the final effects on NPL are positive or negative.

H₁₁: Supervision under the SSM has significant effects on NPL

A dummy variable for membership under the Single Supervisory Mechanism (SSMY) was created based on the available information on the banks' country of origin. The estimated coefficients for the dummy coefficient of membership under the SSM (SSMY) are negative for all exposure types and statistically significant for non-financial corporations and households. Therefore, the empirical evidence supports the statement that Supervision under the SSM has significant effects on NPL generated in the non-financial and household sectors. For the contrary, these effects appear to be insignificant for financial corporations other than credit institutions. Despite the fact that some caution is needed when interpreting the empirical results due to the presence of some asymmetries in the sample, it can be affirmed that regulatory supervision under the Single Supervisory Mechanism (SSMY) contributes favourably to a reduction of NPL in the European banking sector.

H₁₂: Magnitude and direction of the effects on NPL varies across exposure types

In this thesis, several models are estimated across three exposure types: households, non-financial corporations and financial corporations other than credit institutions. As observed when comparing the empirical results across exposure types, the estimated coefficients for households and non-financial corporations show similarities in direction in terms of positive or negative sign, but differences in magnitude in terms of high or low values. For the contrary, the estimated coefficients for financial corporations tends to diverge from the rest of exposure types in both magnitude and direction. Furthermore, exposures in the non-financial sector present the largest estimated coefficients in all models followed by households. In this sense, it can be affirmed that the effects of bank-specific and macroeconomic factors on NPL are greatest for corporates and businesses operating in the non-financial sector, closely followed by the household sector, while exposures in financial corporations other than credit institutions seem to have the lowest impact.

H₁₃: The effects on NPL vary conditional on the initial level of NPL

In this thesis, the use of quantile regression methodology allows the assessment of the determinants of NPL across a sample of banks with heterogeneous levels of non-performing loans in their portfolios. According to the empirical results, the estimated coefficients reach their maxi-

mum values at the upper quantiles and the magnitude of their marginal impact tend to increase along the conditional distribution of NPL for all exposure types. Furthermore, the estimated coefficients present similar levels of significance for lower, median and upper quantiles despite any difference in magnitude at quantile level. Therefore, the empirical evidence suggests that the magnitude of the effects of bank-specific and macroeconomic factors on NPL at bank level will diverge conditional on the initial level of NPL in the portfolio. In this sense, these results characterize the importance of taking into consideration the presence of heterogeneity among European banks when analyzing the determinants of NPL.

Table 2 summarizes these conclusions. This evidence is based on the sign and the statistical significance of the estimated coefficients obtained in the regression models, with the exceptions of H_{12} and H_{13} , whose empirical evidence is based on comparing the estimated coefficients across exposure types and quantiles along the conditional distribution of NPL, respectively.

Table 2: Empirical Evidence for Tested Hypotheses

Hypothesis	Variables	<i>Non-Financial</i>	<i>Households</i>	<i>Financial</i>
H_1 : Joint Significance	ALL	Yes	Yes	Yes
H_2 : Business Cycle	GDP_growth_perc	Yes (+/-)	No (+)	No (+)
	unempl_rate_imf	Yes (+)	Yes (+)	No (+/-)
H_3 : Inflation Rates	infl_perc	Yes (-)	Yes (-)	Yes (-)
H_4 : Public Finance	gross_public_debt	Yes (+)	Yes (+)	Yes (+)
H_5 : Terms of Trade	curr_acc_bal	Yes (+)	Yes (+)	No (+)
H_6 : Bank Size	BankSize_tran	Yes (-)	Yes (-)	Yes (-)
H_7 : Legal Aspects	Stgth_legal_index	No (+)	No (+)	No (+/-)
	Reco_rate_dol	Yes (-)	Yes (-)	No (-)
H_8 : Profitability & Cost	RORAC	Yes (-)	Yes (-)	Yes (-)
	NIM_tr	Yes (+)	Yes (+)	No (+)
	Cost_to_Income	No (-)	No (-)	No (-)
H_9 : Capital Adequacy	CET1_ratio_tr	Yes (-)	Yes (-)	No (-)
	LEVER_tr_ratio	No (+)	No (+)	No (+)
H_{10} : Geo. Diversification	CBDI_index	Yes (+/-)	Yes (+/-)	No (+/-)
H_{11} : SSM	SSMY	Yes (-)	Yes (-)	No (-)
H_{12} : Diff. Exposures	Direction	No	No	Yes
	Magnitude	Yes	Yes	Yes
H_{13} : Diff. NPL Levels	Quantile (%)	Yes	Yes	Yes

Notes: Results are based on signs (+/-) and stat. significance (Yes/no) of estimated coefficients. In H_{12} and H_{13} , results are based on comparisons across exposure types and quantiles.

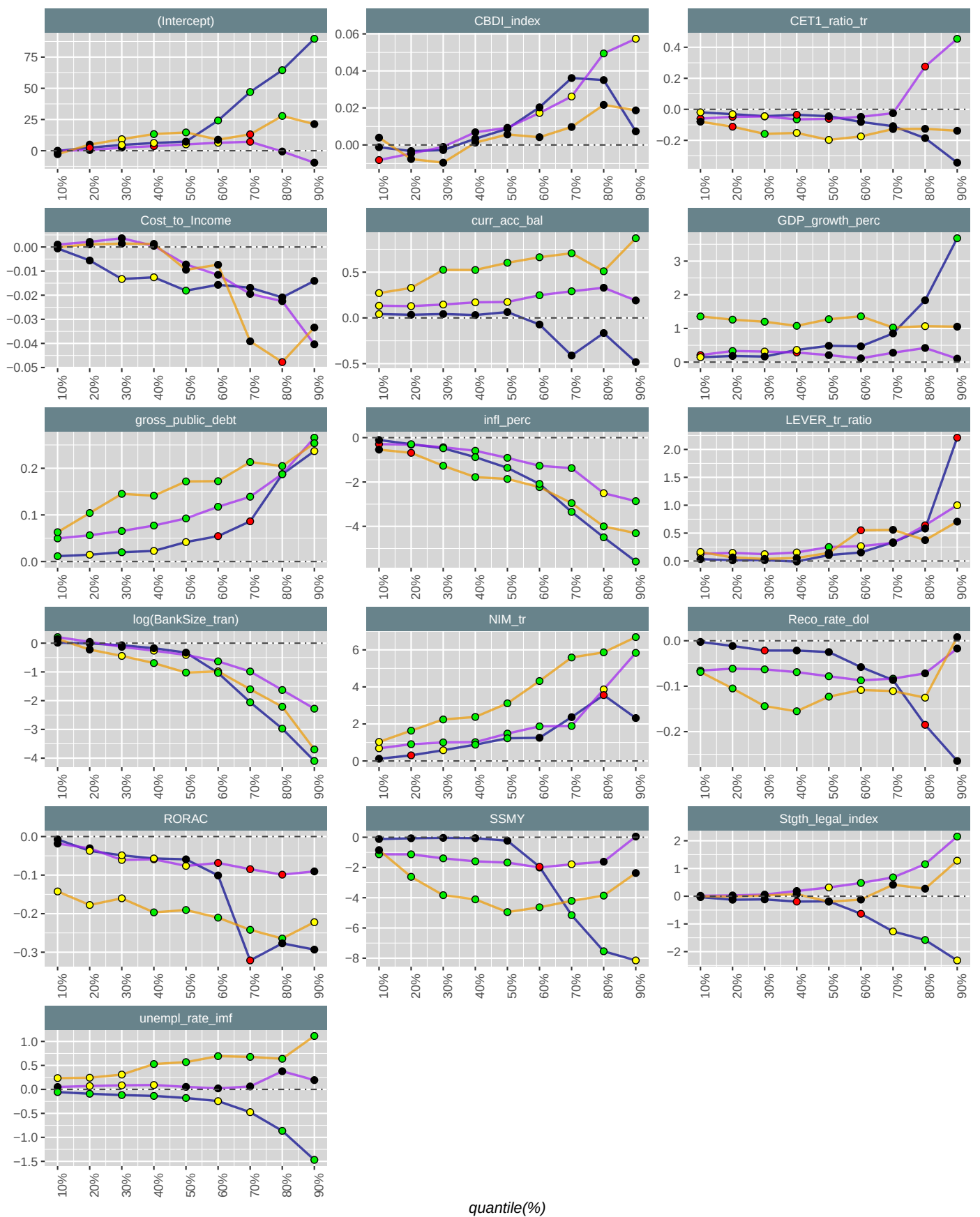
Table 3: Panel Data Models (2014-2019)- All Sample

	<i>panel</i> <i>linear</i>						<i>panel</i> <i>GMM</i>		
	Households		Non-Financial		Financial		Households	Non-Financial	Financial
	RE	FE	RE	FE	RE	FE	SYS	SYS	SYS
lag(NPL_ratio)							1.040*** (0.090)	0.791*** (0.081)	0.542*** (0.153)
CET1_ratio_tr	-0.158*** (0.053)	-0.201*** (0.061)	-0.092 (0.103)	0.080 (0.092)	-0.159 (0.142)	-0.020 (0.314)	0.081 (0.071)	0.001 (0.115)	0.024 (0.114)
LEVER_tr_ratio	0.422** (0.192)	0.153 (0.246)	-0.018 (0.272)	-0.592 (0.418)	0.649 (0.626)	0.260 (0.883)	-0.004 (0.098)	-0.071 (0.160)	0.150 (0.368)
RORAC	-0.036** (0.017)	-0.032** (0.016)	-0.027 (0.043)	-0.001 (0.025)	-0.117** (0.049)	-0.042 (0.036)	-0.073** (0.032)	-0.114* (0.064)	-0.270* (0.158)
log(BankSize_tran)	-2.060*** (0.462)	-4.460*** (1.170)	-2.710*** (0.620)	0.514 (1.500)	-1.760** (0.794)	9.850*** (3.530)	0.161 (0.129)	-0.163 (0.223)	-0.231 (0.560)
Cost_to_Income	-0.004 (0.006)	-0.005 (0.005)	-0.013 (0.012)	-0.013 (0.009)	0.019 (0.022)	0.025 (0.029)	-0.004 (0.005)	-0.012 (0.013)	-0.012** (0.006)
NIM_tr	0.758 (0.535)	0.301 (0.483)	1.340 (0.981)	0.492 (1.070)	0.681 (2.160)	0.589 (2.600)	-0.184 (0.733)	1.050 (0.924)	0.296 (1.190)
Stgth_legal_index	0.766*** (0.285)	0.262 (0.220)	1.070** (0.468)	2.230*** (0.419)	0.020 (0.751)	0.820 (0.738)	0.083 (0.163)	0.094 (0.246)	-0.517 (0.542)
Reco_rate_dol	-0.052** (0.022)	-0.023 (0.020)	-0.130** (0.060)	-0.132** (0.067)	-0.175 (0.122)	-0.167 (0.138)	-0.033* (0.017)	-0.090*** (0.032)	-0.120* (0.071)
infl_perc	-0.213 (0.174)	-0.150 (0.139)	-0.637** (0.325)	0.174 (0.279)	-1.190 (0.895)	-0.262 (0.902)	0.209 (0.508)	-0.927 (0.610)	-1.050 (1.100)
curr_acc_bal	0.102 (0.109)	0.186** (0.093)	0.451*** (0.166)	0.512** (0.200)	0.230 (0.356)	1.060*** (0.384)	0.196*** (0.056)	-0.014 (0.109)	-0.226 (0.244)
GDP_growth_perc	0.244 (0.149)	0.176* (0.102)	0.312 (0.252)	0.043 (0.286)	0.631 (0.725)	-0.149 (0.327)	-0.200 (0.227)	-0.760** (0.330)	0.325 (0.961)
unempl_rate_imf	0.411*** (0.082)	0.413*** (0.071)	1.660*** (0.183)	2.220** (0.184)	0.362 (0.221)	1.070*** (0.266)	0.121** (0.057)	-0.103 (0.114)	-0.264 (0.188)
gross_public_debt	0.153*** (0.025)	0.075*** (0.025)	0.195*** (0.039)	0.191*** (0.061)	0.151*** (0.057)	0.063 (0.093)	0.002 (0.015)	-0.001 (0.028)	0.012 (0.045)
SSMY	-1.970 (1.300)		-5.350*** (1.600)		-2.490 (2.790)				
CBDI_index	0.050*** (0.019)	0.031 (0.019)	0.057** (0.026)	0.119*** (0.039)	0.038 (0.040)	0.039 (0.063)	0.014* (0.008)	-0.005 (0.010)	-0.014 (0.023)
Constant	12.800*** (4.940)		19.500** (8.530)		22.700 (16.300)				
Hausman Test	17.51		1202.57		4.69				
Prob (Hausman Test)	0.2298		0		0.9896				
Sargan Test							39.87	43.28	42.09
Prob (Sargan Test)							0.0527	0.0245	0.0322
Arellano-Bond:AR(1)							-1.94	-0.97	-6.36
Prob (AR(1))							0.0525	0.334	0
Arellano-Bond:AR(2)							0.58	-0.93	0.67
Prob (AR(2))							0.5627	0.3501	0.5019
Observations	619	619	628	628	580	580			
R ²	0.376	0.338	0.532	0.575	0.150	0.124			
Adjusted R ²	0.361	0.155	0.521	0.456	0.127	-0.134			
F Statistic	359.000***	17.700***	687.000***	47.300***	95.400***	4.540***			

Note: HC-robust standard errors are used

*p<0.1; **p<0.05; ***p<0.01

RE:Random Effects, FE:Fixed Effects,SYS: System-GMM, DIFF: Differences-GMM

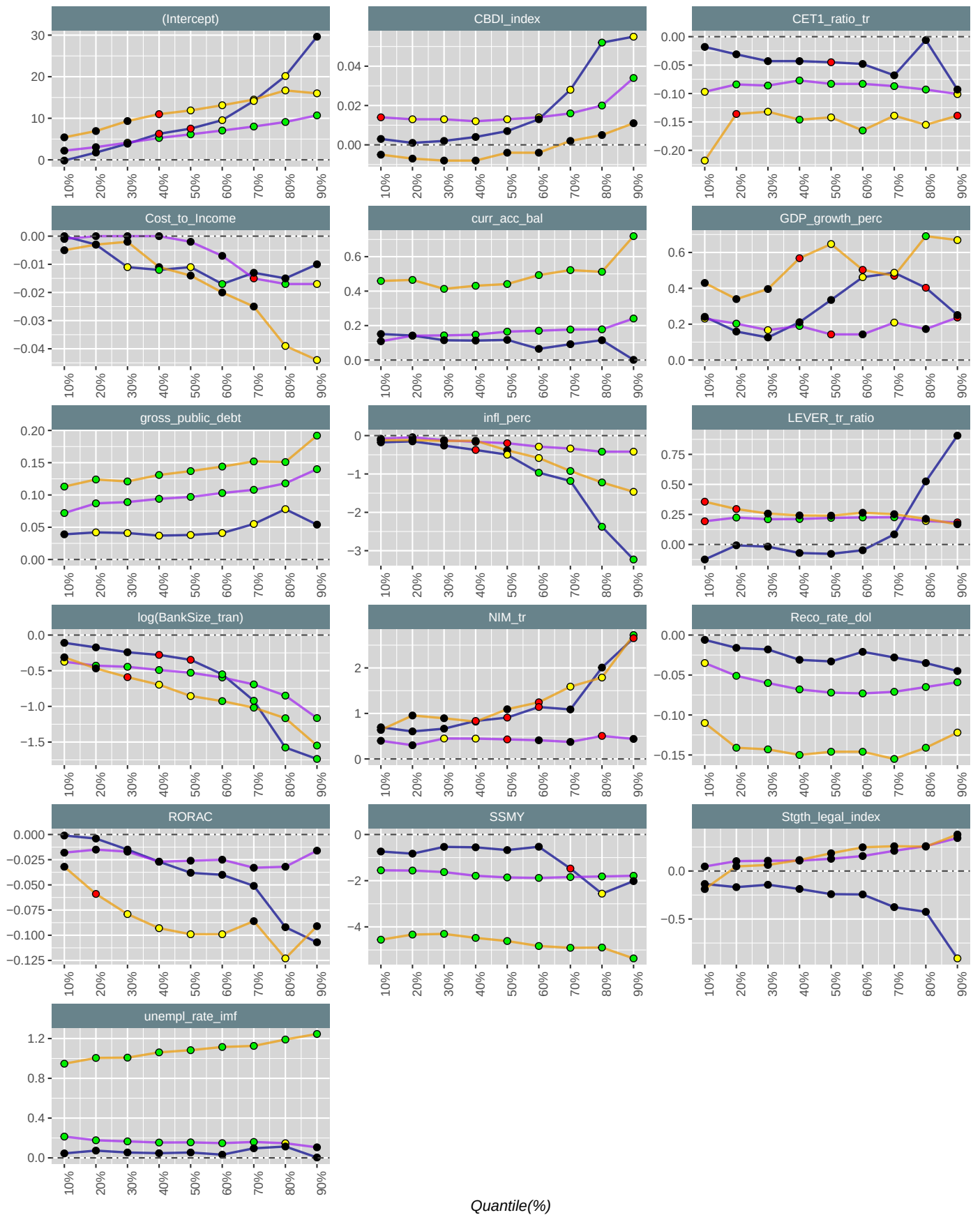


Exposure Financial Corp. Households Non-Financial Corp. P-values: <0.01 <0.05 <0.1 >

Source: EBA Transparency Exercises, authors' calculations

Figure 9: Pooled Quantile Regression Graphics

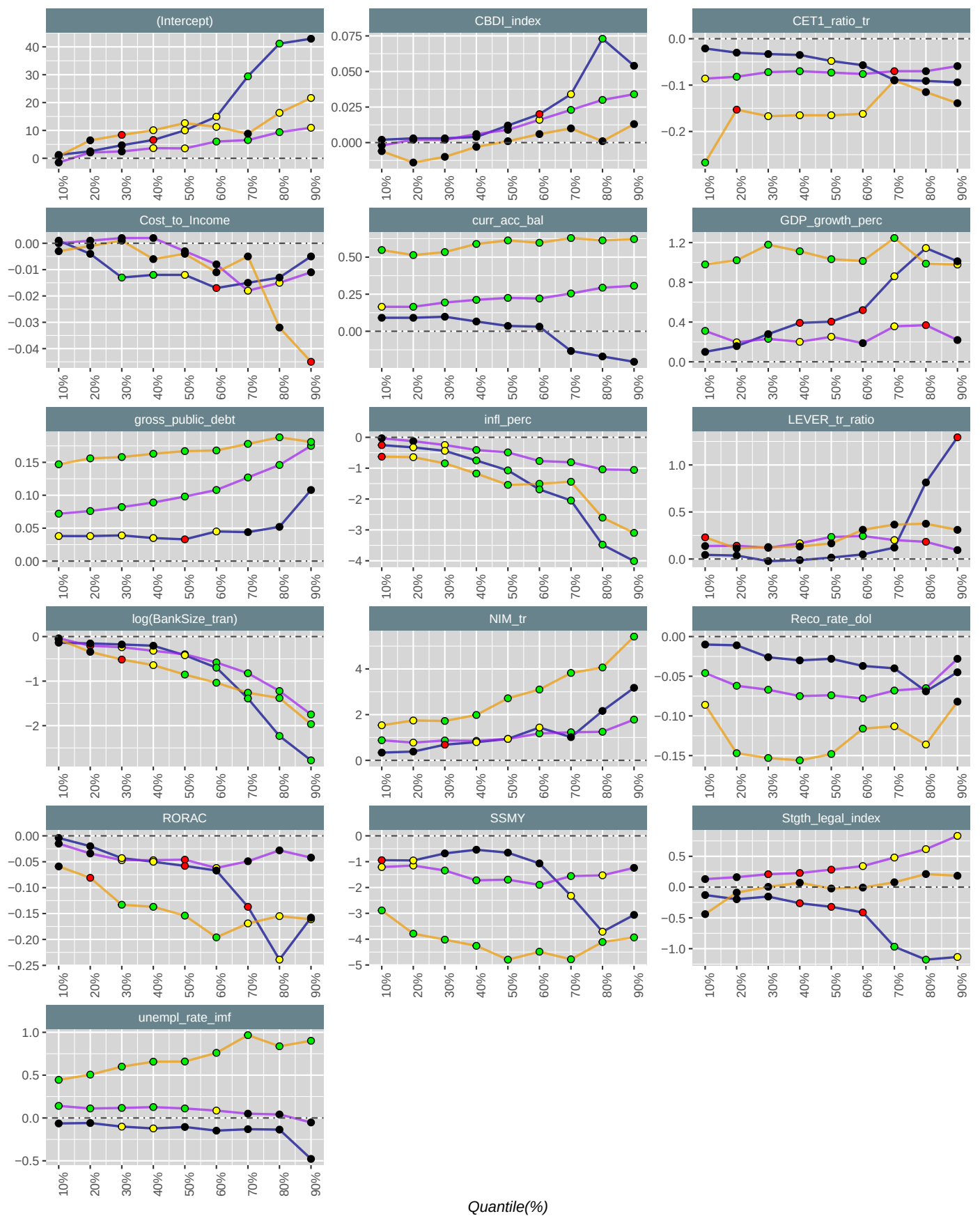
Source: EBA Transparency Exercises, Worldbank and IMF data, authors' calculations



Source: EBA Transparency Exercises, authors' calculations

Figure 10: Fixed Effects Quantile Regression Graphics ($\lambda = 1$)

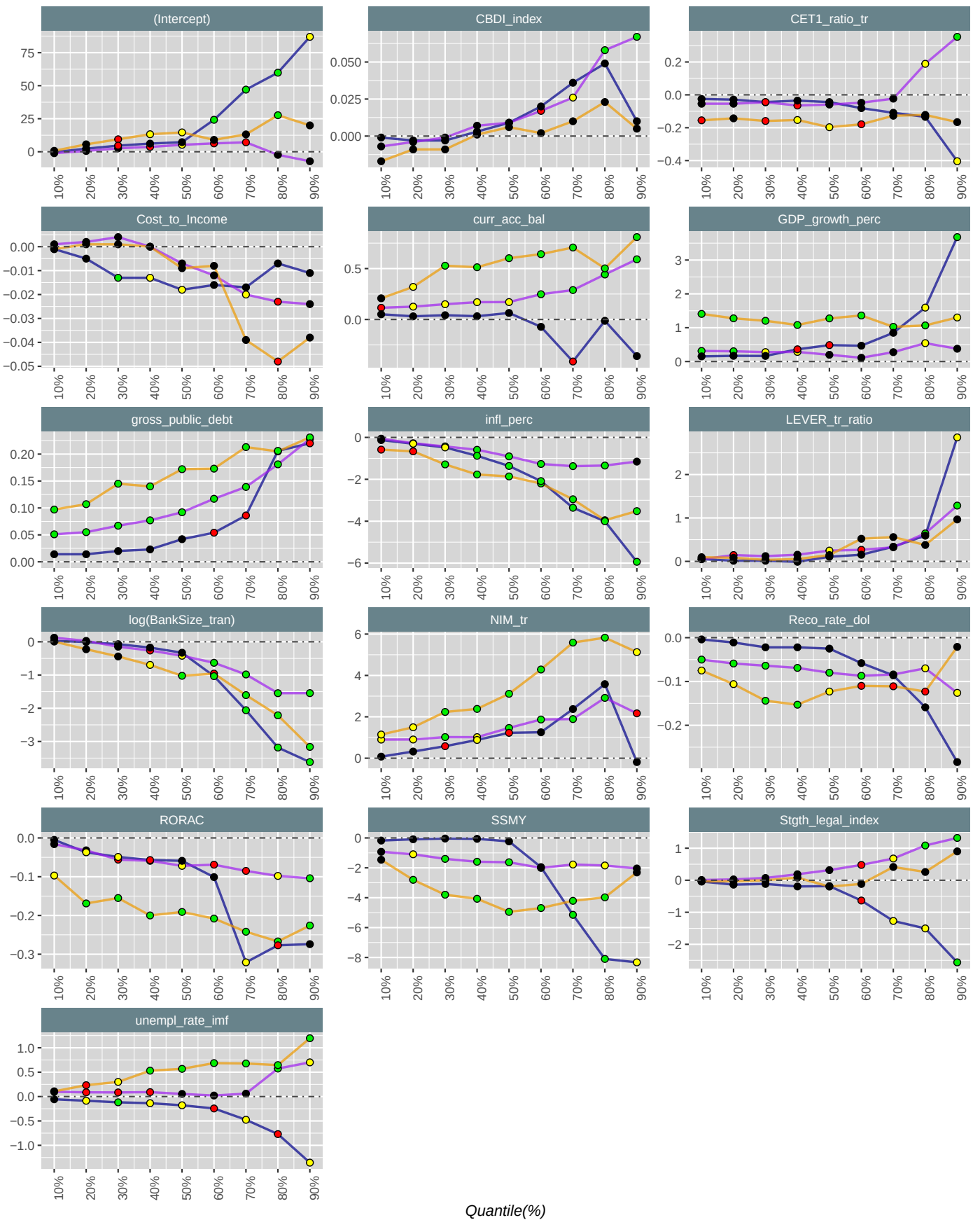
Source: EBA Transparency Exercises, Worldbank and IMF data, authors' calculations



Source: EBA Transparency Exercises, authors' calculations

Figure 11: Fixed Effects Quantile Regression Graphics ($\lambda = 2.5$)

Source: EBA Transparency Exercises, Worldbank and IMF data, authors' calculations



Source: EBA Transparency Exercises, authors' calculations

Figure 12: Fixed Effects Quantile Regression Graphics ($\lambda = 5$)

Source: EBA Transparency Exercises, Worldbank and IMF data, authors' calculations

7 Conclusions

In this thesis, panel data estimation techniques (Random Effects/Fixed Effects/GMM) and quantile regression methods for panel data (Pooled/Penalized Fixed Effects) were applied with the objective of examining the determinants of NPL in the EU-banking sector. For this purpose, the regressions also made use of an extensive, granular dataset on major European banks gathered by the European Banking Authority (EBA) in the context of the EU-wide Transparency Exercise for the period 2014-2019. Particularly, this analysis takes into consideration disaggregated banks' portfolio characteristics in terms of exposure types (households, non-financial corporations, financial corporations other than credit institutions), geographic composition of the exposures (country of destiny worldwide), intensity of cross-border diversification and initial levels of NPL. Moreover, the characteristics of the legal system and the regulatory status under the Single Supervisory Mechanism (SSM) are also analyzed as potential determinants of NPL. In general, a significant correlation between the selected determinants and the development of NPL is observed in the empirical results, pointing out the importance of bank-specific and macroeconomic factors as well as the characteristics of the legal system for explaining the development of NPL in Europe. The signs and significance levels of the estimates for all the considered models and methodologies are also compatible with the various theoretical arguments presented in section 2 following the available literature in NPL, with the exception of the effects of GDP growth on NPL, which appear to play a secondary role in this analysis. In light of the magnitude and statistical significance of the estimated coefficients as well as the general importance and consistency of these estimates across the different models and exposure types, **the following indicators are found to be the "most" important determinants of NPL: Common Equity Tier1 Ratio-CET1- (CET1_ratio_tr) and Returns on Risk adjusted Capital (RORAC) as the main bank-specific indicators, and inflation rates (infl_perc), General government gross debt as percent of GDP (gross_public_debt) and unemployment rates (unempl_rate_imf) as the main macroeconomic indicators. With regard to the characteristics of the legal system, recovery rates (Reco_rate_dol) after NPL resolution proceedings are also found to be an important determinant of NPL.**

Regarding the main bank-specific indicators, Common Equity Tier1 Ratio-CET1- (CET1_ratio_tr) is the main capital adequacy ratio for regulatory and supervisory purposes in the European Banking sector, and its estimated coefficients are negative and statistically significant, and therefore it can be affirmed that an increase in the regulatory capital requirements contributes to a reduction of NPL, pointing out the importance of regulatory capital requirements for controlling NPL at portfolio level. Specially, the magnitude and significance of the estimates greatly increase in models with only bank-specific variables, where the macroeconomic effects are ex-

plicitly excluded from the regressions. Return on Risk adjusted Capital (RORAC) is the main indicator of bank profitability, and its estimated coefficients are found to be statistically significant and negative. In other words, an improvement of a bank's profitability appears to be related with a reduction of NPL at portfolio level.

Regarding the main macroeconomic indicators, inflation rates (*infl_perc*) remain significance in all models, specially in the models applying quantile regression methodology. In this case, the empirical evidence suggests that an increase in the price level contributes to a reduction of problem loans probably by cutting down the real value of outstanding loans. General government gross debt as percent of GDP (*gross_public_debt*) is also found to be very significant and positively related to NPL, suggesting that the performance of the general government also has a large impact on the development of problem loans at bank level. Unemployment rates (*unempl_rate_imf*) show a positive correlation with NPL in all empirical models, supporting the claim that the business cycle also plays an important role in explaining NPL. In this sense, an increase in the unemployment rate in the country where the exposures are located is found to be related with higher NPL for those exposures. With regard to the characteristics of the legal system, Recovery Rate -cents on the dollar- (*Reco_rate_dol*) appears to be a very important determinant of NPL, where an increase in the average recovered amount after the conclusion of proceedings for NPL resolution positively contributes to a reduction of NPL. In other words, given the empirical evidence presented in this thesis, it can be concluded that efficient declaratory procedures of NPL resolution play an important role in explaining NPL at bank level.

In this context, the inclusion of variables about supervisory and legal aspects allowed the identification of interesting empirical relations, where structural and country-specific factors regarding NPL resolution and access to credit are important determinants of NPL. Particular attention is given to the regulatory status under the Single Supervisory Mechanism (SSM), which seems to play an important role in reducing the levels of NPL in Europe. In this sense, **the new empirical evidence presented in this thesis suggests that supervision under the SSM contributes significantly to a reduction of NPL in the European banking system.** These results support the claim that a further integration of the European financial sector within the established European institutions would have a positive effect on the fight against the excessive accumulation of problem loans in the EU-banking sector.

Moreover, the introduction of an index that measures the intensity of cross-border diversification in a bank's portfolio as an additional explanatory variable has delivered some interesting insights in the relationship between cross-border exposures and the development of NPL in the European banking sector. In this respect, **new empirical evidence is found suggesting that the level of geographic diversification in a bank's portfolio is in fact a relevant determinant**

of NPL, although the results tend to differ across regression methods, exposure types and data samples and, therefore, **there is no conclusive evidence of whether the final effects on NPL are positive or negative.**

The results presented in this thesis also provide **some new empirical evidence that highlights the disparities in the quantitative effects of the various determinants of NPL when performing a disaggregated analysis on exposure types.** With regard to the different exposure types, it can be affirmed that the effects of bank-specific and macroeconomic factors on NPL are greatest for corporates and businesses operating in the non-financial sector, closely followed by the household sector, while exposures in financial corporations other than credit institutions present the lowest impact. **In this sense, exposures in non-financial corporations are the most sensitive to the determinants of NPL, closely followed by households, whereas exposures in financial corporations other than credit institutions are poorly explained by these factors.**

Moreover, the results of the quantile regression models also provided **new insights into the determinants of NPL across banks by disaggregating the analysis into groups of banks with different initial levels of NPL in their portfolios.** Given the results of the quantile regression models, **it is identified that the marginal and absolute impact of most determinants of NPL differs from bank to bank, depending heavily on the original amount of NPL in a bank's portfolio.** In particular, these new empirical results suggest that the magnitude of the impact tends to increase for banks with higher levels of NPL.

In summary, although the empirical results for many of the determinants of NPL considered in the present analysis are in line with the general expectations when compared with other results and arguments already available in the academic literature, a serie of "new" and, in this sense, also "unexpected" findings were possible thanks to the novelty and richness of the EBA's data, which allowed to put the focus of this analysis on the disaggregated aspects of NPL in the context of the most recent European banking system.

The "new" findings show that the relative importance of the determinants of NPL depends greatly on the exposure type under consideration, the original amount of NPL as well as the intensity of cross-border diversification in a bank's portfolio. In this sense, this thesis provides some new empirical evidence that further recognizes the heterogeneity in the European banking sector in terms of risk appetite, management strategies and participated markets as well as the need of considering these aspects at a disaggregated level in future empirical research for a proper understanding of the development of NPL in Europe. With regard to the role of the supervisory system, "new" findings are also pro-

vided in support of a further integration of the European financial sector under the Single Supervisory Mechanism (SSM).

In this context, an important contribution also concerns the type of data and the empirical methodology applied for the present analysis. This thesis has benefited from the availability of disaggregated supervisory data disclosed by the EBA on major European banks as part of the EU-wide Transparency Exercise. The use of this data represents quite a novelty given that there are very few published papers applying it in the academic research on NPL. In similar terms, the recent application of quantile regression methodology for panel data in the analysis of NPL also represents quite a novel contribution. In this regard, **the present analysis provides a new example of the potential use of the EBA's data and a new application of the quantile regression methodology for panel data.**

Furthermore, the empirical evidence presented in this thesis may also be relevant in terms of regulation and policy. For instance, the empirical findings indicate that regulatory capital and profitability ratios as defined for banking supervisory purposes are adequate indicators for explaining and predicting the future development of non-performing loans in a bank's portfolio. However, it is advisable for the supervisory authorities to establish additional indicators aimed at better monitoring the managerial performance in the European banking sector. For instance, a deeper monitoring of the business decisions and practices of a bank's management would contribute to the earlier identification of banks presenting an excessive increase of problem loans. In terms of European policy, it is also desirable to foster further integration of the European banking system in the context of the Single Supervisory Mechanism (SSM). **In other words, the supervisory authorities should further concentrate efforts on procedures and systems for improving risk management tools at a disaggregated level, thus accounting for the large heterogeneity in the European banking system, while promoting at the same time further integration of the European financial sector within the established European institutions.**

Additionally, **some empirical results or methods presented in this thesis can also be used for the design and calibration of forecasting models or for bank-stress testing purposes.** In a bank stress testing framework, the calibration may be performed to test alternative scenarios with regard to the development of bank-specific/macroeconomic indicators and their effects on European banks at portfolio level, contributing to a better assessment of the appropriateness of loan loss provisions in the EU banking sector. Moreover, other related performance indicators may also be implemented for microprudential purposes to evaluate the financial risks linked to individual banks showing performance and efficiency problems. Considering that the present analysis is performed on the basis of a disaggregated loan portfolio, the results can also be implemented for stress testing models that focus on different exposure types. Moreover, **the**

presented results are also relevant for the design of recommendations to banks on NPL as published by the ECB and other national supervisory authorities regarding the development and implementation of adequate NPL strategies and Early Warning Indicators (EWI) in the EU banking system.

The present analyses may be extended in several forms such as the inclusion of a broader choice of bank-specific or macroeconomic indicators, the extension of the time period by including most recent available data or a change in the periodicity of the panel data by using narrower time periods on a quarterly or semesterly basis. Furthermore, given the expected increase in NPL due to the dramatic economic consequences of the Covid-19 pandemic that started in 2020, this analysis can also be extended by separately considering the pre-Covid and post-Covid periods and comparing the results to assess any change in the quantitative effects of the determinants of NPL as well as the overall impact of this unprecedented economic shock on the future development of NPL in Europe.

Finally, given the steady increase in the quantity and quality of the financial data publicly provided by the European Banking Authority (EBA) and other European/International institutions in the context of a constantly changing legal and economic environment, the development of new perspectives on the analysis of NPL may direct research towards a deeper study of loan-specific characteristics in a legal and empirical framework.

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A Appendix

A.1 Software

- * **Microsoft Office:** The raw data regarding the EBA transparency exercises was downloaded from the EBA's official website in .csv or/and .xlsx file formats. Therefore, Microsoft Office products were used.
- * **SAS:** SAS software was used to prepare the data sets prior to the empirical analysis, including data collection, cleansing, transformation and storage. SAS is a statistical software suite developed by SAS Institute for data management and advanced analytics. This software is used for some tasks regarding data import, data transformation and relational data in order to set proper data structures.
- * **Rstudio & R programming language:** Final data sets are to be provided in .Rdata file format as the proper empirical analysis is performed in R. RStudio is an integrated development environment for R, a programming language for statistical computing and graphics. The generated code in a .rnw file format will be embedded in a $\text{\LaTeX}$ document using knitr (an engine for dynamic report generation with R). R programming language is used in this thesis for graph/chart generation, descriptive statistics and the estimation and inference of parameters in the econometric models.
- * **TexMaker & PDF:** This master's thesis is presented in PDF format. The document was typed in $\text{\LaTeX}$ (a document preparation system for high-quality typesetting) and compiled into PDF using PDF $\text{\LaTeX}$. TexMaker is used as $\text{\LaTeX}$ editor. TexMaker is a free, modern and cross-platform $\text{\LaTeX}$ editor. This software was used parallel with knitr for the review, enhancement and refinement of $\text{\LaTeX}$ syntax in the .tex files generated by knitr.

A.2 Description of Variables

Bank-Specific variables

* Non-Performing Loans to total gross loans [NPL]

◇ *Definition:*

Harmonized definition of NPL across European countries introduced by the EBA in 2014 for supervisory reporting. In general, Non-Performing Loans (NPL) are loans that satisfy either of the following criteria: (a) material exposures that are more than 90 days past due; and (b) the debtor is assessed as unlikely to pay its credit obligations in full without realization of collateral, regardless of the existence of any past due amount or of the number of days past due (European Banking Authority, 2020a; European Central Banking Supervision, 2014).

◇ *Source:*

EBA Transparency Exercise-Database, author's calculations

Group: cre (Credit Risk and Asset Quality)

Sheet: NPE

Item Number from EBA 2020: 2020605 [Gross carrying amount on Loans and advances]

Dimensions (perf_status) from EBA 2020:

- 0=No breakdown by Performing status
- 1=Performing
- 2=Non Performing
- 3=Performing but past due >30 days and <=90 days
- 4=Non Performing and Defaulted
- 5=Loans and advances subject to impairment

◇ *Calculation* (European Banking Authority, 2020b):

$$\frac{\text{Perf_status}=(\text{'2'})2020605[\text{Gross carrying amount on Loans and advances}]}{\text{Perf_status}=(\text{'0'})2020605[\text{Gross carrying amount on Loans and advances}]}$$

* Single Supervisory Mechanism [SSM]:

◇ *Definition:*

The Single Supervisory Mechanism (SSM) refers to the system of banking supervision in Europe created as an EU policy response to the large accumulations of NPL stocks and the threat of financial instability in many European countries, and it represents a step further in the process of European integration. The SSM is also one of the pillars of the EU banking union along with the Single Resolution Mechanism, and it is composed of the European Central Bank (ECB) and the national supervisory bodies of

the participating EU member states (The Council of the European Union, 2013). In this framework, the European Central Bank (ECB) assumed specific responsibilities in the supervision of the banking sector and also provides guidance on the setting of NPL strategies (European Central Banking Supervision, 2017).

◇ **Source:**

EBA Transparency Exercise-Database (Metadata/Bank Lists)

◇ **Calculation:**

Variable Identification for banks under the framework of the Single Supervisory Mechanism (Y: YES/N: NO)

* **Bank Size (Min EUR)[BankSize_tran]**

◇ **Definition:**

Proxy variable for total bank assets

◇ **Source:**

EBA Transparency Exercise-Database

Group: otr (solvency)

Sheet: Leverage

Item Number from EBA 2020: 2020903 [Total leverage ratio exposures - using a transitional definition of Tier 1 capital]

* **Common Equity Tier 1 Ratio [CET1_ratio_tr]**

◇ **Definition:**

CET1 ratio shows a bank's core capital against its risk-weighted assets to assess its ability to resist financial distress. Common Equity Tier 1 (CET1) is a component of Tier 1 Capital, and it includes the core capital that a bank holds in its capital structure. It encompasses equity capital (ordinary shares) and disclosed reserves such as retained earnings (European Banking Authority, 2019).

◇ **Source:**

EBA Transparency Exercise-Database

Group: otr (solvency)

Sheet: Capital

Item Number from EBA 2020:

2020140 [Common Equity TIER 1 Capital Ratio (transitional period)]

* **Cost to Income [Cost_to_Income]**

◇ **Definition:**

The cost-income ratio is based on consolidated banking data. It shows bank's operating costs as a percentage of net operating income, and it is used as a further indicator of a bank's cost efficiency. Consolidation refers to the elimination of positions and flows within individual banks or banking groups ().

◇ **Source:**

EBA Transparency Exercise-Database, author's calculations

Group: otr (Profitability)

Sheet: P&L

Item Number from EBA 2020:

2020317 [(Administrative expenses)]

2020318 [(Depreciation)]

2020316 [Total Operating Income_Net]

◇ **Calculation** (European Banking Authority, 2020b):

$$\frac{2020317 [(Administrative\ expenses)] + 2020318 [(Depreciation)]}{2020316 [Total\ Operating\ Income_Net]}$$

* **Leverage ratio [LEVER_tr_ratio]**

◇ **Definition:**

the Tier 1 Leverage ratio compares a bank's capital against its total assets. This ratio is used to evaluate how much leverage a bank shows with respect to its consolidated assets. The denominator is composed of Tier 1 assets, which are assets that can be easily liquidated if a bank needs capital in the event of a financial crisis and the numerator is composed of all the components of the Tier 1 capital, which represents a bank's common equity, retained earnings, reserves and certain instruments with discretionary dividends and no maturity (European Banking Authority, 2019).

◇ **Source:**

EBA Transparency Exercise-Database, author's calculations

Group: oth (Solvency)

Sheet: Leverage

Item Number from EBA 2020:

2020905 [Leverage ratio - using a transitional definition of Tier 1 capital]

* **Net Interest Margin [NIM_tr]**

◇ **Definition:**

Net interest margin is calculated by dividing the difference between the interest income generated by a bank's lending operations and the interest expenses paid out to lenders by the bank's average earning assets. It is an industry-specific profitability ratio for banks and other financial institutions that lend out interest-earning assets. Net interest margin is one indicator of a bank's profitability and growth (European Banking Authority, 2019).

◇ **Source:**

EBA Transparency Exercise-Database, author's calculations

Group: oth (Profitability)

Sheet: P&L

Item Number from EBA 2020:

2020301 [Interest income]

2020304 [Interest expenses]

2020903 [Total leverage ratio exposures - using a transitional definition of Tier 1 capital]*

◇ **Calculation** (European Banking Authority, 2020b):

$$\frac{2020301 \text{ [Interest income]} - 2020304 \text{ [Interest expenses]}}{2020903 \text{ [Total leverage ratio exposures - using a transitional definition of Tier 1 capital]}}$$

* **Return on Regulatory (Risk-Adjusted) Capital [RORAC]**

◇ **Definition:**

Return On Risk-Adjusted Capital (RORAC) is a risk-adjusted measure of returns on investment. It accounts for any expected losses and income generated by capital under the assumption that riskier assets are expected to be followed by higher expected returns. The definition of RORAC is closely related to the definition of Return on Equity (ROE), with the exception that the denominator is adjusted to account for the amount of risk (European Banking Authority, 2019).

◇ **Source:**

EBA Transparency Exercise-Database, author's calculations

Group: oth (Profitability)

Sheet: P&L/Capital

Item Number from EBA 2020:

2020335 [Profit or (-) Loss for the year]

2020133 [TIER 1 CAPITAL (net of deductions and after transitional adjustments)]

†

◇ **Calculation:**

$$\frac{2020335 \text{ [Profit or (-) Loss for the year]}}{2020133 \text{ [TIER 1 CAPITAL (net of deductions and after transitional adjustments)]}}$$

* **Cross-Border Diversification Index [CBDI]**

◇ **Definition:**

The *CBDI* index is used to measure the geographic diversification of bank exposures. A Cross-Border Diversification Index (*CBDI*) close to its maximum means that the bank original exposures are very diversified across countries, whereas an index close to zero means that the bank original exposures are very concentrated on certain countries (Morgan and Samolyk, 2016).

◇ **Item Name:** Original Exposure - by exposure class (SA and IRB)

*Proxy for total bank assets

†Tier 1 capital is the most adequate form of a bank's capital

◇ **Source:**

EBA Transparency Exercise-Database, author's calculations

Group: cre (Credit Risk)

Sheet: Credit Risk

Item Number from EBA 2020:

2020502 [Original Exposure - by exposure class (SA_and_IRB)]

◇ **Dimensions:**

- Country_rank:

Taking values from 1 (1st-largest country exposures) up to 10 (10th-largest country exposure)

- Country/ISO_code:

Country identification number and ISO country codes to designate every country as a two-letter combination.

- ◇ **Calculation:** The Cross-Border Diversification Index (*CBDI*) has been calculated as the inverse of the Hirschmann-Herfindahl Index (HHI) for each bank *i* at time *t* as described in the section Data (Morgan and Samolyk, 2016).

Macroeconomic indicators

* **Inflation, consumer prices (annual %) [infl_perc]**

◇ **Definition:**

Inflation as calculated by the consumer price index represents the percentage change on an annual basis of the cost for the average consumer of acquiring a basket of goods and services. The Laspeyres formula is generally used for its calculation.

◇ **Source:**

World Bank Data

◇ **Indicator Code:**

FP.CPI.TOTL.ZG

◇ **Organization:**

International Financial Statistics and the International Monetary Fund (IMF).

* **Strength of legal rights index (0=weak to 12=strong) [Stgth_legal_index]**

◇ **Definition:**

Strength of legal rights index is a measure of the strength of collateral and bankruptcy laws with respect to legal aspects influencing the extension of new credit. The index shows values between 0 and 12, with higher scores indicating that these laws are better designed to expand access to credit (ex-ante).

◇ **Source:**

World Bank Data

◇ **Indicator Code:**

IC.LGL.CRED.XQ

◇ **Organization:**

World Bank, Doing Business project (<http://www.doingbusiness.org/>).

* **GDP growth (annual %) [GDP_growth_perc]**

◇ **Definition:**

Annual percentage growth rate of GDP at market prices on constant local currency. GDP is calculated as the sum of gross value added by all producers that resides in the economy minus subsidies plus taxes.

◇ **Source:**

World Bank Data

◇ **Indicator Code:**

NY.GDP.MKTP.KD.ZG

◇ **Organization:**

World Bank national accounts data, and OECD National Accounts data files.

* **Recovery rate (cents on the dollar) [Reco_rate_dol]**

◇ **Definition:**

The recovery rate indicates amount recovered by creditors as cents on the dollar through debt restructuring, debt execution (foreclosure) or bankruptcy proceedings.

◇ **Source:**

World Bank Data

◇ **Indicator Code:**

RESLV.ISV.RCOV.RT

◇ **Organization:**

World Bank, Doing Business project (<http://www.doingbusiness.org/>).

* **Unemployment rate [unempl_rate_imf]**

◇ **Definition:**

The number of unemployed persons as a percentage of the total labor force.

◇ **Source:**

International Monetary Fund Data (IMF)

◇ **Organization:**

World Economic Outlook Database

* **Current account balance, percent of GDP [curr_acc_bal]**

◇ **Definition:**

The current account includes all transactions in the balance of payments that cover exports and imports of goods and services, payments of income, and current transfers between residents and nonresidents of a country.

◇ **Source:**

International Monetary Fund Data (IMF)

◇ **Organization:**

World Economic Outlook Database

* **General government gross debt, percent of GDP [gross_public_debt]**

◇ **Definition:**

Gross debt consists of all external financial obligations that governments accumulate to finance public expenditures above the available public revenues. This includes debt liabilities such as Special Drawing Rights (SDRs), debt securities, loans, insurances, pensions, currency, deposits and other payables.

◇ **Source:**

International Monetary Fund Data (IMF)

◇ **Organization:**

World Economic Outlook Database

A.3 Original Exposures: Countries of Destination

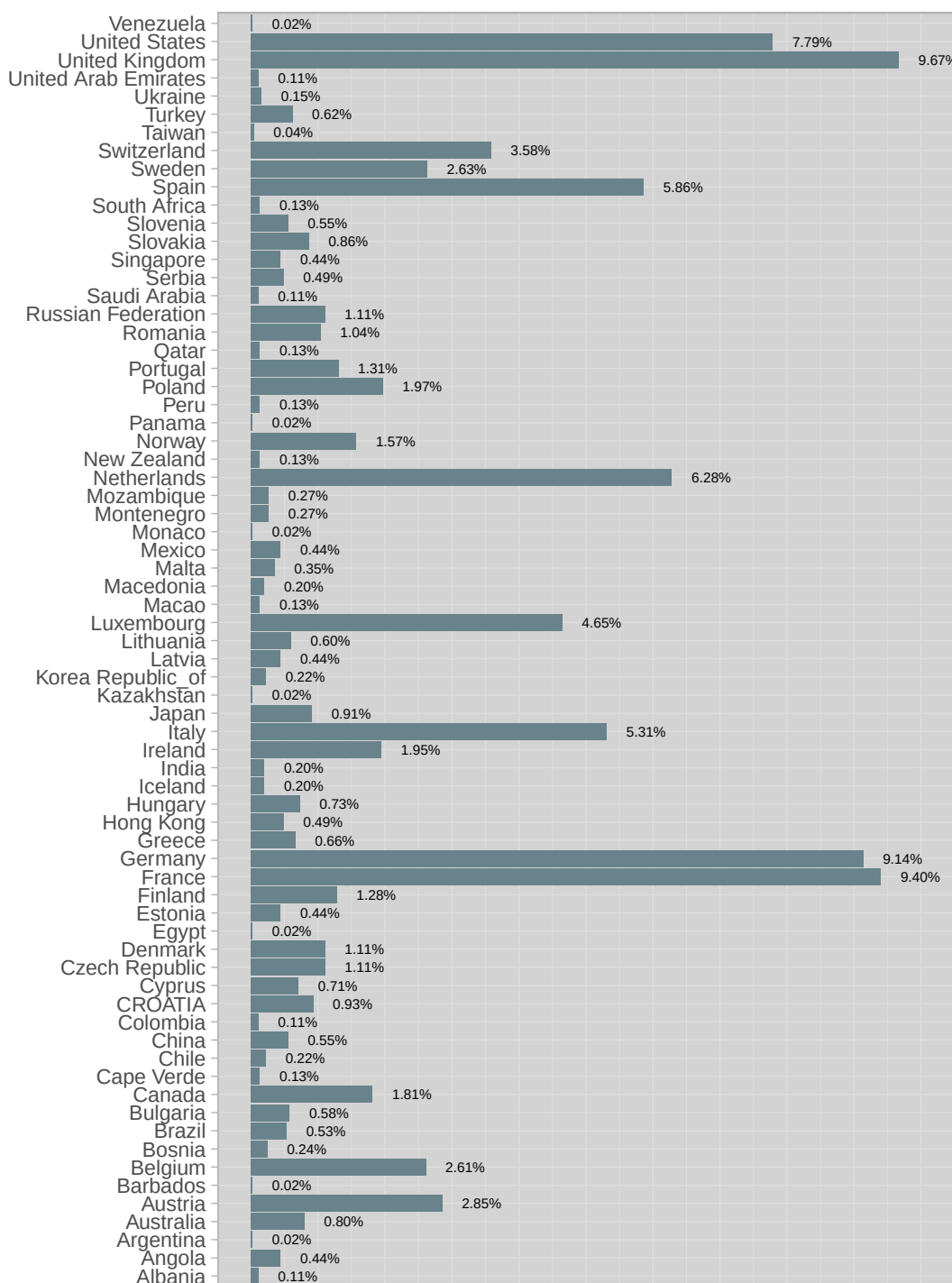


Figure 13: Original Exposures Distribution: Top 10 largest Country of Destination per Bank (2014-2020)

A.4 Distributions and Scatterplots- Explanatory Variables (All Sample)

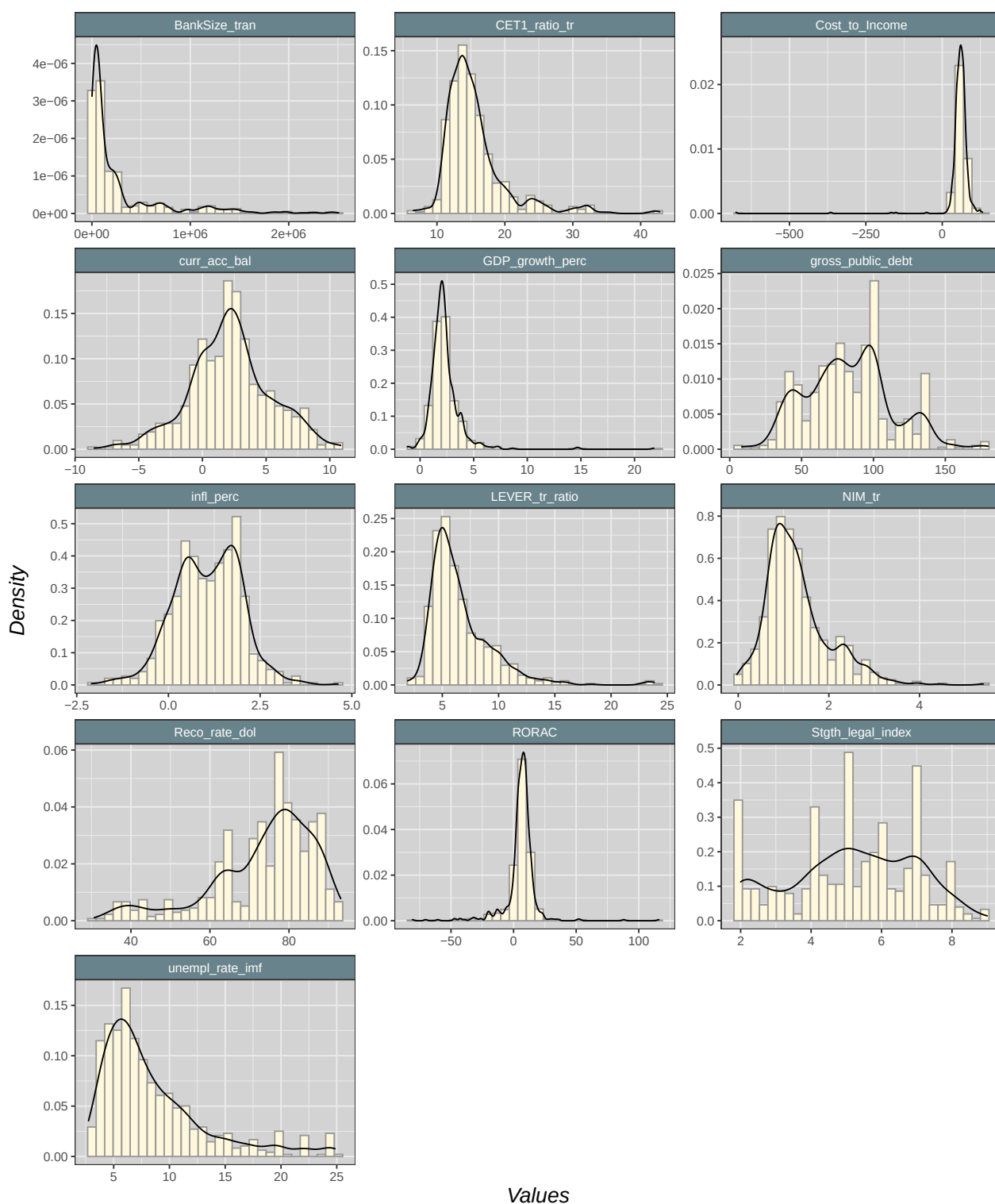


Figure 14: Distributions - Explanatory Variables (All Sample)

Source: EBA Transparency Exercises, Worldbank and IMF data, authors' calculations

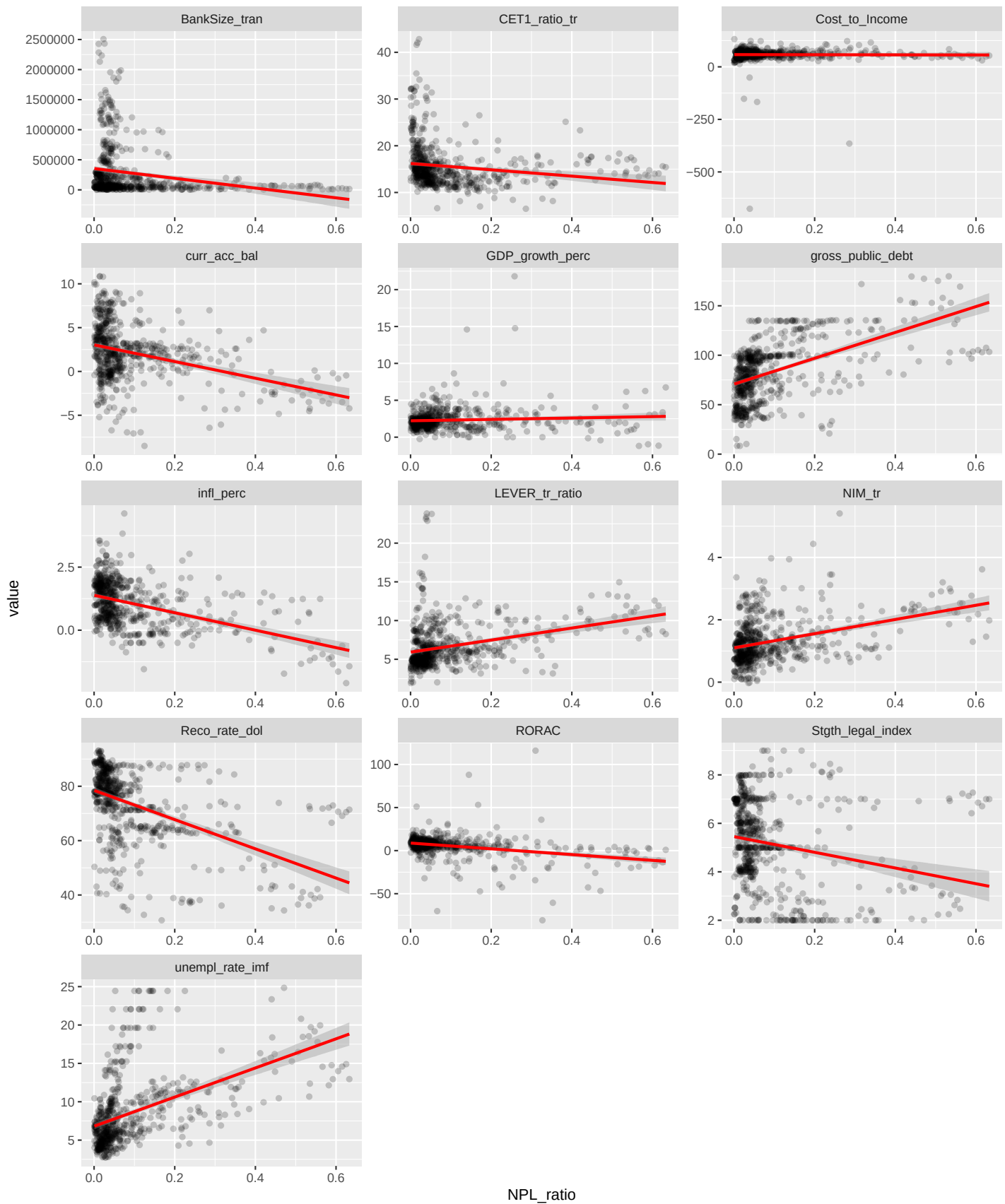


Figure 15: Scatterplots on NPL ratios - Explanatory Variables (All Sample)

Source: EBA Transparency Exercises, Worldbank and IMF data, authors' calculations

A.5 Descriptive Statistics- All Variables

(a) Households

Statistic	Mean	St. Dev.	Min	Pctl(25)	Median	Pctl(75)	Max
NPL_ratio	6.932	10.617	0.089	1.575	3.502	7.119	62.002
CET1_ratio_tr	15.507	4.649	6.516	12.743	14.481	16.475	42.822
LEVER_tr_ratio	6.609	2.797	2.020	4.791	5.878	7.541	23.832
RORAC	5.704	12.502	-80.867	3.265	6.988	10.209	116.080
BankSize_tran	282,745.200	441,229.600	892	39,280.0	80,555.0	271,369.0	2,506,727
Cost_to_Income	57.960	39.749	-675.099	50.045	59.946	69.929	123.119
NIM_tr	1.321	0.707	-0.030	0.845	1.174	1.602	5.410
CBDI_index	34.487	29.447	0.000	0.500	33.066	60.408	87.651
Stgth_legal_index	5.156	1.796	2.000	4.000	5.026	6.804	9.000
Reco_rate_dol	73.526	13.244	30.700	65.276	77.200	82.691	93.100
infl_perc	1.065	0.915	-2.097	0.431	1.131	1.752	4.625
curr_acc_bal	2.158	3.192	-8.500	0.132	2.026	3.905	10.859
GDP_growth_perc	2.315	1.581	-1.187	1.533	2.078	2.697	21.806
unempl_rate_imf	8.575	4.779	2.742	5.265	6.977	10.444	24.845
gross_public_debt	82.754	30.323	8.296	61.910	81.557	99.303	179.814

(b) Non-Financial Corporations

Statistic	Mean	St. Dev.	Min	Pctl(25)	Median	Pctl(75)	Max
NPL_ratio	12.306	14.575	0.020	3.177	5.807	16.603	70.654
CET1_ratio_tr	15.589	4.691	6.516	12.782	14.526	16.625	42.822
LEVER_tr_ratio	6.635	2.889	1.922	4.781	5.884	7.676	23.832
RORAC	5.721	12.427	-80.867	3.273	7.001	10.217	116.080
BankSize_tran	279,818.300	438,725.900	892	39,485.2	79,252.6	270,372.7	2,506,727
Cost_to_Income	57.944	39.625	-675.099	49.813	59.872	69.914	131.823
NIM_tr	1.308	0.711	-0.030	0.831	1.164	1.600	5.410
CBDI_index	34.728	29.549	0.000	0.547	33.146	60.675	87.651
Stgth_legal_index	5.155	1.786	2.000	4.000	5.027	6.790	9.000
Reco_rate_dol	73.551	13.199	30.700	65.356	77.213	82.622	93.100
infl_perc	1.066	0.911	-2.097	0.442	1.130	1.752	4.625
curr_acc_bal	2.159	3.184	-8.500	0.121	2.026	3.905	10.859
GDP_growth_perc	2.307	1.571	-1.187	1.536	2.071	2.678	21.806
unempl_rate_imf	8.553	4.753	2.742	5.274	6.974	10.389	24.845
gross_public_debt	82.776	30.130	8.296	62.706	81.530	99.303	179.814

(c) Financial Corporations

Statistic	Mean	St. Dev.	Min	Pctl(25)	Median	Pctl(75)	Max
NPL_ratio	8.339	17.432	0.001	0.398	1.532	5.659	99.595
CET1_ratio_tr	15.333	5.174	6.516	12.653	14.236	16.352	65.302
LEVER_tr_ratio	6.509	2.374	2.020	4.853	5.905	7.450	16.180
RORAC	5.342	12.753	-80.867	2.975	6.766	10.075	116.080
BankSize_tran	294,187.800	452,817.100	1,652	38,961.3	81,029.5	273,533.5	2,506,727
Cost_to_Income	59.187	40.910	-675.099	51.333	61.076	70.792	123.119
NIM_tr	1.307	0.707	-0.030	0.840	1.171	1.601	5.410
CBDI_index	36.162	29.245	0.000	2.982	35.509	62.667	87.651
Stgth_legal_index	5.183	1.787	2.000	4.044	5.104	6.742	9.000
Reco_rate_dol	73.218	13.139	30.700	65.210	76.956	82.268	93.100
infl_perc	1.033	0.900	-2.097	0.422	1.080	1.721	4.625
curr_acc_bal	2.109	3.082	-8.500	0.149	2.026	3.752	9.934
GDP_growth_perc	2.330	1.606	-1.187	1.571	2.098	2.737	21.806
unempl_rate_imf	8.668	4.794	2.742	5.311	7.018	10.625	24.845
gross_public_debt	84.244	30.031	8.296	63.783	82.759	100.304	179.814

Table 4: Descriptive Statistics (2014-2019)

A.6 Correlation Diagrams

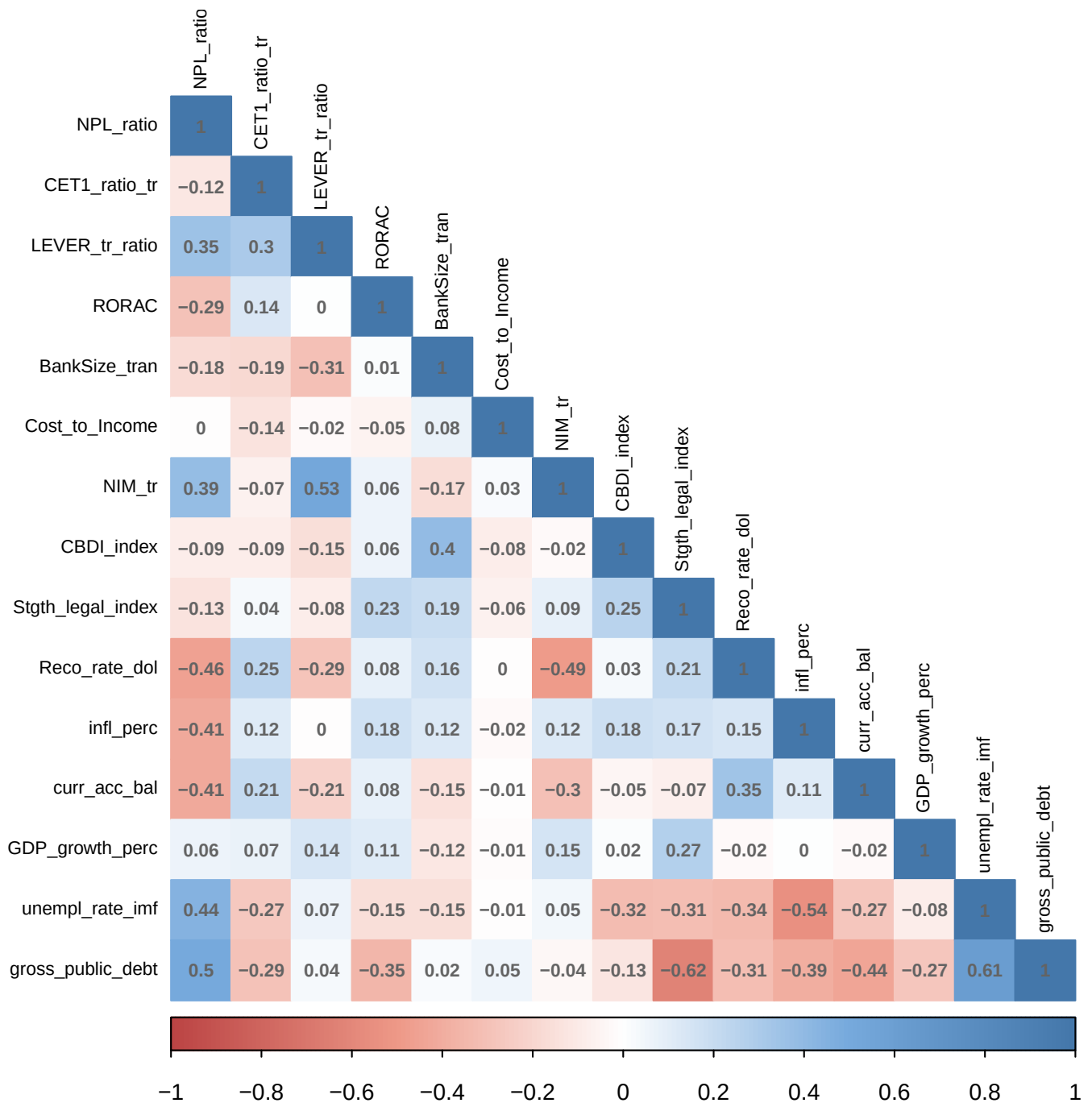


Figure 16: Correlation Matrix: Households

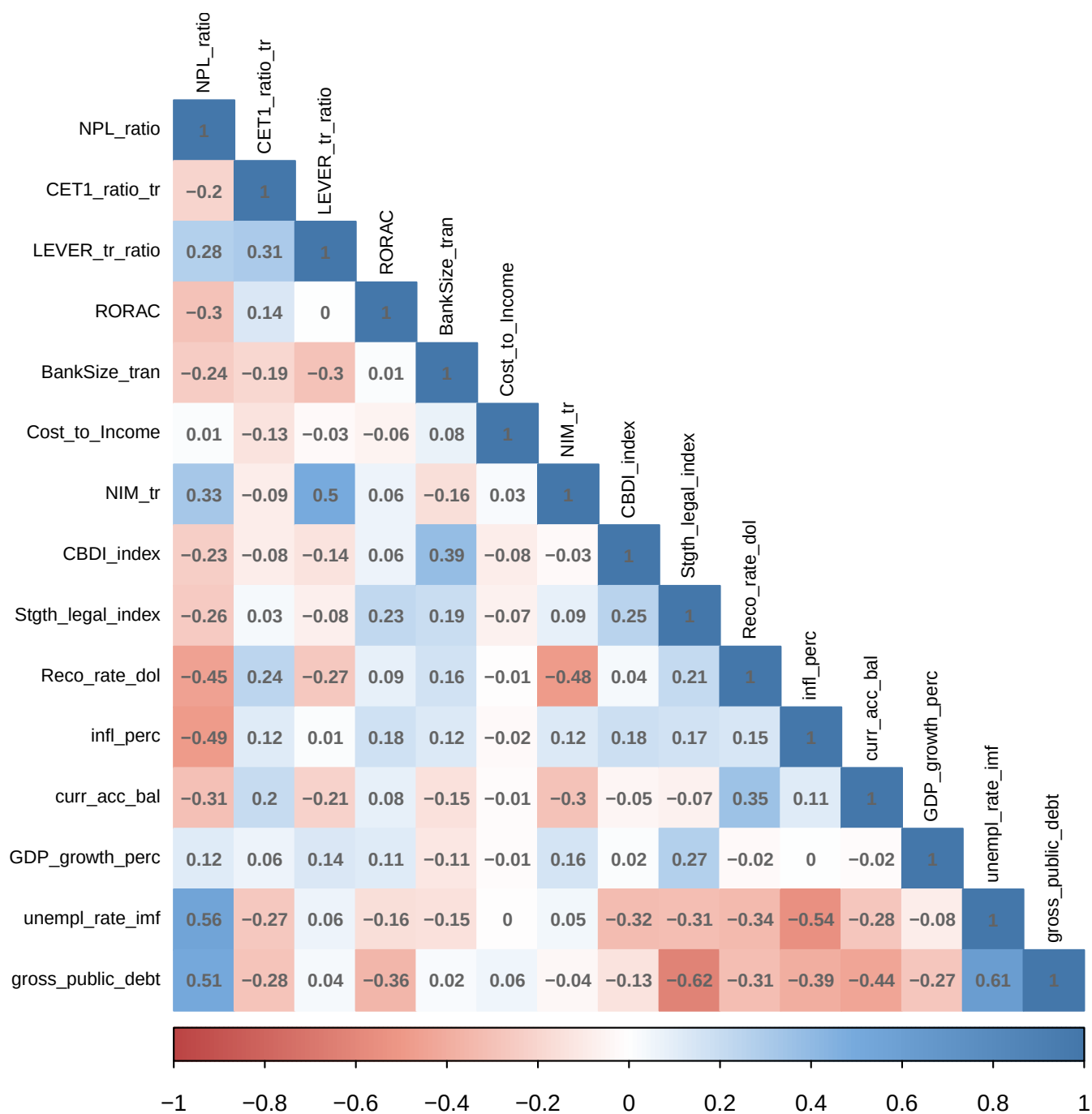


Figure 17: Correlation Matrix: Non-Financial Corporations

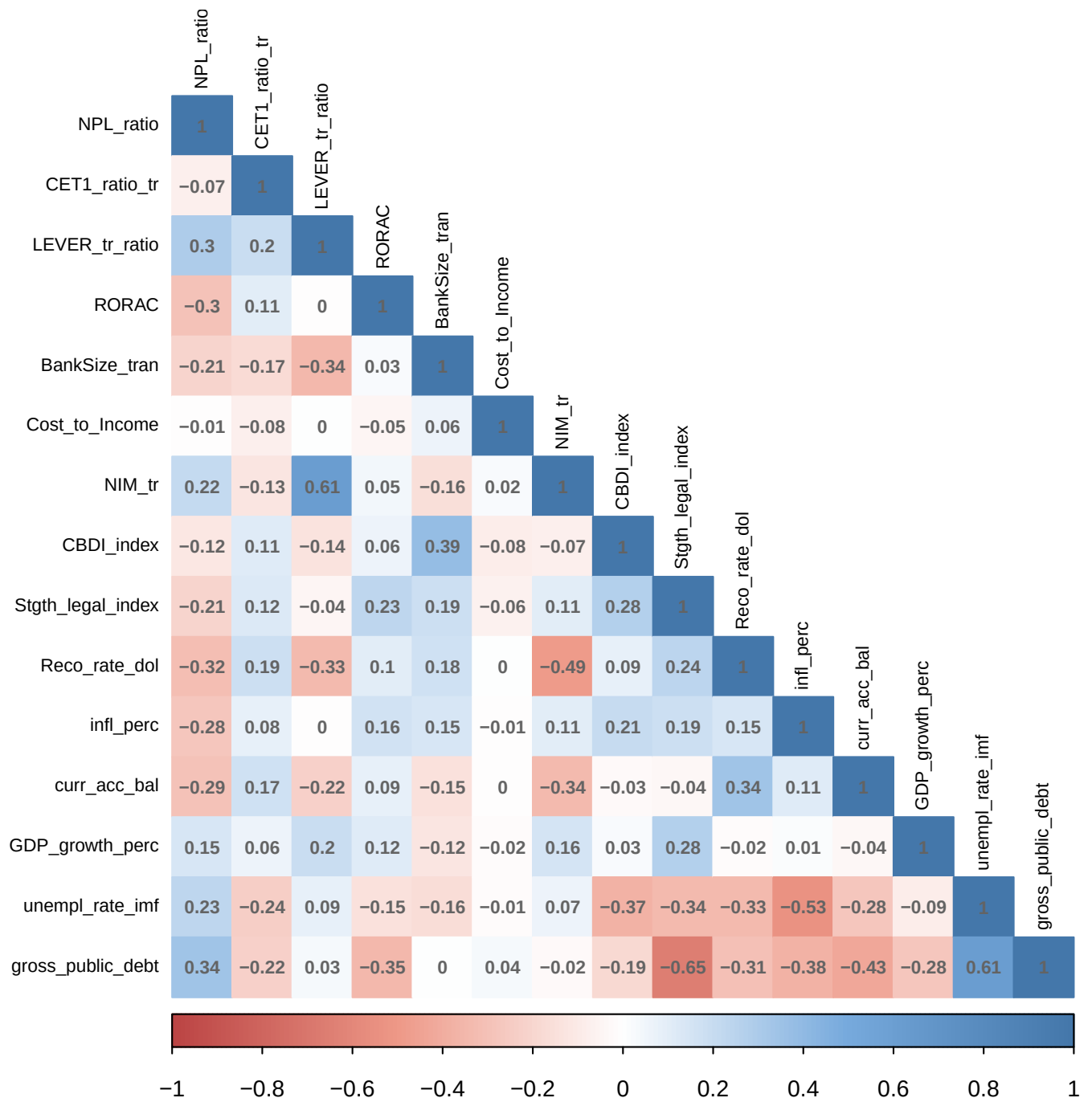


Figure 18: Correlation Matrix: Financial Corporations

A.7 Variance Inflation Factor (VIF)

	VIF	1/VIF		VIF	1/VIF
CET1_ratio_tr	1.517	0.659	CET1_ratio_tr	1.512	0.661
LEVER_tr_ratio	1.968	0.508	LEVER_tr_ratio	1.873	0.534
RORAC	1.183	0.846	RORAC	1.182	0.846
log(BankSize_tran)	1.845	0.542	log(BankSize_tran)	1.817	0.550
Cost_to_Income	1.053	0.950	Cost_to_Income	1.049	0.953
NIM_tr	2.044	0.489	NIM_tr	1.966	0.509
Stgth_legal_index	2.878	0.348	Stgth_legal_index	2.862	0.349
Reco_rate_dol	1.803	0.555	Reco_rate_dol	1.781	0.561
infl_perc	1.545	0.647	infl_perc	1.540	0.649
curr_acc_bal	2.011	0.497	curr_acc_bal	2.000	0.500
GDP_growth_perc	1.202	0.832	GDP_growth_perc	1.201	0.832
unempl_rate_imf	2.392	0.418	unempl_rate_imf	2.389	0.419
gross_public_debt	4.972	0.201	gross_public_debt	4.952	0.202
SSM	2.016	0.496	SSM	2.015	0.496
CBDI_index	1.646	0.608	CBDI_index	1.641	0.609

(a) Households

(b) Non-Financial Corporations

	VIF	1/VIF
CET1_ratio_tr	1.319	0.758
LEVER_tr_ratio	2.289	0.437
RORAC	1.171	0.854
log(BankSize_tran)	1.917	0.522
Cost_to_Income	1.027	0.973
NIM_tr	2.398	0.417
Stgth_legal_index	2.890	0.346
Reco_rate_dol	1.798	0.556
infl_perc	1.508	0.663
curr_acc_bal	1.924	0.520
GDP_growth_perc	1.217	0.822
unempl_rate_imf	2.385	0.419
gross_public_debt	4.757	0.210
SSM	1.987	0.503
CBDI_index	1.663	0.601

(c) Financial Corporations

Table 5: Variance Inflation Vector

A.8 Panel Regression Tests

(a) Households

Method	Statistic	P-Value	Alternative
Lagrange Multiplier Test - (Honda) for unbalanced panels	25.925	1.75479547826439e-148	significant effects
F test for individual effects	47.216	4.37033369289822e-206	significant effects
Hausman Test	17.515	0.229783810376914	one model is inconsistent
Wooldridge's test for unobserved individual effects	4.126	3.69088944446373e-05	unobserved effect
Wooldridge's test for serial correlation in FE panels	141.061	8.33554713205956e-29	serial correlation

(b) Non-Financial Corporations

Method	Statistic	P-Value	Alternative
Lagrange Multiplier Test - (Honda) for unbalanced panels	21.367	1.35551052079772e-101	significant effects
F test for individual effects	20.832	9.00107040667109e-135	significant effects
Hausman Test	1202.573	4.85361065236067e-248	one model is inconsistent
Wooldridge's test for unobserved individual effects	3.97	7.1918537599468e-05	unobserved effect
Wooldridge's test for serial correlation in FE panels	60.579	4.06446580048778e-14	serial correlation

(c) Financial Corporations

Method	Statistic	P-Value	Alternative
Lagrange Multiplier Test - (Honda) for unbalanced panels	16.268	8.33620606248368e-60	significant effects
F test for individual effects	9.493	1.09087787385205e-69	significant effects
Hausman Test	4.693	0.989642232888385	one model is inconsistent
Wooldridge's test for unobserved individual effects	3.776	0.000159090846543112	unobserved effect
Wooldridge's test for serial correlation in FE panels	4.003	0.0460158082569163	serial correlation

Table 6: Breusch-Pagan Lagrange Multiplier, F-Test, Hausman & Wooldridge's Test

A.9 Tables- Quantile Regression (QR) and Slope Tests

	Dependent variable:								
	10%	20%	30%	40%	NPL_ratio 50%	60%	70%	80%	90%
CET1_ratio_tr	−0.060* (0.033)	−0.049* (0.029)	−0.046* (0.024)	−0.065*** (0.022)	−0.061* (0.036)	−0.048 (0.046)	−0.025 (0.093)	0.276* (0.150)	0.455*** (0.156)
LEVER_tr_ratio	0.129* (0.066)	0.146** (0.060)	0.123** (0.062)	0.153** (0.073)	0.251*** (0.078)	0.266** (0.108)	0.325 (0.239)	0.640 (0.395)	1.000** (0.508)
RORAC	−0.018 (0.020)	−0.030 (0.031)	−0.061** (0.029)	−0.059* (0.031)	−0.076* (0.039)	−0.069 (0.044)	−0.085 (0.052)	−0.099* (0.058)	−0.090 (0.063)
log(BankSize_tran)	0.211*** (0.076)	0.044 (0.081)	−0.129 (0.093)	−0.262** (0.120)	−0.410** (0.173)	−0.632*** (0.191)	−0.986*** (0.269)	−1.629*** (0.385)	−2.279*** (0.554)
Cost_to_Income	0.001 (0.007)	0.002 (0.009)	0.004 (0.010)	0.0005 (0.011)	−0.007 (0.011)	−0.012 (0.011)	−0.019 (0.014)	−0.022 (0.025)	−0.040 (0.034)
NIM_tr	0.683** (0.295)	0.907*** (0.218)	1.003*** (0.290)	1.016*** (0.314)	1.479*** (0.328)	1.870*** (0.422)	1.890** (0.756)	3.866*** (1.440)	5.839*** (1.723)
Stgth_legal_index	0.017 (0.081)	0.028 (0.104)	0.059 (0.109)	0.187 (0.116)	0.316** (0.138)	0.478*** (0.171)	0.679*** (0.236)	1.154*** (0.368)	2.153*** (0.502)
Reco_rate_dol	−0.066*** (0.013)	−0.061*** (0.014)	−0.063*** (0.012)	−0.069*** (0.017)	−0.078*** (0.019)	−0.087*** (0.020)	−0.083*** (0.030)	−0.072 (0.060)	−0.017 (0.067)
infl_perc	−0.292* (0.165)	−0.314*** (0.120)	−0.433*** (0.154)	−0.591*** (0.189)	−0.911*** (0.242)	−1.268*** (0.257)	−1.375*** (0.469)	−2.504*** (0.949)	−2.864*** (1.082)
curr_acc_bal	0.133*** (0.051)	0.129** (0.060)	0.146** (0.062)	0.170** (0.072)	0.174*** (0.066)	0.248*** (0.077)	0.291*** (0.103)	0.330 (0.215)	0.191 (0.276)
GDP_growth_perc	0.209* (0.111)	0.329*** (0.114)	0.314** (0.133)	0.284* (0.159)	0.208 (0.220)	0.111 (0.265)	0.279 (0.346)	0.420 (0.382)	0.104 (0.421)
unempl_rate_imf	0.051 (0.036)	0.070** (0.032)	0.085** (0.038)	0.091** (0.040)	0.053 (0.050)	0.021 (0.066)	0.061 (0.145)	0.378 (0.297)	0.194 (0.455)
gross_public_debt	0.050*** (0.010)	0.056*** (0.010)	0.065*** (0.011)	0.077*** (0.013)	0.092*** (0.016)	0.117*** (0.015)	0.139*** (0.023)	0.188*** (0.034)	0.266*** (0.048)
SSMY	−1.125*** (0.378)	−1.130*** (0.345)	−1.397*** (0.422)	−1.593*** (0.520)	−1.673*** (0.474)	−2.003*** (0.466)	−1.788*** (0.636)	−1.617 (1.272)	0.047 (1.670)
CBDI_index	−0.008 (0.005)	−0.005 (0.005)	−0.001 (0.005)	0.007 (0.006)	0.009 (0.008)	0.017** (0.008)	0.026** (0.012)	0.049*** (0.016)	0.057** (0.027)
Constant	−0.073 (1.646)	0.494 (1.991)	2.485 (1.737)	3.669* (2.088)	5.065** (2.398)	6.340** (2.829)	7.162* (4.286)	−0.634 (6.827)	−9.580 (7.974)
Pseudo R ²	0.202	0.232	0.257	0.272	0.286	0.306	0.328	0.38	0.542
Slope-equal test	12.13								
Prob (Slope-equal)	0								
Symmetry (.25-.75) test	4.67								
Prob (Symmetry(.25-.75))	0								
Observations	619	619	619	619	619	619	619	619	619

Note:

*p<0.1; **p<0.05; ***p<0.01

Table 7: Quantile Regression (2014-2019): Households

	Dependent variable:								
	10%	20%	30%	40%	NPL_ratio 50%	60%	70%	80%	90%
CET1_ratio_tr	-0.080 (0.071)	-0.112* (0.065)	-0.158** (0.063)	-0.152** (0.066)	-0.197*** (0.069)	-0.175** (0.077)	-0.127* (0.077)	-0.126 (0.092)	-0.138 (0.173)
LEVER_tr_ratio	0.162** (0.074)	0.064 (0.070)	0.037 (0.085)	0.054 (0.161)	0.145 (0.289)	0.551* (0.328)	0.559 (0.343)	0.374 (0.473)	0.706 (0.662)
RORAC	-0.142** (0.058)	-0.178** (0.069)	-0.160*** (0.061)	-0.197*** (0.059)	-0.191*** (0.064)	-0.210*** (0.080)	-0.242*** (0.087)	-0.265*** (0.087)	-0.222* (0.117)
log(BankSize_tran)	0.145 (0.197)	-0.223 (0.175)	-0.444** (0.222)	-0.693*** (0.241)	-1.023*** (0.287)	-0.981** (0.384)	-1.604*** (0.525)	-2.211*** (0.534)	-3.698*** (1.006)
Cost_to_Income	-0.0001 (0.014)	0.001 (0.016)	0.001 (0.017)	0.001 (0.020)	-0.009 (0.023)	-0.007 (0.028)	-0.039 (0.027)	-0.048 (0.029)	-0.033 (0.027)
NIM_tr	1.030*** (0.397)	1.637*** (0.456)	2.242*** (0.486)	2.378*** (0.619)	3.113*** (0.745)	4.317*** (1.000)	5.588*** (1.322)	5.866*** (1.485)	6.689*** (2.090)
Stgth_legal_index	-0.003 (0.274)	-0.013 (0.168)	0.025 (0.204)	0.073 (0.236)	-0.198 (0.253)	-0.121 (0.293)	0.414 (0.337)	0.271 (0.398)	1.283** (0.628)
Reco_rate_dol	-0.069** (0.027)	-0.105*** (0.033)	-0.144*** (0.032)	-0.155*** (0.033)	-0.123*** (0.041)	-0.108** (0.044)	-0.111** (0.049)	-0.125** (0.063)	0.008 (0.101)
infl_perc	-0.540* (0.323)	-0.685* (0.381)	-1.269*** (0.379)	-1.780*** (0.461)	-1.863*** (0.445)	-2.232*** (0.572)	-2.953*** (0.757)	-4.004*** (0.999)	-4.309*** (1.310)
curr_acc_bal	0.271** (0.118)	0.328*** (0.123)	0.525*** (0.130)	0.524*** (0.119)	0.602*** (0.127)	0.663*** (0.119)	0.707*** (0.145)	0.510*** (0.169)	0.871*** (0.287)
GDP_growth_perc	1.357*** (0.222)	1.261*** (0.221)	1.198*** (0.256)	1.077*** (0.280)	1.275*** (0.297)	1.360*** (0.315)	1.026*** (0.340)	1.067** (0.498)	1.052 (0.996)
unempl_rate_imf	0.236* (0.129)	0.243** (0.114)	0.309** (0.136)	0.529*** (0.116)	0.568*** (0.114)	0.693*** (0.169)	0.678*** (0.147)	0.638** (0.255)	1.113** (0.432)
gross_public_debt	0.063*** (0.024)	0.104*** (0.028)	0.145*** (0.029)	0.141*** (0.024)	0.172*** (0.025)	0.172*** (0.026)	0.213*** (0.029)	0.205*** (0.034)	0.254*** (0.060)
SSMY	-0.848 (0.768)	-2.617*** (0.874)	-3.831*** (1.050)	-4.108*** (0.884)	-4.950*** (0.847)	-4.634*** (0.880)	-4.208*** (0.995)	-3.861*** (1.183)	-2.367 (1.753)
CBDL_index	0.004 (0.010)	-0.008 (0.008)	-0.010 (0.009)	0.001 (0.011)	0.006 (0.013)	0.004 (0.017)	0.010 (0.019)	0.022 (0.021)	0.019 (0.033)
Constant	-2.788 (3.327)	4.894 (3.749)	9.235** (4.403)	13.306*** (4.106)	14.636*** (5.111)	8.821 (5.885)	13.131* (7.530)	27.807*** (8.074)	21.321 (13.292)
Pseudo R ²	0.17	0.216	0.262	0.313	0.365	0.423	0.485	0.528	0.551
Slope-equal test	8.27								
Prob (Slope-equal)	0								
Symmetry (.25-.75) test	65.5								
Prob (Symmetry(.25-.75))	0								
Observations	628	628	628	628	628	628	628	628	628

Note:

*p<0.1; **p<0.05; ***p<0.01

Table 8: Quantile Regression (2014-2019): Non-Financial Corporations

	Dependent variable:								
	10%	20%	30%	40%	NPL_ratio 50%	60%	70%	80%	90%
CET1_ratio_tr	−0.019** (0.009)	−0.031** (0.015)	−0.044** (0.018)	−0.035* (0.020)	−0.044 (0.029)	−0.081 (0.058)	−0.108 (0.122)	−0.186 (0.131)	−0.344 (0.224)
LEVER_tr_ratio	0.033* (0.019)	0.012 (0.049)	0.014 (0.066)	−0.010 (0.090)	0.106 (0.134)	0.153 (0.274)	0.335 (0.482)	0.581 (0.718)	2.211 (1.354)
RORAC	−0.007 (0.014)	−0.037** (0.017)	−0.049** (0.022)	−0.057** (0.027)	−0.059 (0.056)	−0.101 (0.123)	−0.321* (0.173)	−0.277 (0.212)	−0.293 (0.256)
log(BankSize_tran)	0.016 (0.031)	−0.009 (0.062)	−0.072 (0.087)	−0.175 (0.113)	−0.329 (0.214)	−1.034*** (0.366)	−2.056*** (0.577)	−2.973*** (0.563)	−4.099*** (1.180)
Cost_to_Income	−0.001 (0.004)	−0.006 (0.006)	−0.013** (0.006)	−0.013*** (0.005)	−0.018* (0.009)	−0.016 (0.016)	−0.017 (0.036)	−0.021 (0.059)	−0.014 (0.085)
NIM_tr	0.120 (0.098)	0.304 (0.188)	0.580** (0.233)	0.882*** (0.259)	1.226*** (0.442)	1.253 (0.869)	2.366 (1.569)	3.557* (1.890)	2.318 (3.599)
Stgth_legal_index	−0.035 (0.042)	−0.127 (0.078)	−0.115 (0.082)	−0.194* (0.101)	−0.183 (0.198)	−0.633** (0.319)	−1.271** (0.494)	−1.580*** (0.572)	−2.317*** (0.888)
Reco_rate_dol	−0.003 (0.005)	−0.011 (0.010)	−0.022 (0.014)	−0.022 (0.015)	−0.025 (0.028)	−0.058 (0.053)	−0.086 (0.084)	−0.185* (0.109)	−0.265 (0.221)
infl_perc	−0.106 (0.064)	−0.295*** (0.103)	−0.479*** (0.167)	−0.879*** (0.214)	−1.363*** (0.313)	−2.086*** (0.608)	−3.350*** (0.905)	−4.495*** (1.052)	−5.589*** (1.844)
curr_acc_bal	0.042** (0.017)	0.034 (0.025)	0.042 (0.033)	0.032 (0.049)	0.064 (0.102)	−0.072 (0.201)	−0.411 (0.292)	−0.165 (0.347)	−0.483 (0.489)
GDP_growth_perc	0.145* (0.074)	0.181 (0.115)	0.165 (0.143)	0.361** (0.182)	0.485 (0.301)	0.468 (0.545)	0.852 (1.014)	1.835 (1.163)	3.680** (1.471)
unempl_rate_imf	−0.056*** (0.019)	−0.091*** (0.027)	−0.118*** (0.027)	−0.136*** (0.035)	−0.179*** (0.059)	−0.244** (0.112)	−0.475** (0.208)	−0.864*** (0.333)	−1.469*** (0.495)
gross_public_debt	0.012*** (0.004)	0.015** (0.006)	0.020** (0.008)	0.023** (0.011)	0.042** (0.020)	0.054* (0.031)	0.086* (0.052)	0.187** (0.075)	0.237** (0.119)
SSMY	−0.117 (0.121)	−0.066 (0.164)	−0.040 (0.221)	−0.059 (0.322)	−0.231 (0.575)	−1.952 (1.216)	−5.152*** (1.976)	−7.541*** (2.559)	−8.148** (3.497)
CBDL_index	−0.001 (0.002)	−0.003 (0.003)	−0.003 (0.005)	0.003 (0.005)	0.009 (0.009)	0.020 (0.014)	0.036* (0.022)	0.035 (0.035)	0.007 (0.047)
Constant	−0.216 (0.744)	2.433* (1.441)	4.639** (2.067)	6.187** (2.765)	7.266 (5.501)	24.195*** (8.871)	47.036*** (13.834)	64.508*** (14.573)	89.666*** (30.469)
Pseudo R ²	0.016	0.027	0.041	0.06	0.077	0.108	0.168	0.278	0.409
Slope-equal test	15.44								
Prob (Slope-equal)	0								
Symmetry (.25-.75) test	8.03								
Prob (Symmetry(.25-.75))	0								
Observations	580	580	580	580	580	580	580	580	580

Note:

*p<0.1; **p<0.05; ***p<0.01

Table 9: Quantile Regression (2014-2019): Financial Corporations

(a) Households

	Df	Resi.Df	F.stat	P.val
CET1_ratio_tr	8	5563	2.3287	0.0170996**
LEVER_tr_ratio	8	5563	1.5864	0.1232168
RORAC	8	5563	2.2012	0.0245085**
log(BankSize_tran)	8	5563	8.4163	0***
Cost_to_Income	8	5563	0.5867	0.7897477
NIM_tr	8	5563	5.5454	0.0000005***
Stgth_legal_index	8	5563	3.5046	0.0004778***
Reco_rate_dol	8	5563	0.8868	0.5265858
infl_perc	8	5563	4.4616	0.0000208***
curr_acc_bal	8	5563	1.2785	0.2496346
GDP_growth_perc	8	5563	1.1391	0.3330907
unempl_rate_imf	8	5563	1.1153	0.3490547
gross_public_debt	8	5563	7.4650	0***
SSMY	8	5563	1.5215	0.1439789
CBDL_index	8	5563	3.6250	0.0003248***

(b) Non-Financial Corporations

	Df	Resi.Df	F.stat	P.val
CET1_ratio_tr	8	5644	0.9760	0.4525963
LEVER_tr_ratio	8	5644	2.7433	0.0050748***
RORAC	8	5644	1.4448	0.1722868
log(BankSize_tran)	8	5644	8.0284	0***
Cost_to_Income	8	5644	1.1111	0.3519587
NIM_tr	8	5644	5.0522	0.0000028***
Stgth_legal_index	8	5644	2.1936	0.0250352**
Reco_rate_dol	8	5644	3.7041	0.0002517***
infl_perc	8	5644	4.9706	0.0000037***
curr_acc_bal	8	5644	3.0661	0.0018969***
GDP_growth_perc	8	5644	1.0541	0.392523
unempl_rate_imf	8	5644	2.6172	0.0073916***
gross_public_debt	8	5644	7.3197	0***
SSMY	8	5644	5.9118	0.0000001***
CBDL_index	8	5644	1.4505	0.1700455

(c) Financial Corporations

	Df	Resi.Df	F.stat	P.val
CET1_ratio_tr	8	5212	0.9376	0.483783
LEVER_tr_ratio	8	5212	1.1141	0.3498962
RORAC	8	5212	22.0430	0***
log(BankSize_tran)	8	5212	4.0476	0.0000823***
Cost_to_Income	8	5212	1.0990	0.3603319
NIM_tr	8	5212	3.2084	0.0012201***
Stgth_legal_index	8	5212	2.1424	0.0288779**
Reco_rate_dol	8	5212	0.9216	0.4971206
infl_perc	8	5212	7.9523	0***
curr_acc_bal	8	5212	1.8584	0.0620357*
GDP_growth_perc	8	5212	1.8690	0.0603365*
unempl_rate_imf	8	5212	2.5780	0.0083052***
gross_public_debt	8	5212	1.4593	0.1666413
SSMY	8	5212	2.4942	0.0106151**
CBDL_index	8	5212	1.7123	0.0902495*

Table 10: Quantile Regression: Slope-Equality Test

A.10 Tables- Penalized Fixed Effects Quantile Regression (PFEQR)

(a) PFEQR with $\lambda = 1$

	10%	20%	30%	40%	50%	60%	70%	80%	90%
(Intercept)	2.2 (2.624)	3.047 (1.913)	4.126** (1.847)	5.256*** (1.761)	6.136*** (1.688)	7.051*** (1.767)	8.01*** (1.948)	9.1*** (2.34)	10.692*** (3.889)
CET1_ratio_tr	-0.097** (0.045)	-0.084*** (0.027)	-0.086*** (0.023)	-0.077*** (0.021)	-0.083*** (0.02)	-0.083*** (0.02)	-0.087*** (0.023)	-0.093*** (0.028)	-0.101** (0.044)
LEVER_tr_ratio	0.193* (0.115)	0.224*** (0.076)	0.21*** (0.071)	0.212*** (0.068)	0.221*** (0.069)	0.225*** (0.069)	0.226*** (0.069)	0.195** (0.082)	0.184* (0.102)
RORAC	-0.018 (0.021)	-0.015 (0.018)	-0.017 (0.018)	-0.027 (0.018)	-0.026 (0.02)	-0.025 (0.022)	-0.033 (0.024)	-0.032 (0.027)	-0.016 (0.032)
log(BankSize_tran)	-0.374** (0.147)	-0.429*** (0.103)	-0.445*** (0.102)	-0.49*** (0.112)	-0.528*** (0.123)	-0.595*** (0.139)	-0.692*** (0.147)	-0.849*** (0.164)	-1.164*** (0.195)
Cost_to_Income	-0.001 (0.008)	0 (0.007)	0 (0.007)	0 (0.007)	-0.002 (0.008)	-0.007 (0.008)	-0.015* (0.008)	-0.017*** (0.006)	-0.017** (0.008)
NIM_tr	0.399 (0.297)	0.302 (0.25)	0.449** (0.225)	0.447** (0.213)	0.43* (0.227)	0.411 (0.25)	0.374 (0.272)	0.506* (0.306)	0.44 (0.418)
Stgth_legal_index	0.047 (0.133)	0.103 (0.105)	0.107 (0.104)	0.108 (0.106)	0.127 (0.113)	0.156 (0.124)	0.21 (0.144)	0.259 (0.164)	0.343 (0.243)
Reco_rate_dol	-0.035** (0.015)	-0.051*** (0.012)	-0.06*** (0.012)	-0.068*** (0.012)	-0.072*** (0.012)	-0.073*** (0.012)	-0.071*** (0.013)	-0.065*** (0.015)	-0.059*** (0.022)
infl_perc	-0.082 (0.17)	-0.042 (0.124)	-0.12 (0.1)	-0.163* (0.094)	-0.198* (0.102)	-0.288** (0.117)	-0.337** (0.141)	-0.42*** (0.147)	-0.419** (0.18)
curr_acc_bal	0.109 (0.08)	0.141*** (0.045)	0.143*** (0.043)	0.147*** (0.043)	0.165*** (0.043)	0.17*** (0.044)	0.177*** (0.051)	0.178*** (0.064)	0.241*** (0.08)
GDP_growth_perc	0.231** (0.091)	0.203*** (0.072)	0.167** (0.073)	0.19*** (0.071)	0.143* (0.082)	0.143 (0.094)	0.209** (0.104)	0.173 (0.11)	0.237* (0.121)
unempl_rate_imf	0.214*** (0.051)	0.177*** (0.045)	0.166*** (0.047)	0.154*** (0.048)	0.156*** (0.049)	0.148*** (0.051)	0.16*** (0.053)	0.147** (0.061)	0.106 (0.084)
gross_public_debt	0.072*** (0.013)	0.087*** (0.01)	0.089*** (0.01)	0.094*** (0.01)	0.097*** (0.01)	0.103*** (0.011)	0.108*** (0.012)	0.118*** (0.013)	0.14*** (0.018)
SSMY	-1.55*** (0.539)	-1.56*** (0.413)	-1.627*** (0.425)	-1.787*** (0.419)	-1.859*** (0.404)	-1.88*** (0.387)	-1.843*** (0.404)	-1.817*** (0.461)	-1.787*** (0.684)
CBDI_index	0.014* (0.008)	0.013** (0.005)	0.013** (0.005)	0.012** (0.005)	0.013** (0.005)	0.014** (0.006)	0.016*** (0.006)	0.02*** (0.007)	0.034*** (0.01)

(b) PFEQR with $\lambda = 2.5$

	10%	20%	30%	40%	50%	60%	70%	80%	90%
(Intercept)	-1.517 (2.357)	2.087 (2.063)	2.447 (1.864)	3.648** (1.762)	3.568** (1.737)	6.009*** (2.002)	6.537*** (2.373)	9.386*** (3.213)	10.959** (5.153)
CET1_ratio_tr	-0.086** (0.044)	-0.082*** (0.031)	-0.072*** (0.025)	-0.07*** (0.023)	-0.073*** (0.024)	-0.076*** (0.027)	-0.07* (0.037)	-0.07 (0.046)	-0.059 (0.069)
LEVER_tr_ratio	0.138 (0.104)	0.141* (0.081)	0.119 (0.079)	0.167** (0.075)	0.235*** (0.077)	0.245*** (0.08)	0.201** (0.092)	0.183* (0.11)	0.095 (0.163)
RORAC	-0.015 (0.023)	-0.034 (0.023)	-0.047** (0.023)	-0.047** (0.024)	-0.046* (0.026)	-0.062** (0.028)	-0.049 (0.031)	-0.028 (0.033)	-0.042 (0.034)
log(BankSize_tran)	-0.039 (0.132)	-0.205* (0.12)	-0.239** (0.121)	-0.319** (0.134)	-0.398*** (0.151)	-0.581*** (0.174)	-0.823*** (0.202)	-1.222*** (0.239)	-1.748*** (0.23)
Cost_to_Income	0 (0.007)	0.001 (0.008)	0.002 (0.008)	0.002 (0.009)	-0.003 (0.009)	-0.008 (0.009)	-0.018*** (0.008)	-0.015** (0.007)	-0.011 (0.012)
NIM_tr	0.879*** (0.321)	0.776** (0.311)	0.87*** (0.29)	0.853*** (0.285)	0.939*** (0.298)	1.179*** (0.328)	1.229*** (0.369)	1.25*** (0.443)	1.782*** (0.566)
Stgth_legal_index	0.13 (0.131)	0.161 (0.117)	0.209* (0.122)	0.229* (0.133)	0.283* (0.153)	0.341** (0.172)	0.48*** (0.199)	0.616** (0.238)	0.832** (0.33)
Reco_rate_dol	-0.046*** (0.014)	-0.062*** (0.013)	-0.067*** (0.015)	-0.075*** (0.015)	-0.074*** (0.016)	-0.078*** (0.016)	-0.068*** (0.019)	-0.065*** (0.023)	-0.028 (0.032)
infl_perc	-0.03 (0.155)	-0.126 (0.132)	-0.249** (0.119)	-0.412*** (0.129)	-0.489*** (0.155)	-0.768*** (0.191)	-0.808*** (0.229)	-1.04*** (0.274)	-1.057*** (0.338)
curr_acc_bal	0.165** (0.073)	0.165*** (0.053)	0.194*** (0.052)	0.212*** (0.051)	0.225*** (0.055)	0.221*** (0.054)	0.255*** (0.063)	0.294*** (0.079)	0.307*** (0.116)
GDP_growth_perc	0.311*** (0.099)	0.196** (0.085)	0.231*** (0.084)	0.201** (0.088)	0.252** (0.105)	0.188 (0.133)	0.357** (0.165)	0.616** (0.188)	0.219 (0.222)
unempl_rate_imf	0.141*** (0.047)	0.111*** (0.039)	0.117*** (0.039)	0.127*** (0.039)	0.111*** (0.038)	0.086** (0.044)	0.05 (0.056)	0.041 (0.08)	-0.052 (0.159)
gross_public_debt	0.072*** (0.012)	0.076*** (0.012)	0.082*** (0.012)	0.089*** (0.012)	0.098*** (0.012)	0.108*** (0.012)	0.127*** (0.014)	0.146*** (0.016)	0.175*** (0.021)
SSMY	-1.208** (0.528)	-1.149** (0.49)	-1.341*** (0.474)	-1.722** (0.454)	-1.697*** (0.446)	-1.897*** (0.449)	-1.557*** (0.501)	-1.527** (0.591)	-1.238 (0.822)
CBDI_index	-0.002 (0.008)	0.002 (0.006)	0.002 (0.006)	0.006 (0.006)	0.009 (0.006)	0.016** (0.007)	0.023*** (0.008)	0.03*** (0.01)	0.034*** (0.013)

(c) PFEQR with $\lambda = 5$

	10%	20%	30%	40%	50%	60%	70%	80%	90%
(Intercept)	-1.08 (2.102)	0.58 (2.314)	2.649 (2.127)	3.669* (2.224)	5.261** (2.622)	6.34* (3.254)	7.144* (3.774)	-2.326 (4.966)	-7.174 (7.799)
CET1_ratio_tr	-0.054 (0.037)	-0.054 (0.035)	-0.045 (0.03)	-0.065* (0.035)	-0.059 (0.041)	-0.048 (0.049)	-0.023 (0.059)	0.189** (0.085)	0.353*** (0.135)
LEVER_tr_ratio	0.049 (0.087)	0.145* (0.085)	0.121 (0.095)	0.153 (0.104)	0.248** (0.116)	0.266* (0.136)	0.324* (0.171)	0.644*** (0.224)	1.285*** (0.46)
RORAC	-0.016 (0.023)	-0.032 (0.03)	-0.056* (0.031)	-0.059* (0.032)	-0.072** (0.034)	-0.069* (0.04)	-0.085* (0.044)	-0.098** (0.044)	-0.104*** (0.036)
log(BankSize_tran)	0.124 (0.121)	0.027 (0.119)	-0.148 (0.124)	-0.262* (0.153)	-0.42** (0.192)	-0.632*** (0.226)	-0.983*** (0.269)	-1.547*** (0.309)	-1.545*** (0.33)
Cost_to_Income	0.001 (0.008)	0.002 (0.008)	0.004 (0.009)	0 (0.01)	-0.007 (0.01)	-0.012 (0.01)	-0.02** (0.01)	-0.023* (0.013)	-0.024 (0.02)
NIM_tr	0.9** (0.374)	0.904** (0.361)	1.02*** (0.373)	1.016** (0.414)	1.462*** (0.425)	1.87*** (0.479)	1.891*** (0.521)	2.916*** (0.611)	2.168* (1.281)
Stgth_legal_index	0.008 (0.121)	0.026 (0.132)	0.071 (0.146)	0.187 (0.184)	0.316 (0.223)	0.478* (0.268)	0.678** (0.285)	1.085*** (0.332)	1.32*** (0.44)
Reco_rate_dol	-0.05*** (0.017)	-0.059*** (0.017)	-0.064*** (0.019)	-0.069*** (0.021)	-0.08*** (0.022)	-0.087*** (0.024)	-0.084*** (0.026)	-0.07** (0.035)	-0.126** (0.058)
infl_perc	-0.061 (0.144)	-0.278** (0.139)	-0.431*** (0.163)	-0.591*** (0.186)	-0.906*** (0.245)	-1.268*** (0.299)	-1.372*** (0.338)	-1.343*** (0.431)	-1.155 (0.784)
curr_acc_bal	0.115* (0.068)	0.126** (0.062)	0.15** (0.068)	0.17** (0.071)	0.171** (0.07)	0.248*** (0.072)	0.289*** (0.083)	0.442*** (0.111)	0.591*** (0.191)
GDP_growth_perc	0.312*** (0.112)	0.303*** (0.102)	0.278** (0.114)	0.284** (0.135)	0.199 (0.174)	0.111 (0.201)	0.277 (0.245)	0.543** (0.264)	0.378 (0.276)
unempl_rate_imf	0.097** (0.046)	0.087* (0.046)	0.085* (0.048)	0.091* (0.048)	0.054 (0.052)	0.021 (0.065)	0.06 (0.102)	0.574*** (0.191)	0.701** (0.33)
gross_public_debt	0.051*** (0.012)	0.055*** (0.012)	0.067*** (0.014)	0.077*** (0.016)	0.092*** (0.016)	0.117*** (0.017)	0.139*** (0.017)	0.181*** (0.021)	0.228*** (0.029)
SSMY	-0.927 (0.575)	-1.093** (0.548)	-1.4*** (0.533)	-1.593*** (0.558)	-1.627*** (0.58)	-2.003*** (0.646)	-1.778** (0.712)	-1.843** (0.926)	-2.044 (1.387)
CBDI_index	-0.007 (0.007)	-0.004 (0.007)	-0.001 (0.007)	0.007 (0.007)	0.009 (0.009)	0.017* (0.01)	0.026** (0.011)	0.058*** (0.013)	0.067*** (0.019)

Table 11: Fixed Effects Quantile Regression (2014-2019): Households

(a) PFEQR with $\lambda = 1$

	10%	20%	30%	40%	50%	60%	70%	80%	90%
(Intercept)	5.407 (7.377)	6.925 (6.38)	9.329 (6.166)	10.988* (5.792)	11.883** (5.502)	13.132** (5.499)	14.498** (5.821)	16.691** (6.468)	15.997** (8.064)
CET1_ratio_tr	-0.218** (0.093)	-0.136* (0.069)	-0.132** (0.058)	-0.146*** (0.054)	-0.142** (0.056)	-0.165*** (0.059)	-0.139** (0.064)	-0.155** (0.066)	-0.139* (0.081)
LEVER_tr_ratio	0.356* (0.195)	0.295* (0.169)	0.257 (0.168)	0.242 (0.17)	0.24 (0.183)	0.264 (0.196)	0.252 (0.207)	0.213 (0.225)	0.168 (0.249)
RORAC	-0.032 (0.027)	-0.059* (0.035)	-0.079** (0.036)	-0.093** (0.039)	-0.099** (0.04)	-0.099** (0.046)	-0.086 (0.055)	-0.123** (0.061)	-0.091 (0.07)
log(BankSize_tran)	-0.312 (0.351)	-0.468 (0.33)	-0.59* (0.33)	-0.696** (0.333)	-0.855** (0.333)	-0.926*** (0.326)	-1.018*** (0.322)	-1.166*** (0.327)	-1.549*** (0.39)
Cost_to_Income	-0.005 (0.013)	-0.003 (0.014)	-0.002 (0.016)	-0.011 (0.018)	-0.014 (0.019)	-0.02 (0.019)	-0.025 (0.019)	-0.039** (0.02)	-0.044** (0.019)
NIM_tr	0.637 (0.693)	0.955 (0.585)	0.894 (0.593)	0.82 (0.63)	1.092 (0.676)	1.245* (0.708)	1.589** (0.742)	1.789** (0.789)	2.722*** (0.985)
Stgth_legal_index	-0.188 (0.246)	0.048 (0.22)	0.063 (0.204)	0.114 (0.214)	0.185 (0.219)	0.247 (0.214)	0.258 (0.225)	0.254 (0.251)	0.38 (0.352)
Reco_rate_dol	-0.11** (0.046)	-0.141*** (0.044)	-0.143*** (0.041)	-0.15*** (0.04)	-0.146*** (0.039)	-0.146*** (0.039)	-0.155*** (0.043)	-0.141*** (0.049)	-0.122** (0.06)
infl_perc	-0.121 (0.122)	-0.117 (0.286)	-0.139 (0.243)	-0.135 (0.248)	-0.384 (0.253)	-0.584** (0.294)	-0.923*** (0.328)	-1.221*** (0.423)	-1.465** (0.586)
curr_acc_bal	0.459*** (0.444)	0.465*** (0.106)	0.413*** (0.106)	0.431*** (0.106)	0.441*** (0.112)	0.493*** (0.112)	0.522*** (0.122)	0.512*** (0.145)	0.719*** (0.16)
GDP_growth_perc	0.43 (0.318)	0.34 (0.313)	0.396 (0.316)	0.568* (0.303)	0.647** (0.284)	0.503* (0.276)	0.47* (0.28)	0.691*** (0.228)	0.669** (0.288)
unempl_rate_imf	0.947*** (0.197)	1.005*** (0.188)	1.008*** (0.2)	1.061*** (0.206)	1.083*** (0.203)	1.115*** (0.195)	1.126*** (0.21)	1.19*** (0.22)	1.246*** (0.282)
gross_public_debt	0.113*** (0.024)	0.124*** (0.021)	0.121*** (0.023)	0.131*** (0.025)	0.137*** (0.024)	0.144*** (0.024)	0.152*** (0.026)	0.151*** (0.03)	0.192*** (0.036)
SSMY	-4.556*** (1.104)	-4.334*** (0.867)	-4.304*** (0.891)	-4.478*** (0.907)	-4.613*** (0.846)	-4.829*** (0.842)	-4.907*** (0.886)	-4.895*** (0.962)	-5.366*** (1.088)
CBDI_index	-0.005 (0.017)	-0.007 (0.013)	-0.008 (0.013)	-0.008 (0.013)	-0.004 (0.013)	-0.004 (0.012)	0.002 (0.012)	0.005 (0.014)	0.011 (0.017)

(b) PFEQR with $\lambda = 2.5$

	10%	20%	30%	40%	50%	60%	70%	80%	90%
(Intercept)	0.779 (6.156)	6.449 (5.509)	8.397* (4.804)	10.078** (4.862)	12.633** (5.116)	11.279** (5.523)	8.826 (6.111)	16.309** (7.588)	21.654** (10.025)
CET1_ratio_tr	-0.267*** (0.094)	-0.153* (0.086)	-0.167** (0.069)	-0.165** (0.067)	-0.165** (0.067)	-0.162** (0.075)	-0.09 (0.082)	-0.115 (0.083)	-0.139 (0.103)
LEVER_tr_ratio	0.23* (0.129)	0.113 (0.144)	0.127 (0.156)	0.133 (0.183)	0.166 (0.23)	0.311 (0.28)	0.366 (0.327)	0.375 (0.378)	0.31 (0.448)
RORAC	-0.059 (0.036)	-0.081* (0.043)	-0.133*** (0.045)	-0.137*** (0.049)	-0.154*** (0.055)	-0.196*** (0.06)	-0.169** (0.068)	-0.155** (0.071)	-0.161** (0.077)
log(BankSize_tran)	-0.064 (0.295)	-0.343 (0.276)	-0.516* (0.27)	-0.641** (0.282)	-0.853*** (0.297)	-1.037*** (0.329)	-1.261*** (0.329)	-1.38*** (0.449)	-1.965*** (0.611)
Cost_to_Income	-0.003 (0.016)	-0.001 (0.015)	0.001 (0.018)	-0.006 (0.02)	-0.004 (0.021)	-0.011 (0.023)	-0.005 (0.023)	-0.032 (0.023)	-0.045* (0.023)
NIM_tr	1.536** (0.633)	1.743** (0.68)	1.724*** (0.643)	1.986*** (0.74)	2.716*** (0.836)	3.098*** (0.883)	3.825*** (0.904)	4.064*** (1.027)	5.415*** (1.488)
Stgth_legal_index	-0.439 (0.282)	-0.089 (0.217)	0.004 (0.201)	0.071 (0.213)	-0.023 (0.235)	-0.008 (0.262)	0.078 (0.282)	0.212 (0.303)	0.184 (0.471)
Reco_rate_dol	-0.086** (0.035)	-0.147*** (0.038)	-0.153*** (0.035)	-0.156*** (0.036)	-0.148*** (0.04)	-0.116*** (0.044)	-0.113** (0.051)	-0.136** (0.057)	-0.082 (0.065)
infl_perc	-0.627* (0.359)	-0.643** (0.313)	-0.844*** (0.266)	-1.174*** (0.285)	-1.541*** (0.326)	-1.506*** (0.413)	-1.441*** (0.522)	-2.603*** (0.643)	-3.096*** (0.714)
curr_acc_bal	0.548*** (0.149)	0.514*** (0.117)	0.534*** (0.109)	0.589*** (0.109)	0.613*** (0.109)	0.597*** (0.107)	0.629*** (0.122)	0.613*** (0.132)	0.622*** (0.201)
GDP_growth_perc	0.98*** (0.289)	1.022*** (0.291)	1.179*** (0.302)	1.113*** (0.311)	1.033*** (0.306)	1.015*** (0.291)	1.246*** (0.275)	0.988*** (0.257)	0.979** (0.422)
unempl_rate_imf	0.445*** (0.162)	0.506*** (0.126)	0.599*** (0.108)	0.657*** (0.112)	0.659*** (0.117)	0.76*** (0.143)	0.967*** (0.16)	0.837*** (0.178)	0.901*** (0.331)
gross_public_debt	0.147*** (0.029)	0.156*** (0.022)	0.158*** (0.021)	0.163*** (0.023)	0.167*** (0.023)	0.168*** (0.023)	0.178*** (0.025)	0.188*** (0.029)	0.181*** (0.045)
SSMY	-2.888*** (1.103)	-3.787*** (0.941)	-4.019*** (0.847)	-4.258*** (0.813)	-4.793*** (0.825)	-4.486*** (0.83)	-4.781*** (0.878)	-4.11*** (1.001)	-3.932*** (1.434)
CBDI_index	-0.006 (0.012)	-0.014 (0.011)	-0.01 (0.01)	-0.003 (0.011)	0.001 (0.012)	0.006 (0.013)	0.01 (0.015)	0.001 (0.017)	0.013 (0.023)

(c) PFEQR with $\lambda = 5$

	10%	20%	30%	40%	50%	60%	70%	80%	90%
(Intercept)	0.724 (5.721)	5.601 (5.455)	9.433* (5.322)	13.25** (5.617)	14.636** (5.849)	9.003 (7.058)	13.131 (8.177)	27.653*** (9.356)	19.958 (13.175)
CET1_ratio_tr	-0.155* (0.086)	-0.143 (0.091)	-0.159* (0.082)	-0.153** (0.077)	-0.197** (0.088)	-0.179* (0.103)	-0.127 (0.1)	-0.122 (0.101)	-0.166 (0.141)
LEVER_tr_ratio	0.096 (0.089)	0.083 (0.116)	0.034 (0.154)	0.053 (0.23)	0.145 (0.328)	0.521 (0.419)	0.559 (0.5)	0.378 (0.533)	0.966 (0.67)
RORAC	-0.097** (0.041)	-0.169*** (0.058)	-0.155*** (0.058)	-0.2** (0.056)	-0.191*** (0.062)	-0.208*** (0.073)	-0.242*** (0.081)	-0.267*** (0.085)	-0.226*** (0.083)
log(BankSize_tran)	0.007 (0.258)	-0.224 (0.279)	-0.441 (0.294)	-0.695** (0.341)	-1.023*** (0.37)	-0.957* (0.494)	-1.604*** (0.613)	-2.213*** (0.726)	-3.163*** (1.047)
Cost_to_Income	-0.001 (0.014)	0.001 (0.018)	0.001 (0.02)	0 (0.023)	-0.009 (0.024)	-0.008 (0.026)	-0.039 (0.027)	-0.048* (0.027)	-0.038 (0.025)
NIM_tr	1.143** (0.574)	1.494** (0.608)	2.235*** (0.647)	2.382*** (0.802)	3.113*** (0.994)	4.287*** (1.181)	5.588*** (1.412)	5.826*** (1.534)	5.127** (2.05)
Stgth_legal_index	-0.036 (0.276)	-0.03 (0.256)	-0.001 (0.271)	0.081 (0.297)	-0.198 (0.332)	-0.116 (0.369)	0.414 (0.399)	0.258 (0.422)	0.901 (0.652)
Reco_rate_dol	-0.075** (0.038)	-0.106** (0.043)	-0.144*** (0.04)	-0.153*** (0.045)	-0.123** (0.052)	-0.11* (0.057)	-0.111* (0.066)	-0.123* (0.068)	-0.021 (0.083)
infl_perc	-0.583* (0.34)	-0.665* (0.344)	-1.289*** (0.341)	-1.771*** (0.384)	-1.863*** (0.43)	-2.2*** (0.559)	-2.953*** (0.705)	-3.953*** (0.833)	-3.518*** (1)
curr_acc_bal	0.208 (0.16)	0.32** (0.145)	0.528*** (0.142)	0.513*** (0.146)	0.602*** (0.138)	0.642*** (0.148)	0.707*** (0.146)	0.502*** (0.16)	0.809*** (0.25)
GDP_growth_perc	1.408*** (0.263)	1.277*** (0.238)	1.204*** (0.227)	1.08*** (0.241)	1.275*** (0.273)	1.361*** (0.274)	1.026*** (0.279)	1.067*** (0.275)	1.302** (0.63)
unempl_rate_imf	0.107 (0.163)	0.233* (0.141)	0.3** (0.132)	0.531*** (0.133)	0.568*** (0.133)	0.686*** (0.163)	0.678*** (0.187)	0.645*** (0.214)	1.196*** (0.401)
gross_public_debt	0.097*** (0.034)	0.107*** (0.029)	0.145*** (0.025)	0.14*** (0.027)	0.172*** (0.028)	0.173*** (0.032)	0.213*** (0.034)	0.205*** (0.038)	0.231*** (0.06)
SSMY	-1.46 (1.223)	-2.803*** (1.063)	-3.8*** (0.957)	-4.071*** (0.994)	-4.95*** (1.074)	-4.691*** (1.167)	-4.208*** (1.177)	-3.975*** (1.211)	-2.308 (1.831)
CBDI_index	-0.017 (0.011)	-0.009 (0.011)	-0.009 (0.011)	0.001 (0.013)	0.006 (0.016)	0.002 (0.02)	0.01 (0.023)	0.023 (0.025)	0.005 (0.033)

Table 12: Fixed Effects Quantile Regression (2014–2019): Non-Financial Corporations

(a) PFEQR with $\lambda = 1$

	10%	20%	30%	40%	50%	60%	70%	80%	90%
(Intercept)	-0.181 (4.569)	1.747 (3.723)	3.879 (3.728)	6.265* (3.759)	7.516* (3.884)	9.538** (4.57)	14.158** (6.414)	20.161** (8.732)	29.625 (21.739)
CET1_ratio_tr	-0.018 (0.061)	-0.031 (0.049)	-0.043 (0.038)	-0.043 (0.03)	-0.045* (0.027)	-0.048 (0.031)	-0.068 (0.05)	-0.006 (0.092)	-0.093 (0.127)
LEVER_tr_ratio	-0.125 (0.189)	-0.007 (0.112)	-0.018 (0.089)	-0.071 (0.089)	-0.078 (0.095)	-0.048 (0.129)	0.084 (0.25)	0.525 (0.42)	0.906 (0.61)
RORAC	-0.001 (0.018)	-0.004 (0.018)	-0.015 (0.02)	-0.027 (0.023)	-0.038 (0.027)	-0.04 (0.037)	-0.051 (0.055)	-0.092 (0.063)	-0.107 (0.069)
log(BankSize_tran)	-0.109 (0.206)	-0.173 (0.167)	-0.241 (0.155)	-0.278* (0.162)	-0.347* (0.181)	-0.55*** (0.208)	-0.921*** (0.281)	-1.577*** (0.342)	-1.737*** (0.59)
Cost_to_Income	0 (0.009)	-0.003 (0.008)	-0.011** (0.004)	-0.012*** (0.004)	-0.011** (0.004)	-0.017*** (0.006)	-0.013 (0.009)	-0.015 (0.014)	-0.01 (0.014)
NIM_tr	0.695 (0.444)	0.605 (0.481)	0.665 (0.48)	0.835* (0.493)	0.909* (0.513)	1.141* (0.584)	1.088 (0.925)	2.008 (1.332)	2.653* (1.534)
Stgh_legal_index	-0.135 (0.296)	-0.168 (0.204)	-0.143 (0.179)	-0.187 (0.165)	-0.241 (0.166)	-0.244 (0.192)	-0.376 (0.257)	-0.425 (0.303)	-0.91** (0.44)
Reco_rate_dol	-0.006 (0.036)	-0.016 (0.027)	-0.018 (0.026)	-0.031 (0.026)	-0.033 (0.027)	-0.021 (0.031)	-0.028 (0.043)	-0.035 (0.064)	-0.045 (0.175)
infl_perc	-0.176 (0.249)	-0.15 (0.194)	-0.26 (0.189)	-0.374* (0.204)	-0.497** (0.233)	-0.967*** (0.288)	-1.182*** (0.453)	-2.376*** (0.621)	-3.229*** (0.928)
curr_acc_bal	0.151 (0.232)	0.142 (0.104)	0.115 (0.087)	0.113 (0.08)	0.117 (0.077)	0.065 (0.088)	0.092 (0.119)	0.115 (0.166)	0.001 (0.318)
GDP_growth_perc	0.241 (0.281)	0.159 (0.199)	0.126 (0.185)	0.211 (0.205)	0.335 (0.213)	0.462** (0.227)	0.487** (0.246)	0.403* (0.229)	0.251 (0.408)
unempl_rate_imf	0.045 (0.103)	0.073 (0.083)	0.054 (0.085)	0.047 (0.09)	0.054 (0.1)	0.032 (0.113)	0.096 (0.145)	0.114 (0.186)	0.004 (0.256)
gross_public_debt	0.039 (0.026)	0.042** (0.018)	0.041** (0.017)	0.037** (0.017)	0.038** (0.016)	0.041** (0.018)	0.055** (0.024)	0.078** (0.036)	0.054 (0.073)
SSMY	-0.736 (0.791)	-0.829 (0.595)	-0.535 (0.561)	-0.554 (0.531)	-0.674 (0.53)	-0.53 (0.6)	-1.474* (0.835)	-2.561** (1.14)	-2.018 (1.909)
CBDI_index	0.003 (0.014)	0.001 (0.008)	0.002 (0.008)	0.004 (0.008)	0.007 (0.008)	0.013 (0.01)	0.028** (0.013)	0.052*** (0.017)	0.055** (0.028)

(b) PFEQR with $\lambda = 2.5$

	10%	20%	30%	40%	50%	60%	70%	80%	90%
(Intercept)	1.223 (2.835)	2.517 (2.662)	4.686 (3.084)	6.583* (3.453)	10.022** (4.603)	14.927** (5.856)	29.428*** (7.641)	41.164*** (10.23)	42.912 (28.027)
CET1_ratio_tr	-0.021 (0.036)	-0.03 (0.028)	-0.033 (0.026)	-0.035 (0.024)	-0.048** (0.024)	-0.057 (0.037)	-0.089 (0.067)	-0.091 (0.116)	-0.094 (0.155)
LEVER_tr_ratio	0.044 (0.077)	0.037 (0.075)	-0.022 (0.079)	-0.012 (0.087)	0.015 (0.105)	0.049 (0.182)	0.121 (0.328)	0.814 (0.564)	1.294* (0.74)
RORAC	-0.004 (0.014)	-0.02 (0.015)	-0.043** (0.019)	-0.05** (0.024)	-0.058* (0.032)	-0.067 (0.057)	-0.137* (0.083)	-0.239** (0.1)	-0.158 (0.112)
log(BankSize_tran)	-0.137 (0.116)	-0.154 (0.116)	-0.177 (0.129)	-0.205 (0.147)	-0.419** (0.181)	-0.698*** (0.253)	-1.389*** (0.35)	-2.232*** (0.487)	-2.779*** (0.703)
Cost_to_Income	0.001 (0.005)	-0.004 (0.006)	-0.013*** (0.004)	-0.012*** (0.004)	-0.012** (0.005)	-0.017* (0.009)	-0.015 (0.015)	-0.013 (0.021)	-0.005 (0.022)
NIM_tr	0.339 (0.298)	0.384 (0.332)	0.686* (0.381)	0.797** (0.405)	0.939** (0.475)	1.436** (0.723)	1.016 (1.198)	2.162 (1.741)	3.174 (2.197)
Stgh_legal_index	-0.129 (0.192)	-0.197 (0.147)	-0.154 (0.136)	-0.262* (0.141)	-0.321* (0.165)	-0.412* (0.226)	-0.968*** (0.319)	-1.178*** (0.433)	-1.135** (0.554)
Reco_rate_dol	-0.01 (0.019)	-0.011 (0.019)	-0.026 (0.021)	-0.03 (0.023)	-0.028 (0.028)	-0.037 (0.034)	-0.04 (0.052)	-0.069 (0.085)	-0.045 (0.245)
infl_perc	-0.253* (0.153)	-0.327** (0.166)	-0.438** (0.195)	-0.751*** (0.227)	-1.072*** (0.329)	-1.689*** (0.429)	-2.048*** (0.626)	-3.479*** (0.853)	-4.01*** (1.156)
curr_acc_bal	0.091 (0.146)	0.091 (0.075)	0.098 (0.07)	0.066 (0.064)	0.036 (0.072)	0.031 (0.105)	-0.133 (0.141)	-0.17 (0.24)	-0.206 (0.396)
GDP_growth_perc	0.1 (0.173)	0.158 (0.16)	0.279 (0.184)	0.392* (0.207)	0.404* (0.23)	0.52* (0.293)	0.861** (0.344)	1.145** (0.451)	1.012 (0.948)
unempl_rate_imf	-0.064 (0.059)	-0.059 (0.048)	-0.101** (0.051)	-0.122** (0.054)	-0.105 (0.067)	-0.148 (0.1)	-0.131 (0.161)	-0.136 (0.276)	-0.478 (0.382)
gross_public_debt	0.038** (0.018)	0.038** (0.016)	0.039** (0.016)	0.035** (0.016)	0.033* (0.017)	0.045** (0.022)	0.044 (0.03)	0.052 (0.049)	0.108 (0.084)
SSMY	-0.943* (0.535)	-0.954** (0.481)	-0.679 (0.493)	-0.537 (0.496)	-0.651 (0.544)	-1.067 (0.815)	-2.322** (1.145)	-3.713** (1.621)	-3.06 (2.341)
CBDI_index	0.002 (0.009)	0.003 (0.006)	0.003 (0.006)	0.004 (0.007)	0.012 (0.008)	0.02* (0.012)	0.034** (0.016)	0.073*** (0.024)	0.054 (0.034)

(c) PFEQR with $\lambda = 5$

	10%	20%	30%	40%	50%	60%	70%	80%	90%
(Intercept)	-0.304 (1.727)	2.349 (2.101)	4.639* (2.667)	6.187 (3.821)	7.266 (6.191)	24.195*** (8.616)	47.032*** (11.316)	59.85*** (13.302)	86.97** (35.251)
CET1_ratio_tr	-0.025 (0.021)	-0.029 (0.021)	-0.044* (0.023)	-0.035 (0.027)	-0.044 (0.032)	-0.081 (0.055)	-0.109 (0.109)	-0.134 (0.149)	-0.404** (0.2)
LEVER_tr_ratio	0.053 (0.056)	0.014 (0.071)	0.014 (0.081)	-0.01 (0.1)	0.106 (0.146)	0.153 (0.258)	0.338 (0.453)	0.591 (0.705)	2.856** (1.169)
RORAC	-0.005 (0.015)	-0.037** (0.016)	-0.049** (0.021)	-0.057* (0.03)	-0.059 (0.056)	-0.101 (0.091)	-0.321** (0.138)	-0.277* (0.153)	-0.274 (0.183)
log(BankSize_tran)	0.028 (0.083)	-0.006 (0.099)	-0.072 (0.121)	-0.175 (0.183)	-0.329 (0.28)	-1.034*** (0.391)	-2.059*** (0.576)	-3.183*** (0.661)	-3.62*** (0.986)
Cost_to_Income	-0.001 (0.005)	-0.005 (0.006)	-0.013*** (0.004)	-0.013** (0.005)	-0.018** (0.008)	-0.016 (0.014)	-0.017 (0.026)	-0.007 (0.034)	-0.011 (0.043)
NIM_tr	0.075 (0.221)	0.321 (0.272)	0.58* (0.337)	0.882** (0.415)	1.226* (0.645)	1.253 (1.052)	2.374 (1.652)	3.58 (2.201)	-0.188 (3.203)
Stgh_legal_index	-0.041 (0.102)	-0.137 (0.114)	-0.115 (0.121)	-0.194 (0.167)	-0.183 (0.245)	-0.633* (0.329)	-1.27** (0.495)	-1.504** (0.586)	-2.563*** (0.818)
Reco_rate_dol	-0.004 (0.011)	-0.011 (0.015)	-0.022 (0.018)	-0.022 (0.023)	-0.025 (0.032)	-0.058 (0.046)	-0.086 (0.068)	-0.159 (0.106)	-0.284 (0.287)
infl_perc	-0.128 (0.114)	-0.301** (0.149)	-0.479** (0.205)	-0.879*** (0.315)	-1.363*** (0.433)	-2.086*** (0.656)	-3.358*** (0.947)	-4.002*** (1.044)	-5.937*** (1.535)
curr_acc_bal	0.051 (0.055)	0.031 (0.057)	0.042 (0.061)	0.032 (0.072)	0.064 (0.109)	-0.072 (0.162)	-0.413* (0.24)	-0.013 (0.351)	-0.36 (0.51)
GDP_growth_perc	0.152 (0.116)	0.168 (0.143)	0.165 (0.177)	0.361* (0.217)	0.485* (0.28)	0.468 (0.393)	0.849 (0.568)	1.594** (0.787)	3.678*** (1.381)
unempl_rate_imf	-0.056 (0.035)	-0.086** (0.036)	-0.118*** (0.04)	-0.136** (0.056)	-0.179** (0.088)	-0.244* (0.147)	-0.476** (0.238)	-0.77* (0.401)	-1.355** (0.556)
gross_public_debt	0.014 (0.012)	0.014 (0.013)	0.02 (0.016)	0.023 (0.021)	0.042 (0.026)	0.054* (0.032)	0.086* (0.046)	0.206*** (0.069)	0.22* (0.117)
SSMY	-0.175 (0.344)	-0.089 (0.38)	-0.04 (0.463)	-0.059 (0.636)	-0.231 (0.974)	-1.952 (1.243)	-5.144*** (1.81)	-8.098*** (2.437)	-8.327** (3.422)
CBDI_index	-0.001 (0.005)	-0.003 (0.005)	-0.003 (0.006)	0.003 (0.008)	0.009 (0.012)	0.02 (0.015)	0.036 (0.022)	0.049 (0.03)	0.01 (0.044)

Table 13: Fixed Effects Quantile Regression (2014–2019): Financial Corporations

A.11 Robustness Checks- Bank Specific Variables

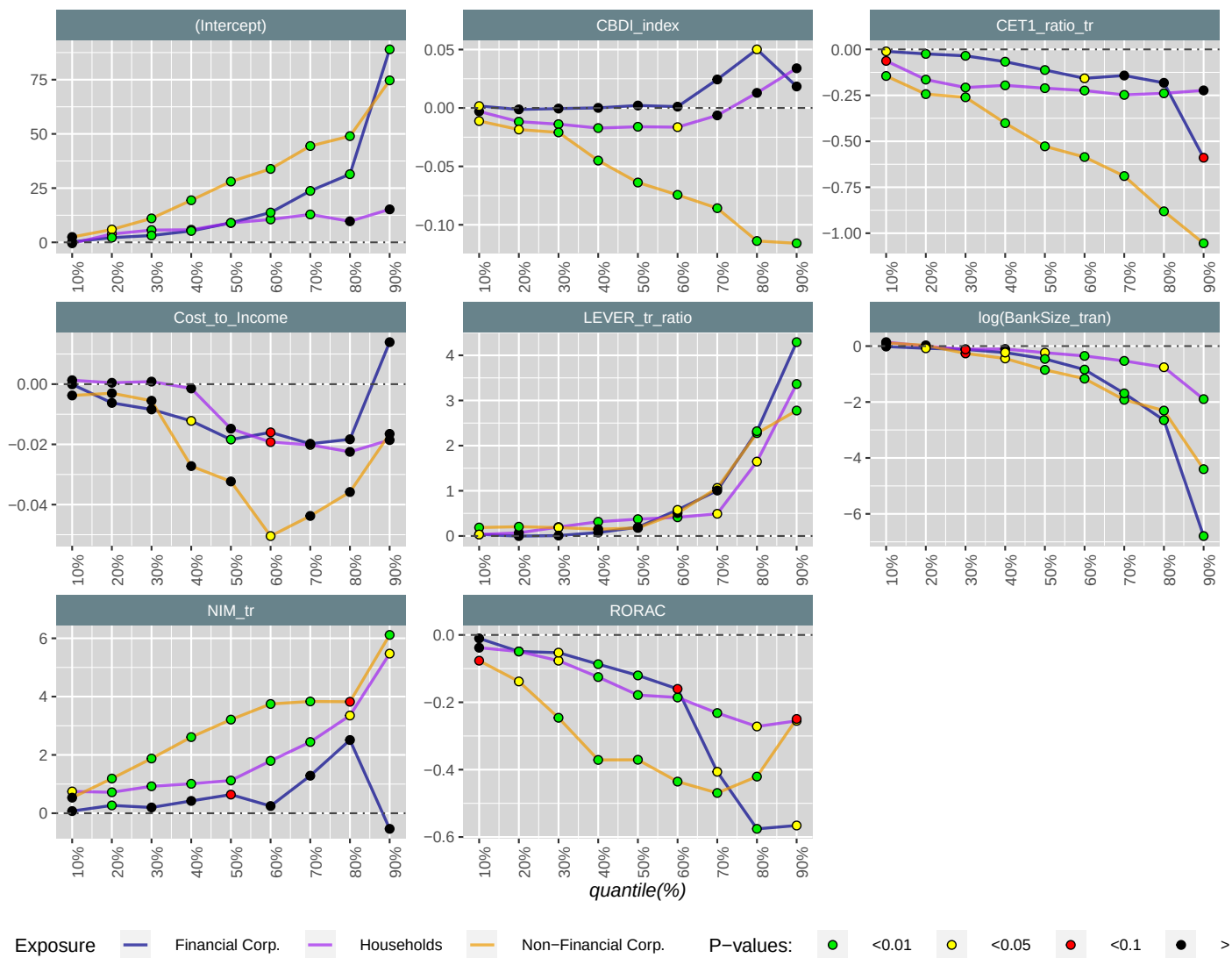
Table 14: Panel Data Models (2014-2019)- Model with only Bank Specific Variables

	<i>panel</i> <i>linear</i>						<i>panel</i> <i>GMM</i>		
	Households		Non-Financial		Financial		Households	Non-Financial	Financial
	RE	FE	RE	FE	RE	FE	SYS	SYS	SYS
lag(NPL_ratio)							1.010*** (0.068)	0.746*** (0.068)	0.658*** (0.229)
CET1_ratio_tr	-0.322*** (0.068)	-0.284*** (0.072)	-0.657*** (0.120)	-0.348*** (0.120)	-0.433*** (0.156)	-0.336 (0.317)	0.017 (0.056)	-0.185* (0.112)	-0.044 (0.127)
LEVER_tr_ratio	0.202 (0.247)	-0.149 (0.280)	0.014 (0.344)	-1.780*** (0.616)	0.805 (0.637)	-0.079 (0.903)	0.036 (0.090)	-0.133 (0.130)	0.141 (0.426)
RORAC	-0.047** (0.019)	-0.042** (0.018)	-0.077 (0.054)	-0.042 (0.036)	-0.153*** (0.057)	-0.078* (0.044)	-0.069** (0.030)	-0.135 (0.086)	-0.199 (0.144)
log(BankSize_tran)	-2.500*** (0.556)	-6.010*** (1.390)	-2.450*** (0.651)	-5.340** (2.140)	-1.830** (0.762)	7.150** (3.320)	0.114 (0.104)	-0.359* (0.199)	-0.269 (0.443)
Cost.to_Income	-0.006 (0.007)	-0.006 (0.006)	-0.018 (0.013)	-0.017 (0.012)	0.017 (0.022)	0.023 (0.029)	-0.008 (0.006)	-0.016 (0.013)	-0.009 (0.006)
NIM_tr	0.319 (0.619)	-0.014 (0.520)	0.884 (1.320)	0.205 (1.430)	0.076 (1.780)	0.330 (2.330)	0.144 (0.551)	0.838 (0.769)	1.010 (0.890)
CBDI_index	-0.006 (0.020)	-0.009 (0.018)	-0.072*** (0.028)	-0.057 (0.041)	-0.014 (0.034)	-0.028 (0.053)	0.005 (0.007)	-0.006 (0.010)	-0.004 (0.018)
Constant	39.600*** (6.800)		53.400*** (9.080)		31.300*** (11.000)				
Hausman Test	28.63		11.38		328.98				
Prob (Hausman Test)	2e-04		0.123		0				
Sargan Test							28.95	29.49	26.81
Prob (Sargan Test)							0.0888	0.0785	0.1406
Arellano-Bond:AR(1)							-1.7	-0.88	-2.84
Prob (AR(1))							0.0895	0.3769	0.0045
Arellano-Bond:AR(2)							0.88	-0.39	0.78
Prob (AR(2))							0.3813	0.6961	0.4349
Observations	619	619	628	628	580	580			
R ²	0.140	0.181	0.121	0.100	0.064	0.033			
Adjusted R ²	0.130	-0.031	0.111	-0.136	0.052	-0.233			
F Statistic	96.500***	15.500*** (df = 7; 491)	80.000***	7.870*** (df = 7; 497)	35.700***	2.210** (df = 7; 454)			

Note: HC-robust standard errors are used

*p<0.1; **p<0.05; ***p<0.01

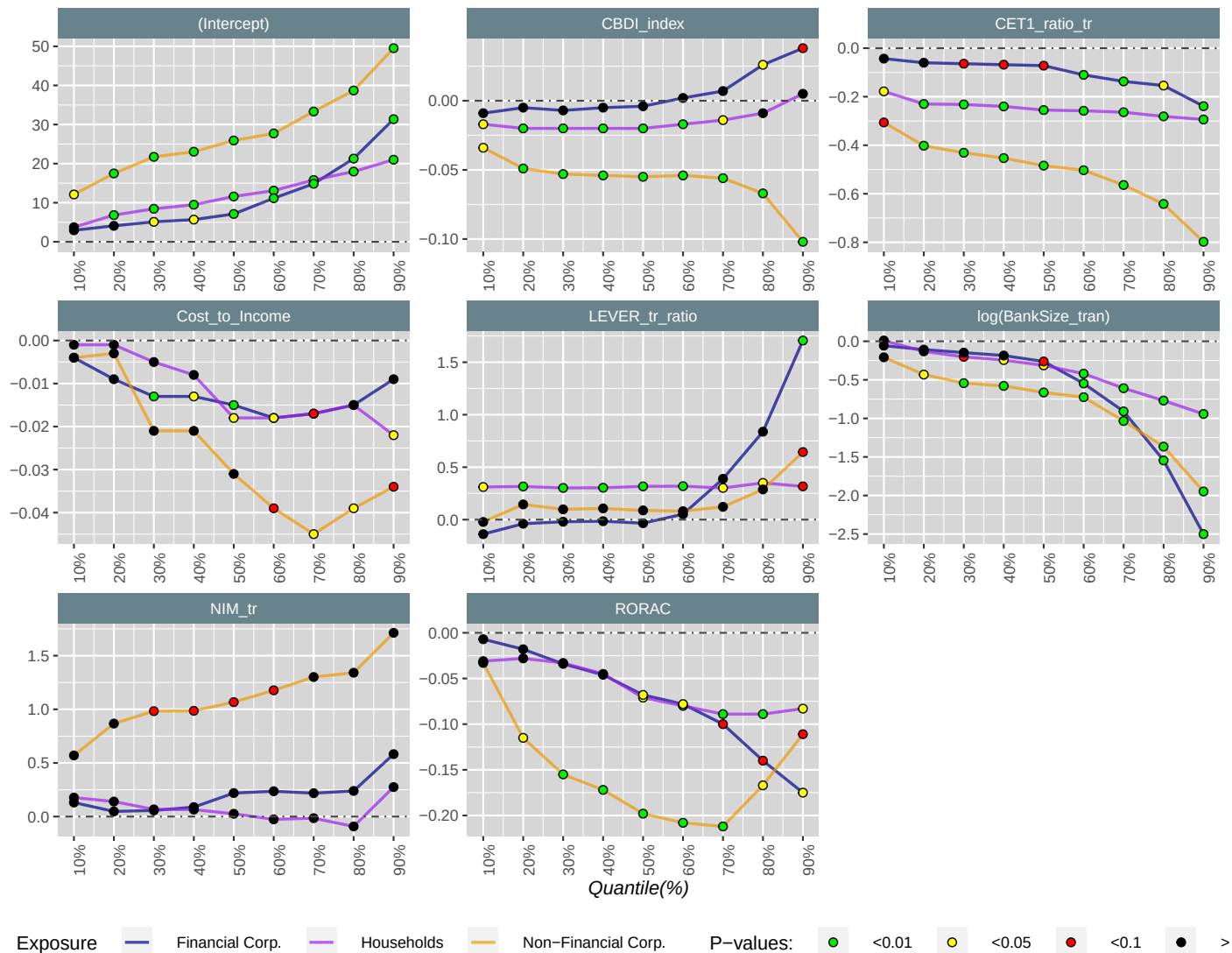
RE: Random Effects, FE: Fixed Effects, SYS: System-GMM, DIFF: Differences-GMM



Source: EBA Transparency Exercises, authors' calculations

Figure 19: Pooled Quantile Regression Graphics- Model with only Bank-specific Variables

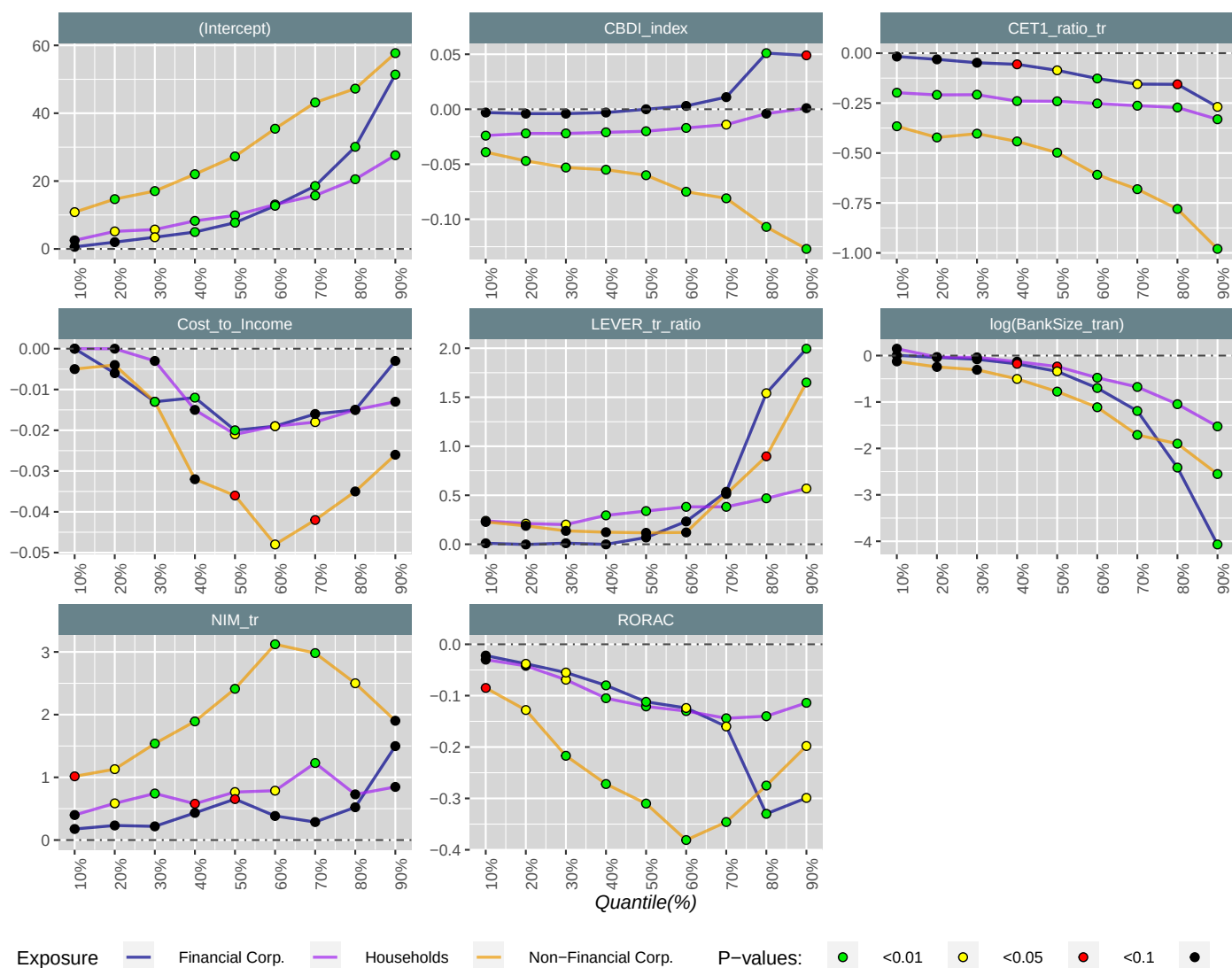
Source: EBA Transparency Exercises, Worldbank and IMF data, authors' calculations



Source: EBA Transparency Exercises, authors' calculations

Figure 20: Fixed Effects Quantile Regression Graphics ($\lambda = 1$) - Model with only Bank-specific Variables

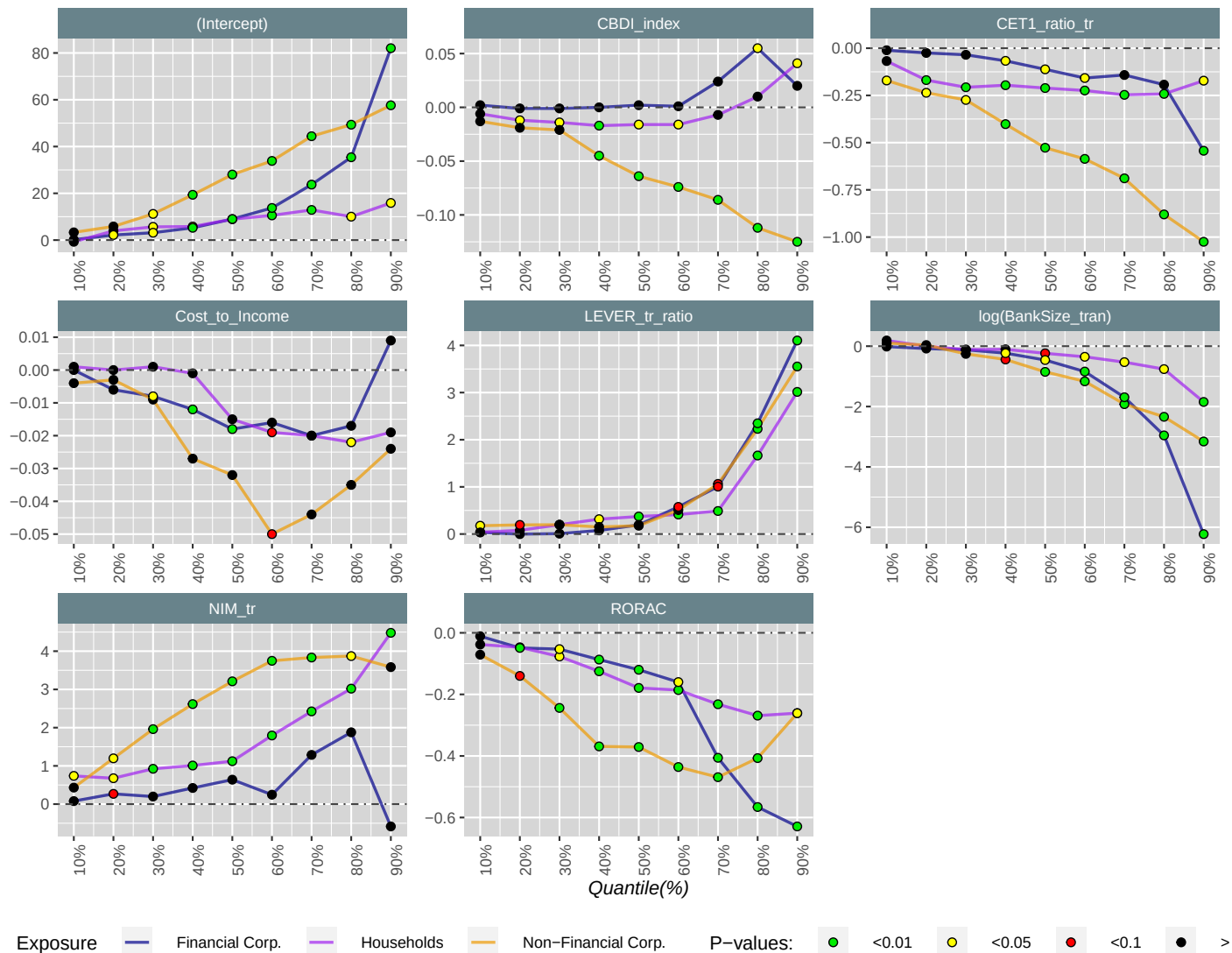
Source: EBA Transparency Exercises, Worldbank and IMF data, authors' calculations



Source: EBA Transparency Exercises, authors' calculations

Figure 21: Fixed Effects Quantile Regression Graphics ($\lambda = 2.5$) - Model with only Bank-specific Variables

Source: EBA Transparency Exercises, Worldbank and IMF data, authors' calculations



Source: EBA Transparency Exercises, authors' calculations

Figure 22: Fixed Effects Quantile Regression Graphics ($\lambda = 5$) - Model with only Bank-specific Variables

Source: EBA Transparency Exercises, Worldbank and IMF data, authors' calculations

A.12 Robustness Checks- Domestic Exposure $\geq 75\%$

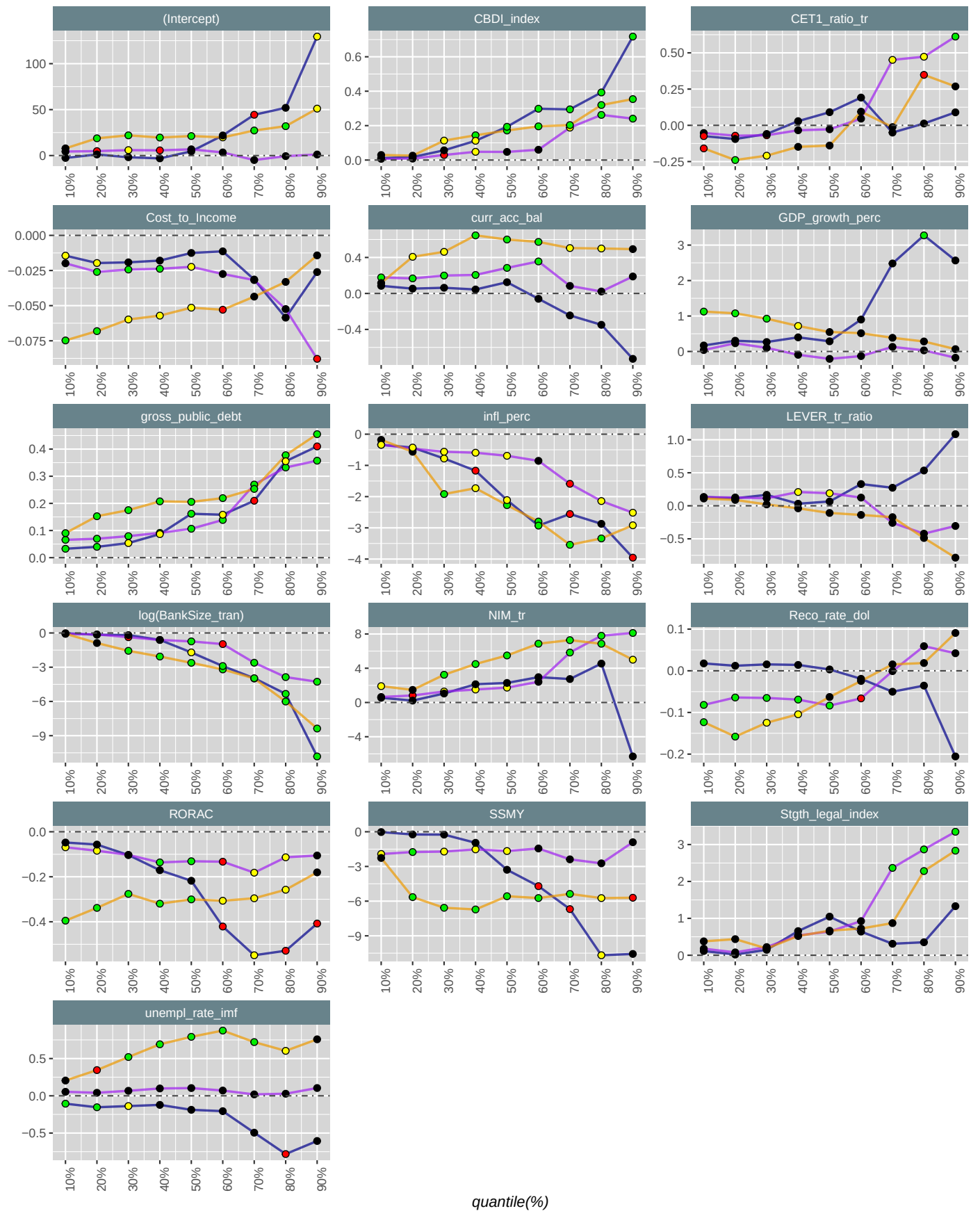
Table 15: Panel Data Models (2014-2019)- Domestic Exposure $\geq 75\%$

	<i>panel linear</i>						<i>panel GMM</i>		
	Households		Non-Financial		Financial		Households	Non-Financial	Financial
	RE	FE	RE	FE	RE	FE	SYS	SYS	SYS
lag(NPL_ratio)							0.465*** (0.123)	0.423*** (0.100)	0.541*** (0.124)
CET1_ratio_tr	0.106 (0.092)	0.009 (0.107)	-0.295** (0.130)	-0.399*** (0.148)	-0.348 (0.239)	-0.410* (0.231)	0.093 (0.162)	-0.127 (0.164)	-0.221 (0.242)
LEVER_tr_ratio	0.391** (0.154)	0.534*** (0.184)	0.292 (0.196)	0.452* (0.235)	1.330* (0.690)	1.670** (0.818)	0.093 (0.140)	-0.019 (0.248)	0.708 (0.624)
RORAC	-0.093** (0.038)	-0.074** (0.036)	-0.098 (0.081)	-0.060 (0.059)	-0.249*** (0.096)	-0.194** (0.085)	-0.165*** (0.035)	-0.166* (0.091)	-0.244** (0.122)
log(BankSize_tran)	-2.380*** (0.498)	-2.480*** (0.718)	-4.290*** (0.668)	-4.390*** (0.840)	-3.490*** (1.050)	-2.920** (1.420)	-1.260** (0.591)	-2.450** (0.991)	-1.300* (0.758)
Cost_to_Income	-0.026*** (0.008)	-0.023*** (0.008)	-0.033** (0.014)	-0.031** (0.013)	-0.008 (0.021)	0.002 (0.018)	-0.029*** (0.006)	-0.042** (0.021)	-0.008 (0.019)
NIM_tr	3.930*** (0.912)	2.330*** (0.852)	3.800*** (1.360)	2.940* (1.520)	-0.964 (3.360)	-2.240 (3.620)	3.830*** (1.270)	4.870*** (1.470)	1.290 (2.370)
Stgth_legal_index	3.370*** (0.499)	3.680*** (0.541)	1.480** (0.599)	1.380* (0.745)	0.806 (0.953)	0.490 (0.956)	1.310 (0.821)	1.090 (0.687)	1.070 (0.739)
Reco_rate_dol	-0.051 (0.038)	-0.089** (0.043)	-0.058 (0.055)	-0.111** (0.055)	-0.073 (0.136)	-0.139 (0.118)	0.011 (0.045)	0.016 (0.055)	-0.068 (0.088)
infl_perc	-1.440*** (0.497)	-1.050** (0.415)	-2.960*** (0.611)	-2.610*** (0.567)	-2.620* (1.390)	-2.710* (1.500)	-1.490** (0.726)	-2.500*** (0.766)	-1.690 (1.450)
curr_acc_bal	0.319** (0.147)	0.312** (0.142)	0.366** (0.174)	0.384* (0.196)	-0.079 (0.439)	0.551 (0.448)	0.150 (0.131)	0.255 (0.241)	0.043 (0.283)
GDP_growth_perc	0.049 (0.253)	-0.106 (0.211)	0.830** (0.413)	0.745* (0.417)	1.570* (0.911)	0.934 (0.849)	-0.276 (0.210)	-0.530 (0.536)	0.809 (0.856)
unempl_rate_imf	-0.129 (0.108)	-0.265** (0.113)	0.495*** (0.164)	0.463*** (0.168)	-0.644*** (0.231)	-0.863** (0.347)	0.183 (0.133)	0.337 (0.274)	-0.283 (0.206)
gross_public_debt	0.358*** (0.035)	0.408*** (0.044)	0.296*** (0.044)	0.293*** (0.059)	0.249*** (0.066)	0.216** (0.087)	0.111* (0.063)	0.138* (0.073)	0.101* (0.056)
SSMY	-2.040 (1.630)	-3.770* (2.040)	-5.340*** (2.010)	-5.820** (2.600)	-4.540 (3.680)	-3.130 (5.380)			
CBDI_index	0.172*** (0.035)	0.153*** (0.036)	0.172*** (0.047)	0.168*** (0.043)	0.227*** (0.085)	0.154* (0.091)	0.133*** (0.048)	0.143** (0.060)	0.162** (0.072)
Constant	-15.000** (6.840)		32.100*** (9.410)		34.100 (21.900)				
Hausman Test	43.91		537.05		28.29				
Prob (Hausman Test) $1e-04$			0		0.0198				
Sargan Test							36.97	39.08	26.93
Prob (Sargan Test)							0.0957	0.0623	0.4674
Arellano-Bond:AR(1)							-12.92	-2.52	-3.03
Prob (AR(1))							0	0.0118	0.0024
Arellano-Bond:AR(2)							0.82	-1.08	-0.36
Prob (AR(2))							0.4102	0.2787	0.7161
Observations	374	374	379	379	339	339			
R ²	0.657	0.632	0.653	0.636	0.308	0.236			
Adjusted R ²	0.643	0.522	0.639	0.527	0.276	-0.009			
F Statistic	668.000***	32.800*** (df = 15; 287)	665.000***	33.900*** (df = 15; 291)	142.000***	5.260*** (df = 15; 256)			

Note: HC-robust standard errors are used

*p<0.1; **p<0.05; ***p<0.01

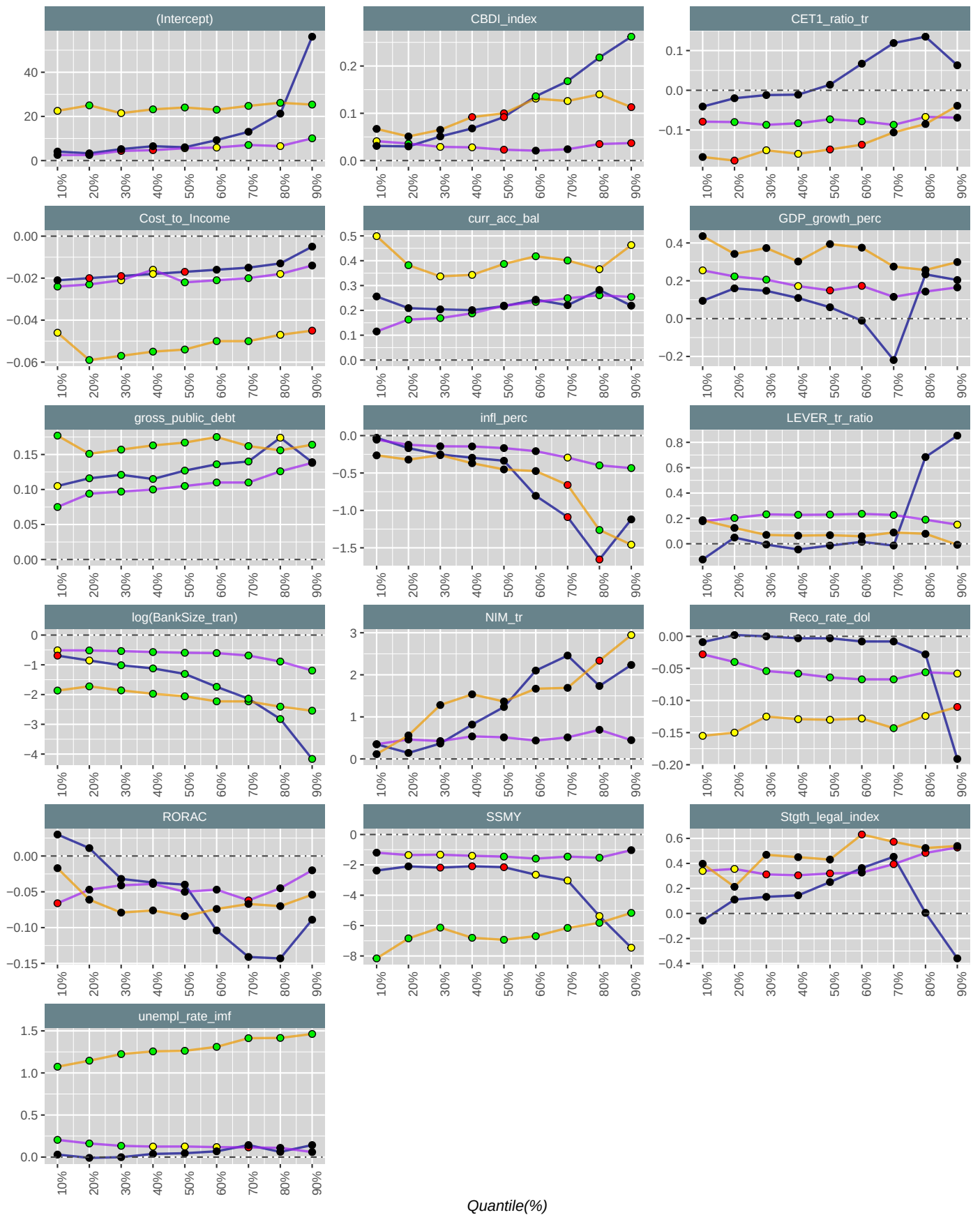
RE:Random Effects, FE:Fixed Effects,SYS: System-GMM, DIFF: Differences-GMM



Source: EBA Transparency Exercises, authors' calculations

Figure 23: Pooled Quantile Regression Graphics- Domestic Exposure $\geq 75\%$

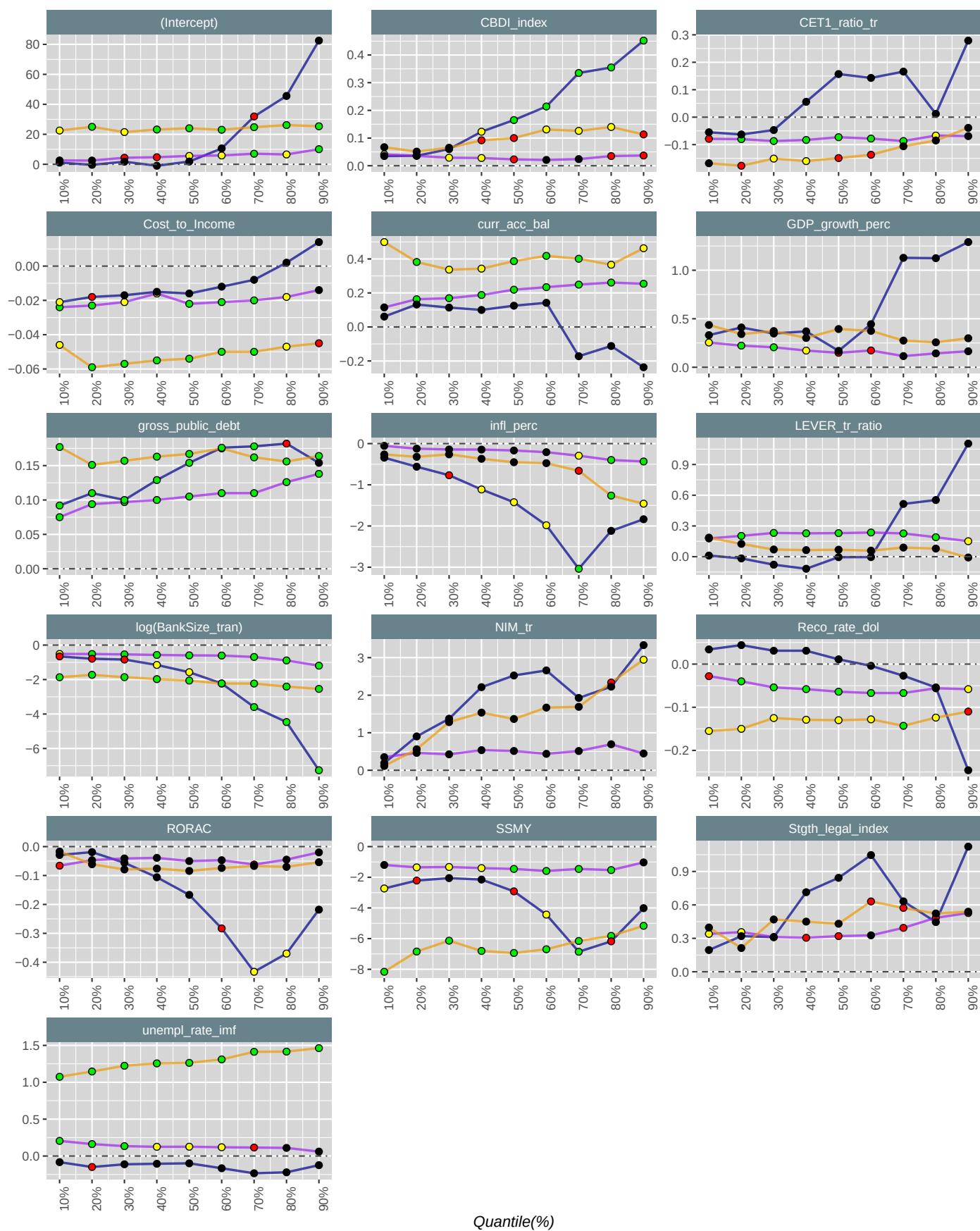
Source: EBA Transparency Exercises, Worldbank and IMF data, authors' calculations



Source: EBA Transparency Exercises, authors' calculations

Figure 24: Fixed Effects Quantile Regression Graphics ($\lambda = 1$) - Domestic Exposure $\geq 75\%$

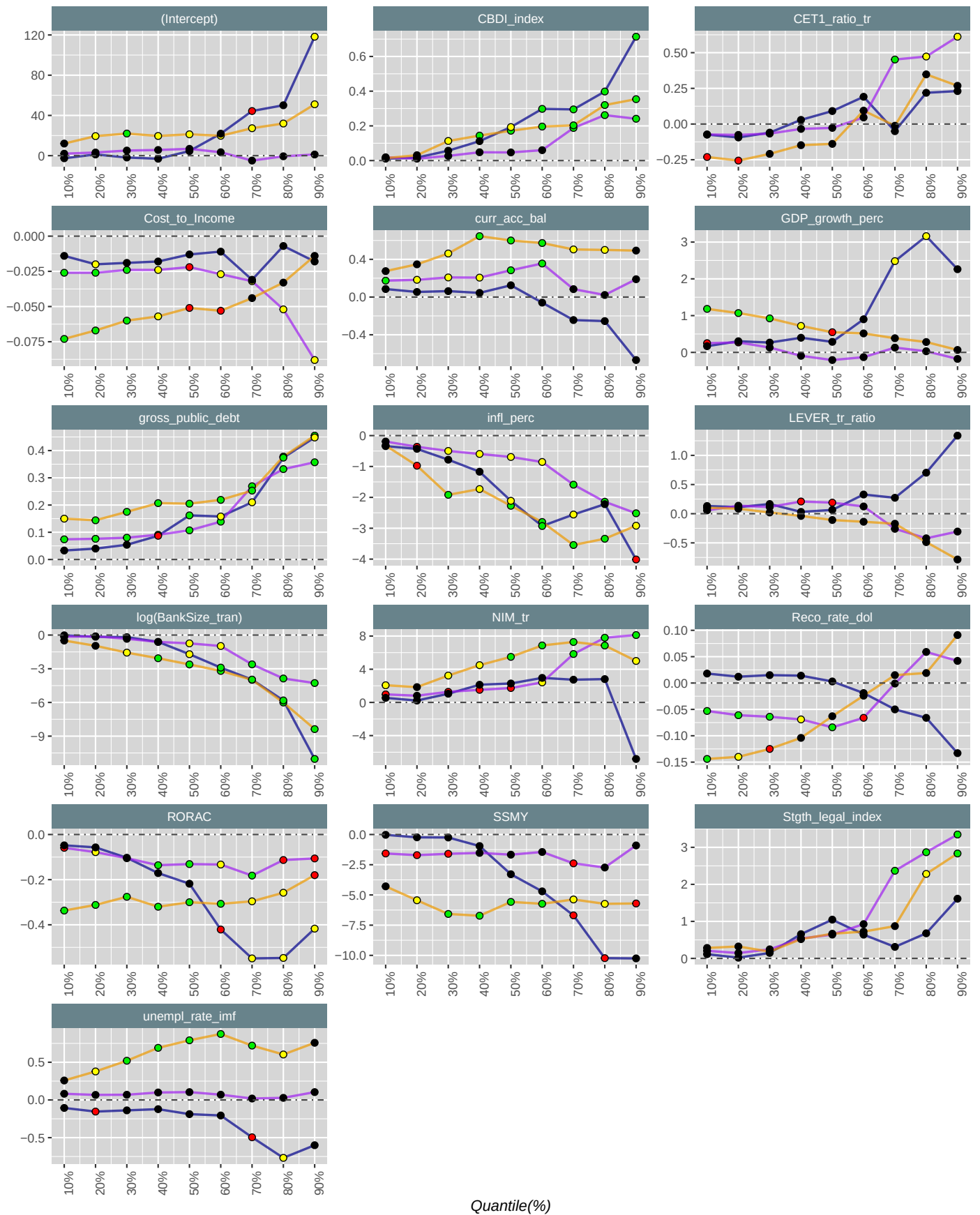
Source: EBA Transparency Exercises, Worldbank and IMF data, authors' calculations



Source: EBA Transparency Exercises, authors' calculations

Figure 25: Fixed Effects Quantile Regression Graphics ($\lambda = 2.5$) - Domestic Exposure $\geq 75\%$

Source: EBA Transparency Exercises, Worldbank and IMF data, authors' calculations



Source: EBA Transparency Exercises, authors' calculations

Figure 26: Fixed Effects Quantile Regression Graphics ($\lambda = 5$) - Domestic Exposure $\geq 75\%$

Source: EBA Transparency Exercises, Worldbank and IMF data, authors' calculations